

Question ID f1bfbed3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: f1bfbed3

Marta Coll and colleagues’ 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri’s 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers’ decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____

Which choice most logically completes the text?

- Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of

A. invertebrate species that had not been described at the time of Bianchi and Morri’s census.
- some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri

B. but treated as indicative of distinct species by Coll and colleagues.
- Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical

C. species of microorganism than Coll and colleagues were.
- the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and

D. colleagues underestimating the number of microorganism species.

ID: f1bfbed3 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically completes the text's discussion of the different counts of species in the Mediterranean Sea. The text states that Coll and colleagues reported almost double the number of species that Bianchi and Morri reported in their study ten years earlier. According to the text, this difference can only be partly attributed to new invertebrate species being described in the years between the two studies, which means there must be an additional factor that made Coll and colleagues' count so much higher than Bianchi and Morri's count. The text goes on to explain that factor: researchers have a relatively poor understanding of microorganisms' morphological variability, or the differences in microorganisms' structure and form. This poor understanding makes it hard to classify microorganisms by species and means that researchers' decisions about classifying microorganisms can have a large effect on the overall species counts that researchers report. Additionally, the text says that the two censuses reported similar numbers of vertebrate, plant, and algal species, which means that the difference in overall species did not come from differences in those categories. Given all this information, it most logically follows that Coll and colleagues may have treated some of the differences among microorganisms as indicative of the microorganisms being different species, whereas Bianchi and Morri treated those differences as variations within species, resulting in Coll and colleagues reporting many more species than Bianchi and Morri did.

Choice A is incorrect because the text explicitly addresses this issue by stating that the description of new invertebrate species in the years between the two studies can explain only part of the difference in the number of species reported by the studies. The focus of the text is on explaining the difference between Coll and colleagues' count and Bianchi and Morri's count that cannot be accounted for by the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's study. Choice C is incorrect because nothing in the text suggests that Bianchi and Morri may have been less sensitive to how much the form and structure of microorganisms vary within the same species than Coll and colleagues were. If Bianchi and Morri had been less sensitive to within-species variation than Coll and colleagues were, Bianchi and Morri would likely have reported more species than Coll and colleagues did, since less sensitivity to within-species variation would lead researchers to classify as different species microorganisms that more sensitive researchers would classify as variations within the same species. The text indicates, however, that Bianchi and Morri reported far fewer species than Coll and colleagues did; since the text also excludes other explanations for this difference, it suggests that in fact Bianchi and Morri were more sensitive to within-species variation than Coll and colleagues were, leading Bianchi and Morri to report fewer overall species. Choice D is incorrect because the text is focused on explaining why Coll and colleagues reported many more species than Bianchi and Morri did, and an underestimate of the number of microorganism species by Coll and colleagues would not explain that difference—it would suggest, in fact, that the difference in the number of species should have been even larger.

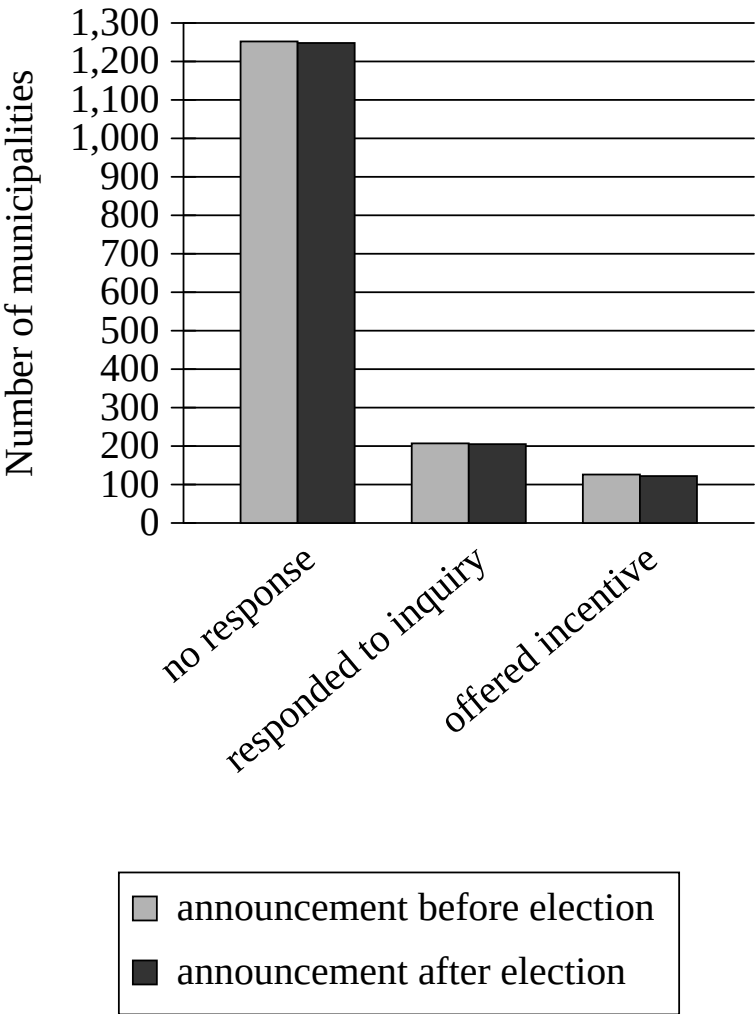
Question Difficulty: Hard

Question ID a15b3219

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: a15b3219

Municipalities’ Responses to Inquiries about Potential Incentives for Firm



In the United States, firms often seek incentives from municipal governments to expand to those municipalities. A team of political scientists hypothesized that municipalities are much more likely to respond to firms and offer incentives if expansions can be announced in time to benefit local elected officials than if they can’t. The team contacted officials in thousands of municipalities, inquiring about incentives for a firm looking to expand and indicating that the firm would announce its expansion on a date either just before or just after the next election.

Which choice best describes data from the graph that weaken the team’s hypothesis?

- A large majority of the municipalities that received an inquiry mentioning plans for an announcement before the next election didn’t respond to the inquiry.
- The proportion of municipalities that responded to the inquiry or offered incentives didn’t substantially differ across the announcement timing conditions.

Only around half the municipalities that responded to inquiries mentioning plans for an announcement before the next election offered incentives.

Of the municipalities that received an inquiry mentioning plans for an announcement date after the next election, more than 1,200 didn't respond and only around 100 offered incentives.

ID: a15b3219 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The lighter bars show what happened when the announcement was to come before the election, and the darker bars show what happened when the announcement was to come after the election. For all three of the outcomes, the light and dark bars are virtually the same, demonstrating that the announcement timing didn't actually make a difference.

Choice A is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement after election" data for comparison. Choice C is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement after election" data for comparison. Choice D is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement before election" data for comparison.

Question Difficulty: Hard

Question ID ed314256

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: ed314256

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play’s refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue’s immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More*’s use of its performance space?

- The choice of a New York City venue likely enabled the play’s creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.

Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.

The production’s dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.

Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play’s narrative than audience members who depart from that itinerary.

ID: ed314256 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The text says that the production’s use of its large, winding space has a very specific effect on the audience. Given that the space itself is so important to creating this effect, it would be difficult to reproduce the production in a different space.

Choice A is incorrect. The fact that the venue is in New York City isn’t connected to the experimental nature of the performance. It’s the size of the venue, not its location in New York, that affects the theatrical experience. Choice B is incorrect. The text never suggests that audience members are disappointed because they can’t see the entire production. In fact, it suggests that that’s an important part of the experience. Choice D is incorrect. The text doesn’t mention a recommended itinerary for audience members.

Question Difficulty: Hard

Question ID 458b4a11

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 458b4a11

To understand how temperature change affects microorganism-mediated cycling of soil nutrients in alpine ecosystems, Eva Kaštovská et al. collected plant-soil cores in the Tatra Mountains at elevations around 2,100 meters and transplanted them to elevations of 1,700–1,800 meters, where the mean air temperature was warmer by 2°C. Microorganism-mediated nutrient cycling was accelerated in the transplanted cores; crucially, microorganism community composition was unchanged, allowing Kaštovská et al. to attribute the acceleration to temperature-induced increases in microorganism activity.

It can most reasonably be inferred from the text that the finding about the microorganism community composition was important for which reason?

- It provided preliminary evidence that microorganism-mediated nutrient cycling was accelerated in the transplanted A. cores.
- It suggested that temperature-induced changes in microorganism activity may be occurring at increasingly high B. elevations.
- C. It ruled out a potential alternative explanation for the acceleration in microorganism-mediated nutrient cycling.
- It clarified that microorganism activity levels in the plant-soil cores varied depending on which microorganisms D. comprised the community.

ID: 458b4a11 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it accurately describes why the finding about the microorganism community composition was important. The text describes an experiment by Eva Kaštovská and her team in which they collected plant-soil cores at one elevation and transplanted them to sites at a lower elevation, where the mean air temperature was warmer. Kaštovská and her team observed that microorganism-mediated nutrient cycling was accelerated in the transplanted cores and that "crucially, microorganism community composition was unchanged," which allowed the team to attribute the acceleration to changes in microorganism activity brought about by the difference in temperature. This strongly implies that the team wouldn't have been able to make that attribution otherwise, meaning that a change in microorganism composition represented another possible explanation for the acceleration that had to be ruled out.

Choice A is incorrect. Although the text says microorganism-mediated cycling of soil nutrients increased in the transplanted cores, this is unrelated to what's important about the finding that the microorganism composition didn't change—that it allowed the team to attribute the change in activity solely to the change in temperature. Choice B is incorrect. Although the text compares activity in one core at two different elevations, the text doesn't address changes in activity at various elevations over time. Choice D is incorrect. Although different microorganisms likely exhibit different levels of activity, the text indicates that there was no change in microorganism composition, and there is nothing in the text about different microorganisms having different activity levels.

Question Difficulty: Hard

Question ID 1ba5ad7a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 1ba5ad7a

Many literary theorists distinguish between *fabula*, a narrative’s content, and *syuzhet*, a narrative’s arrangement and presentation of events. In the film *The Godfather Part II*, the *fabula* is the story of the Corleone family, and the *syuzhet* is the presentation of the story as it alternates between two timelines in 1901 and 1958. But literary theorist Mikhail Bakhtin maintained that *fabula* and *syuzhet* are insufficient to completely describe a narrative—he held that systematic categorizations of artistic phenomena discount the subtle way in which meaning is created by interactions between the artist, the work, and the audience.

Which choice best states the main idea of the text?

- Literary theorist Mikhail Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.
- Literary theorist Mikhail Bakhtin claimed that meaning is not inherent in a narrative but is created when an audience encounters a narrative so that narratives are interpreted differently by different people.
- The storytelling methods used in *The Godfather Part II* may seem unusually complicated, but they can be easily understood when two concepts from literary theory are utilized.
- Narratives that are told out of chronological order are more difficult for audiences to understand than are narratives presented chronologically.

ID: 1ba5ad7a Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. The text begins by explaining that many literary theorists rely on the concepts of *fabula* (a narrative's content) and *syuzhet* (a narrative's arrangement and presentation of events) and illustrates these concepts by explaining how they can be applied to the film *The Godfather Part II*. The text then discusses how Mikhail Bakhtin, a literary theorist, argued that *fabula* and *syuzhet* can't fully describe a narrative, since systematic categorizations such as these fail to account for all the ways in which interactions between the artist, the work, and the audience produce meaning. Thus, the main idea is that Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.

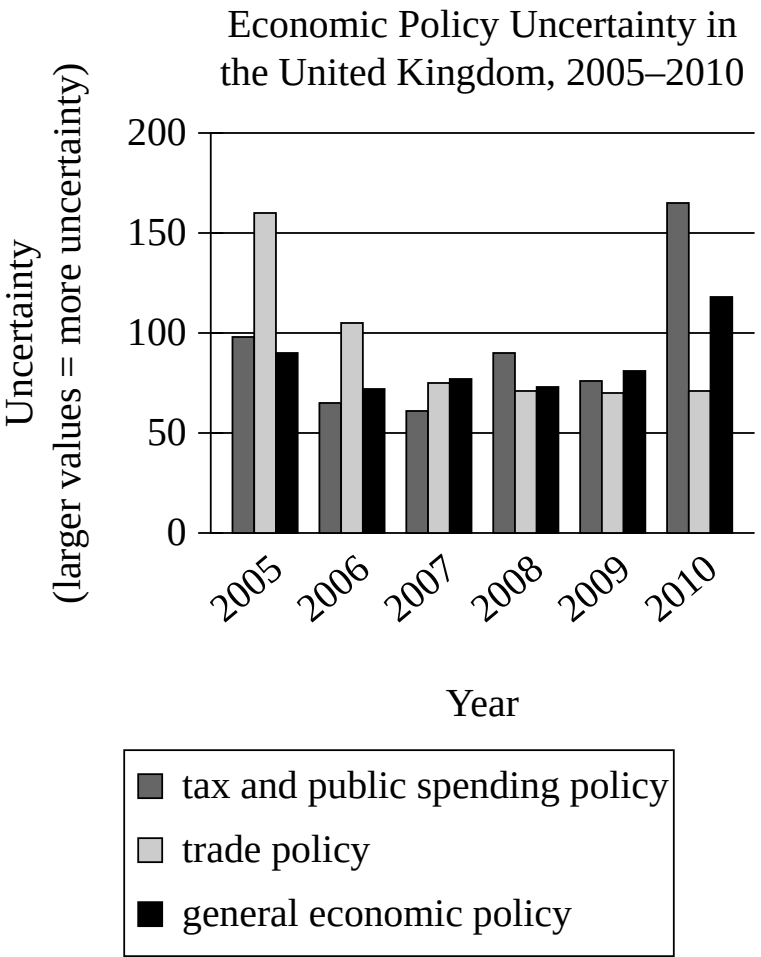
Choice B is incorrect because according to the text, Mikhail Bakhtin believed that meaning was created through the interactions of the artist, narrative, and audience, not simply through the interaction between the audience and narrative; moreover, the text doesn't address whether Bakhtin focused on the ways in which different people interpret narratives differently. Choice C is incorrect. Although the text implies that the storytelling methods used in *The Godfather Part II* are complicated, it discusses the film only to illustrate how the concepts of *fabula* and *syuzhet* may be applied to a narrative. The film's storytelling methods aren't the primary focus of the text. Choice D is incorrect. The text discusses *The Godfather Part II*, whose narrative doesn't adhere to a single chronological order, only to illustrate the concepts of *fabula* (a narrative's content) and *syuzhet* (a narrative's arrangement and presentation of events). The primary focus of this text isn't the structure of this film or of other narratives that are told out of chronological order; moreover, the text doesn't consider whether such structures make it harder for audiences to understand narratives.

Question Difficulty: Hard

Question ID 702eb7e3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 702eb7e3



High levels of public uncertainty about which economic policies a country will adopt can make planning difficult for businesses, but measures of such uncertainty have not tended to be very detailed. Recently, however, economist Sandile Hlatshwayo analyzed trends in news reports to derive measures not only for general economic policy uncertainty but also for uncertainty related to specific areas of economic policy, like tax or trade policy. One revelation of her work is that a general measure may not fully reflect uncertainty about specific areas of policy, as in the case of the United Kingdom, where general economic policy uncertainty _____

Which choice most effectively uses data from the graph to illustrate the claim?

- aligned closely with uncertainty about tax and public spending policy in 2005 but differed from uncertainty about tax and public spending policy by a large amount in 2009.
- was substantially lower than uncertainty about tax and public spending policy each year from 2005 to 2010.
- reached its highest level between 2005 and 2010 in the same year that uncertainty about trade policy and tax and public spending policy reached their lowest levels.

was substantially lower than uncertainty about trade policy in 2005 and substantially higher than uncertainty about trade D. policy in 2010.

ID: 702eb7e3 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it uses data from the graph to effectively illustrate the text's claim about general economic policy uncertainty in the United Kingdom. The graph presents values for economic policy uncertainty in tax and public spending policy, trade policy, and general economic policy in the UK from 2005 to 2010. The graph shows that in 2005, the value for general economic policy uncertainty (approximately 90) was substantially lower than the value for uncertainty about trade policy specifically (approximately 160). It also shows that in 2010, the value for general economic policy uncertainty (approximately 120) was substantially higher than the value for uncertainty about trade policy (approximately 70). The substantial differences between these values in 2005 and 2010 support the claim that a general measure may not fully reflect uncertainty about specific areas of policy.

Choice A is incorrect because the graph shows that the level of general economic policy uncertainty was similar to the level of uncertainty about tax and public spending policy in both 2005 (with values of approximately 90 and 100, respectively) and 2009 (with values of approximately 80 and 75, respectively). Choice B is incorrect because the graph shows that general economic policy uncertainty was higher than uncertainty about tax and public spending policy in 2006, 2007, and 2009, not that it was lower each year from 2005 to 2010. Choice C is incorrect because the graph shows that general economic policy uncertainty reached its highest level in 2010, which was when uncertainty about tax and public spending policy also reached its highest level, not its lowest level.

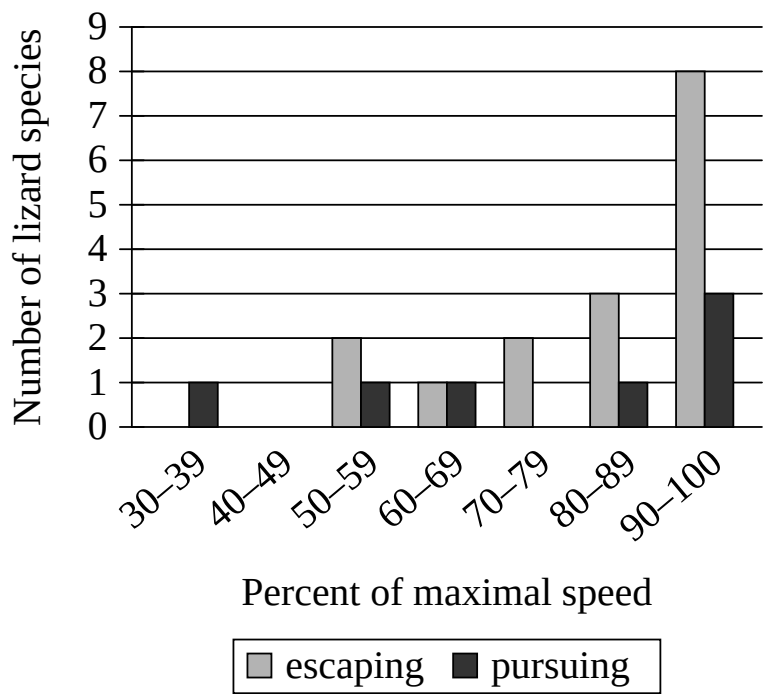
Question Difficulty: Hard

Question ID 1281dfd5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 1281dfd5

Number of Lizard Species by Average Percent of Maximal Speed Used When Pursuing Prey or Escaping Predators



It may seem that the optimal strategy for an animal pursuing prey or escaping predators is to move at maximal speed, but the energy expense of exploiting full speed capacity can disfavor such a strategy even in escape contexts, as evidenced by the fact that _____

Which choice most effectively uses data from the graph to complete the text?

- most lizard species use about the same percentage of their maximal speed when escaping predation as they do when pursuing prey.
- multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.
- more lizard species use, on average, 90%–100% of their maximal speed while escaping predation than use any other percentage of their maximal speed.
- at least 4 lizard species use, on average, less than 100% of their maximal speed while pursuing prey.

ID: 1281dfd5 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that complete the text's discussion of lizard species' use of maximal speed when escaping predators. According to the text, moving at maximal speed (the highest speed possible) requires so much energy that it is not always an effective strategy for animals, even when they are escaping predators. The graph displays data on the average percent of maximal speed used by lizard species while either escaping predators or pursuing prey. The graph categorizes the data for both pursuing and escaping by the number of species using 30%–39% of maximal speed, 40%–49% of maximal speed, 50%–59% of maximal speed, 60%–69% of maximal speed, 70%–79% of maximal speed, 80%–89% of maximal speed, and 90%–100% of maximal speed, respectively. In the graph, there is at least one species in each of the following percent categories for maximal speed while escaping predators: 50%–59%, 60%–69%, 70%–79%, and 80%–89%. Thus, the data in the graph show that multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.

Choice A is incorrect because the data in the graph isn't organized in such a way that a comparison of the percentage of maximal speed used when escaping predation with the percentage used when pursuing prey is possible at the level of individual species. Choice C is incorrect. It is true that in the graph, the percent category with the largest number of species using maximal speed while escaping predators is 90%–100% (8 species total). However, these data don't complete the text, which is concerned instead with how animals are discouraged from using maximal speed even when escaping predators because of the amount of energy required to use it. Choice D is incorrect because these data from the graph pertain to maximal speed while pursuing prey and therefore don't complete the text's discussion of lizard species' use of maximal speed when escaping predators.

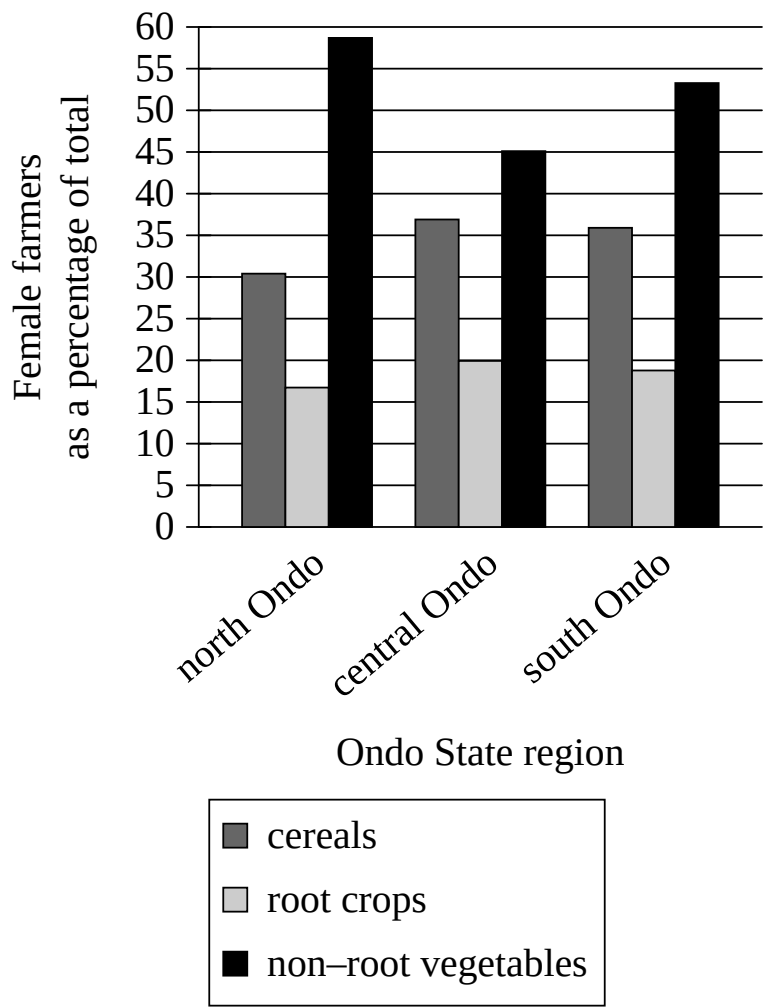
Question Difficulty: Hard

Question ID 403fb4e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 403fb4e4

Percentage of Ondo State
Small-Scale Farmers Who Are
Female, by Main Crop Grown



Geographer Adebayo Oluwole Eludoyin and his colleagues surveyed small-scale farmers in three locations in Ondo State, Nigeria—which has mountainous terrain in the north, an urbanized center, and coastal terrain in the south—to learn more about their practices, like the types of crops they mainly cultivated. In some regions, female farmers were found to be especially prominent in the cultivation of specific types of crops and even constituted the majority of farmers who cultivated those crops; for instance, _____

Which choice most effectively uses data from the graph to complete the example?

- most of the farmers who mainly cultivated cereals and most of the farmers who mainly cultivated non-root vegetables in south Ondo were women.
- more women in central Ondo mainly cultivated root crops than mainly cultivated cereals.

C. most of the farmers who mainly cultivated non–root vegetables in north and south Ondo were women.

D. a relatively equal proportion of women across the three regions of Ondo mainly cultivated cereals.

ID: 403fb4e4 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it uses data from the graph to effectively complete the example of Eludoyin and his colleagues’ findings concerning female farmers in some regions of Ondo State, Nigeria. The graph presents values for the percentage of Ondo State small-scale farmers who are female, by type of crop and region. The graph shows that of the farmers mainly cultivating non-root vegetables, approximately 57% in north Ondo and approximately 54% in south Ondo are female; in other words, most of those farmers are female, which exemplifies the idea that female farmers make up the majority (more than half) of the farmers cultivating specific types of crops in some regions.

Choice A is incorrect because it inaccurately cites data from the graph: the graph shows that in south Ondo, most of the farmers mainly cultivating non-root vegetables are women (approximately 54%), but that only about 35% (less than half) of the farmers mainly cultivating cereals are women. Choice B is incorrect because it inaccurately cites data from the graph: the graph shows that more women in central Ondo mainly cultivate cereals than mainly cultivate root crops (approximately 36% and 20%, respectively). Additionally, it doesn’t effectively complete the example because the graph shows that female farmers don’t make up the majority (more than half) of the farmers for any type of crop in central Ondo. Choice D is incorrect because it doesn’t effectively complete the example; it simply states that a relatively equal proportion of women across the three regions mainly cultivate cereals, which doesn’t address the value for that proportion and thus doesn’t show that a majority (more than half) of the farmers cultivating certain crops are female.

Question Difficulty: Hard

Question ID 94c54577

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 94c54577

While attending school in New York City in the 1980s, Okwui Enwezor encountered few works by African artists in exhibitions, despite New York’s reputation as one of the best places to view contemporary art from around the world. According to an arts journalist, later in his career as a renowned curator and art historian, Enwezor sought to remedy this deficiency, not by focusing solely on modern African artists, but by showing how their work fits into the larger context of global modern art and art history.

Which finding, if true, would most directly support the journalist’s claim?

As curator of the Haus der Kunst in Munich, Germany, Enwezor organized a retrospective of Ghanaian sculptor El Anatsui’s work entitled *El Anatsui: Triumphant Scale*, one of the largest art exhibitions devoted to a Black artist in Europe’s A. history.

In the exhibition *Postwar: Art Between the Pacific and the Atlantic, 1945–1965*, Enwezor and cocurator Katy Siegel brought works by African artists such as Malangatana Ngwenya together with pieces by major figures from other B. countries, like US artist Andy Warhol and Mexico’s David Siqueiros.

Enwezor’s work as curator of the 2001 exhibition *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* showed how African movements for independence from European colonial powers following the Second World War profoundly influenced work by African artists of the period, such as Kamala Ibrahim Ishaq and Thomas C. Mukarobgwa.

Enwezor organized the exhibition *In/sight: African Photographers, 1940 to the Present* not to emphasize a particular aesthetic trend but to demonstrate the broad range of ways in which African artists have approached the medium of D. photography.

ID: 94c54577 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the arts journalist’s claim about Enwezor’s work as a curator and art historian. In the text, the arts journalist asserts that Enwezor wished not just to focus on modern African artists but also to show “how their work fits into the larger context of global modern art and art history,” or how their work relates to artistic developments and work by other artists elsewhere in the world. The description of *Postwar: Art Between the Pacific and the Atlantic, 1945–1965* indicates that Enwezor and Siegel’s exhibition brought works by African artists together with works by artists from other countries, thus supporting the arts journalist’s claim that Enwezor sought to show works by African artists in a context of global modern art and art history.

Choice A is incorrect because it describes a retrospective that wouldn’t support the arts journalist’s claim that Enwezor wanted to show how works by modern African artists fit into the larger context of global modern art and art history. The description of *El Anatsui: Triumphant Scale* indicates that the retrospective focused only on the work of a single African artist, El Anatsui. The description doesn’t suggest that the exhibition showed how El Anatsui’s works fit into a global artistic context. Choice C is incorrect because it describes an exhibition that wouldn’t support the arts journalist’s claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* indicates that the exhibition showed how African artists were influenced by movements for independence from European colonial powers following the Second World War. Although this suggests that Enwezor intended the exhibition to place works by African artists in a political context, it doesn’t indicate that the works were placed in a global artistic context. Choice D is incorrect because it describes an exhibition that wouldn’t support the arts journalist’s claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *In/sight: African Photographers, 1940 to the Present* indicates that the exhibition was intended to reveal the broad range of approaches taken by African photographers, not that the exhibition showed how photography by African artists fits into a global artistic context.

Question Difficulty: Hard

Question ID ce4448b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: ce4448b7

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers’ reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____

Which choice most logically completes the text?

- A. find the disruptions more irritating as time went on.
- B. rate the listening experience as more negative than those whose listening experience was uninterrupted.
- C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.
- D. perceive the volume of the construction noise as growing softer over time.

ID: ce4448b7 Answer

Correct Answer: B

Rationale

Choice B is the best answer. It most logically completes the text. The text tells us that disruptions to an enjoyable experience increase viewers’ enjoyment. It also says that researchers suspect the opposite is true for disruptions to an unpleasant experience. Thus, we can infer that the researchers expect to find that the interrupted unpleasant experience was worse for listeners than the uninterrupted unpleasant experience.

Choice A is incorrect. It doesn’t logically complete the text. The text never makes any claims about how irritating the disruptions themselves are perceived to be. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences. Choice C is incorrect. It doesn’t logically complete the text. The text never makes any claims about how long any experience is perceived to be. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences. Choice D is incorrect. It doesn’t logically complete the text. The text never makes any claims about how interruptions affect the perceived volume of the unpleasant or pleasant experience. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences.

Question Difficulty: Hard

Question ID 7a1877be

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 7a1877be

Nucleobase Concentrations from Murchison Meteorite and Soil Samples in Parts per Billion

Nucleobase	Murchison meteorite sample 1	Murchison meteorite sample 2	Murchison soil sample
Isoguanine	0.5	0.04	not detected
Purine	0.2	0.02	not detected
Xanthine	39	3	1
Adenine	15	1	40
Hypoxanthine	24	1	2

Employing high-performance liquid chromatography—a process that uses pressurized water to separate material into its component molecules—astrochemist Yashiro Oba and colleagues analyzed two samples of the Murchison meteorite that landed in Australia as well as soil from the landing zone of the meteorite to determine the concentrations of various organic molecules. By comparing the relative concentrations of types of molecules known as nucleobases in the Murchison meteorite with those in the soil, the team concluded that there is evidence that the nucleobases in the Murchison meteorite formed in space and are not the result of contamination on Earth.

Which choice best describes data from the table that support the team’s conclusion?

- A. Isoguanine and purine were detected in both meteorite samples but not in the soil sample.
- B. Adenine and xanthine were detected in both of the meteorite samples and in the soil sample.
- C. Hypoxanthine and purine were detected in both the Murchison meteorite sample 2 and in the soil sample.
- D. Isoguanine and hypoxanthine were detected in the Murchison meteorite sample 1 but not in sample 2.

ID: 7a1877be Answer

Correct Answer: A

Rationale

Choice A is the best answer. The researchers concluded that the meteorite’s nucleobases weren’t the result of soil contamination. Presence of nucleobases in the meteorite and not in soil provides evidence that those nucleobases likely didn’t come from the soil.

Choice B is incorrect. This choice doesn’t justify the conclusion. The researchers concluded that the meteorite’s nucleobases weren’t the result of soil contamination. If the nucleobases are present in both the soil and meteorite, then it’s possible that these nucleobases came from the soil. Choice C is incorrect. This choice misreads the table. Purine was not detected in the soil sample. Choice D is incorrect. This choice misreads the table. Both isoguanine and hypoxanthine were detected in both Murchison meteorite samples.

Question Difficulty: Hard

Question ID a68fd3e7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: a68fd3e7

Many of William Shakespeare’s tragedies address broad themes that still appeal to today’s audiences. For instance, *Romeo and Juliet*, which is set in the Italy of Shakespeare’s time, tackles the themes of parents versus children and love versus hate, and the play continues to be read and produced widely around the world. But understanding Shakespeare’s so-called history plays can require a knowledge of several centuries of English history. Consequently, _____

Which choice most logically completes the text?

- A. many theatergoers and readers today are likely to find Shakespeare’s history plays less engaging than the tragedies.
- B. some of Shakespeare’s tragedies are more relevant to today’s audiences than twentieth-century plays.
- C. *Romeo and Juliet* is the most thematically accessible of all Shakespeare’s tragedies.
- D. experts in English history tend to prefer Shakespeare’s history plays to his other works.

ID: a68fd3e7 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of the relative appeal of different kinds of plays by Shakespeare to today’s audiences. According to the text, Shakespeare’s tragedies address broad themes that continue to appeal to today’s audiences. Indeed, the text suggests that these themes are timeless, as illustrated by the example of *Romeo and Juliet*, which the text states is still read and widely performed despite being set in the Italy of Shakespeare’s time. In contrast, the text indicates that audiences and readers may need to be familiar with several centuries of English history in order to understand Shakespeare’s history plays. Because many theatergoers and readers are unlikely to possess such extensive historical knowledge, it follows that they are likely to find Shakespeare’s history plays less engaging than his more accessible tragedies.

Choice B is incorrect because the text never introduces a comparison between Shakespeare’s tragedies and twentieth-century plays, only between Shakespeare’s tragedies and his history plays. Since twentieth-century plays aren’t mentioned, there’s no basis in the text for the idea that some of Shakespeare’s tragedies are more relevant than twentieth-century plays to today’s audiences. Choice C is incorrect. Although the text indicates that *Romeo and Juliet* is thematically accessible to today’s audiences, it doesn’t suggest that *Romeo and Juliet* is more accessible than Shakespeare’s other tragedies. Rather, the text presents *Romeo and Juliet* as an example to support the idea that Shakespeare’s tragedies hold continued appeal for today’s readers and theatergoers. Choice D is incorrect. Although experts in English history would likely possess the knowledge needed to understand Shakespeare’s history plays, the text never mentions such experts or suggests that they would enjoy the history plays more than Shakespeare’s other works.

Question Difficulty: Hard

Question ID bc7b1a04

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: bc7b1a04

Three Studies' Estimated Average Velocity of LMC

Researchers	Study year	Estimated average velocity
Murai and Fujimoto	1980	344 km/s
Kallivayalil and colleagues	2006	378 km/s
Gardiner and colleagues	1994	297 km/s

In 2006, Nitya Kallivayalil and colleagues calculated the most accurate estimate yet of the average velocity (in kilometers per second) of the Large Magellanic Cloud (LMC) galaxy. Before the 2006 study, estimates of the average velocity were low enough for the LMC to maintain an orbit around the Milky Way galaxy, but according to an analysis by Gurtina Besla and colleagues, the estimated velocity from the 2006 study is too high for the LMC to maintain such an orbit. Therefore, if Besla and colleagues are correct, the maximum average velocity for the LMC that would allow it to maintain orbit around the Milky Way is likely _____

Which choice most effectively uses data from the table to complete the statement?

- A. above 344 km/s but below 378 km/s.
- B. above 297 km/s but below 344 km/s.
- C. above 378 km/s.
- D. below 297 km/s.

ID: bc7b1a04 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it uses data from the table to give the range of velocities for the LMC from the 1980 value (344 km/s) to the 2006 value (378 km/s), thereby effectively completing the text. The text indicates that before 2006, all the estimated velocities of the LMC were within the range necessary to maintain orbit around the Milky Way galaxy. It then indicates that, according to Besla and colleagues, the 2006 estimate of 378 km/s is too high to maintain that orbit. This strongly implies that if the 1980 value (344 km/s) is below the orbital threshold, and if Besla and colleagues are correct that the 2006 value (378 km/s) is above that threshold, the maximum orbital velocity for the LMC must be somewhere in the range from above 344 km/s to below 378 km/s.

Choice B is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first estimate that exceeded the velocity needed to maintain orbit around the Milky Way. Thus, the 1980 estimate of 344 km/s and the 1994 estimate of 297 km/s must both be below the maximum possible orbital velocity for the LMC. Choice C is incorrect. The text states that Besla and colleagues' analysis found that the velocity from the 2006 study (378 km/s) was too high for the LMC to maintain orbit around the Milky Way. Therefore, if a velocity of 378 km/s is too high, an even higher velocity will also be too high. Choice D is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first to exceed the velocity range required to maintain orbit around the Milky Way. Thus, the 1994 estimate of 297 km/s must be below the maximum possible orbital velocity for the LMC.

Question Difficulty: Hard

Question ID 3d91c973

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 3d91c973

Mosasaurus were large marine reptiles that lived in the Late Cretaceous period, approximately 100 million to 66 million years ago. Celina Suarez, Alberto Pérez-Huerta, and T. Lynn Harrell Jr. examined oxygen-18 isotopes in mosasaur tooth enamel in order to calculate likely mosasaur body temperatures and determined that mosasaurs were endothermic—that is, they used internal metabolic processes to maintain a stable body temperature in a variety of ambient temperatures. Suarez, Pérez-Huerta, and Harrell claim that endothermy would have enabled mosasaurs to include relatively cold polar waters in their range.

Which finding, if true, would most directly support Suarez, Pérez-Huerta, and Harrell’s claim?

- Mosasaurus’ likely body temperatures are easier to determine from tooth enamel oxygen-18 isotope data than the body
- A. temperatures of nonendothermic Late Cretaceous marine reptiles are.
- Fossils of both mosasaurs and nonendothermic marine reptiles have been found in roughly equal numbers in regions
- B. known to be near the poles during the Late Cretaceous, though in lower concentrations than elsewhere.
- Several mosasaur fossils have been found in regions known to be near the poles during the Late Cretaceous, while
- C. relatively few fossils of nonendothermic marine reptiles have been found in those locations.
- During the Late Cretaceous, seawater temperatures were likely higher throughout mosasaurs’ range, including near the
- D. poles, than seawater temperatures at those same latitudes are today.

ID: 3d91c973 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents the finding that, if true, would best support Suarez, Pérez-Huerta, and Harrell's claim about mosasaurs. The text states that Suarez, Pérez-Huerta, and Harrell's research on mosasaur tooth enamel led them to conclude that mosasaurs were endothermic, which means that they could live in waters at many different temperatures and still maintain a stable body temperature. The researchers claim that endothermy enabled mosasaurs to live in relatively cold waters near the poles. If several mosasaur fossils have been found in areas that were near the poles during the period when mosasaurs were alive and fossils of nonendothermic marine reptiles are rare in such locations, that would support the researchers' claim: it would show that mosasaurs inhabited polar waters but nonendothermic marine mammals tended not to, suggesting that endothermy may have been the characteristic that enabled mosasaurs to include polar waters in their range.

Choice A is incorrect because finding that it's easier to determine mosasaur body temperatures from tooth enamel data than it is to determine nonendothermic reptile body temperatures wouldn't support the researchers' claim. Whether one research process is more difficult than another indicates nothing about the results of those processes and therefore is irrelevant to the issue of where mosasaurs lived and what enabled them to live in those locations. Choice B is incorrect because finding roughly equal numbers of mosasaur and nonendothermic marine reptile fossils in areas that were near the poles in the Late Cretaceous would suggest that endothermy didn't give mosasaurs any particular advantage when it came to expanding their range to include relatively cold polar waters, thereby weakening the researchers' claim rather than supporting it. Choice D is incorrect because finding that the temperature of seawater in the Late Cretaceous was warmer than seawater today wouldn't weaken the researchers' claim. Seawater in the Late Cretaceous could have been warmer than seawater today but still cold enough for endothermy to be advantageous to mosasaurs, so this finding wouldn't provide enough information to either support or weaken the researchers' claim.

Question Difficulty: Hard

Question ID 58e9e497

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: 58e9e497

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____

Which choice most logically completes the text?

- A. those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.

the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to

B. Haudenosaunee farming techniques.

C. Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.

Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the

D. nineteenth century.

ID: 58e9e497 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of Euro-American farmers’ use of Haudenosaunee agricultural techniques. According to the text, some Euro-American farmers were using these techniques in the early nineteenth century despite few of the farmers having seen Haudenosaunee farms. One explanation for these facts might be that the farmers developed techniques on their own that already had been invented centuries earlier by the Haudenosaunee people, but the text explicitly bars, or rules out, this explanation. If Euro-American farmers didn’t learn these techniques from direct observation of Haudenosaunee practices and didn’t invent the techniques independently, then the most logical explanation is that they learned the techniques from other people who were more directly influenced by Haudenosaunee practices than the farmers themselves were. Once they learned about Haudenosaunee agricultural practices, Euro-American farmers could then apply those practices to their own farming.

Choice B is incorrect because the fact that some Euro-American farmers in the northeastern United States were using Haudenosaunee techniques suggests that the techniques were likely useful for the crops the farmers raised, not that the crops typically cultivated by the farmers were not well suited to Haudenosaunee farming techniques. If the farmers’ crops were ill suited to the techniques, it’s unlikely that the farmers would have used those techniques. Choice C is incorrect because the text indicates only that Haudenosaunee agricultural techniques were used by Euro-American farmers in the northeastern United States, not that these techniques were widely used outside this region. Choice D is incorrect because the text states that some Euro-American farmers were using Haudenosaunee farming techniques early in the nineteenth century. This suggests that some Euro-American farmers were beginning to recognize the benefits of these techniques near the start of the century, not that such farmers only began to recognize the benefits of the techniques much later.

Question Difficulty: Hard

Question ID 1a2b29c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 1a2b29c9

The following text is adapted from María Cristina Mena’s 1914 short story “The Vine-Leaf.”

It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop. The doctor’s hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

- A. Many have come to tolerate him despite his disheveled appearance.
- B. Few feel concerned that he will divulge their confidences.
- C. Some dislike how freely he discusses his own family.
- D. Most would be unimpressed by him were it not for his professional expertise.

ID: 1a2b29c9 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The text describes a saying in the capital that Malsufrido keeps more secrets than an archbishop. It also says that when he takes off his hat, “the family secrets do not fly out... but remain nicely bottled up,” suggesting that he will not betray his confidences.

Choice A is incorrect. This choice doesn’t reflect the text. While his hat is large and appears to have been brushed in the wrong direction, Dr. Malsufrido does not seem to be regarded as ill-dressed or disheveled. Choice C is incorrect. This choice is the opposite of what the text says. The secrets of families (his and others’) remain “bottled up” in his head. Choice D is incorrect. This choice isn’t supported by the text. His professional expertise is not discussed in the passage.

Question Difficulty: Hard

Question ID 71904085

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 71904085

Linguist Deborah Tannen has cautioned against framing contentious issues in terms of two highly competitive perspectives, such as pro versus con. According to Tannen, this debate-driven approach can strip issues of their complexity and, when used in front of an audience, can be less informative than the presentation of multiple perspectives in a noncompetitive format. To test Tannen’s hypothesis, students conducted a study in which they showed participants one of three different versions of local news commentary about the same issue. Each version featured a debate between two commentators with opposing views, a panel of three commentators with various views, or a single commentator.

Which finding from the students’ study, if true, would most strongly support Tannen’s hypothesis?

- On average, participants perceived commentators in the debate as more knowledgeable about the issue than
- A. commentators in the panel.
- On average, participants perceived commentators in the panel as more knowledgeable about the issue than the single
- B. commentator.
- On average, participants who watched the panel correctly answered more questions about the issue than those who
- C. watched the debate or the single commentator did.
- On average, participants who watched the single commentator correctly answered more questions about the issue than
- D. those who watched the debate did.

ID: 71904085 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents the finding that, if true, would most strongly support Tannen's hypothesis. According to the text, Tannen's hypothesis is that multiple perspectives presented in a noncompetitive format is more informative than a debate between opposing viewpoints is. If participants who saw a panel of three commentators with various views about an issue answered more questions about the issue correctly than did participants who saw a debate, that would support Tannen's hypothesis since it would show that participants who heard multiple varied perspectives were better informed than were participants who heard a debate between opposing viewpoints.

Choice A is incorrect because finding that participants perceived commentators in the debate as more knowledgeable than commentators in the panel is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists. Choice B is incorrect because finding that participants perceived commentators in the panel as more knowledgeable than a single commentator is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists, and Tannen's hypothesis says nothing about how informative single commentators are. Choice D is incorrect because finding that participants who watched a single commentator answered more questions correctly than participants who watched the debate did wouldn't be relevant to Tannen's hypothesis, which is that hearing multiple varying perspectives is more informative than hearing a debate. Tannen's hypothesis says nothing about how informative single commentators are.

Question Difficulty: Hard

Question ID 04cbeca3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 04cbeca3

In 1534 CE, King Henry VIII of England split with the Catholic Church and declared himself head of the Church of England, in part because Pope Clement VII refused to annul his marriage to Catherine of Aragon. Two years later, Henry VIII introduced a policy titled the Dissolution of the Monasteries that by 1540 had resulted in the closure of all Catholic monasteries in England and the confiscation of their estates. Some historians assert that the enactment of the policy was primarily motivated by perceived financial opportunities.

Which quotation from a scholarly article best supports the assertion of the historians mentioned in the text?

“At the time of the Dissolution of the Monasteries, about 2 percent of the adult male population of England were monks; A. by 1690, the proportion of the adult male population who were monks was less than 1 percent.”

“A contemporary description of the Dissolution of the Monasteries, Michael Sherbrook’s *Falle of the Religious Howses*, recounts witness testimony that monks were allowed to keep the contents of their cells and that the monastery timber B. was purchased by local yeomen.”

“In 1535, the year before enacting the Dissolution of the Monasteries, Henry commissioned a survey of the value of church holdings in England—the work, performed by sheriffs, bishops, and magistrates, began that January and was C. swiftly completed by the summer.”

“The October 1536 revolt known as the Pilgrimage of Grace had several economic motives: high food prices due to a poor harvest the prior year; the Dissolution of the Monasteries, which closed reliable sources of food and shelter for D. many; and rents and taxes throughout Northern England that were not merely high but predatory.”

ID: 04cbeca3 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The fact that Henry VIII commissioned a survey of church holdings just before enacting the Dissolution of the Monasteries suggests that he was interested in the potential profits from claiming their assets. This supports the historians’ assertion.

Choice A is incorrect. This choice describes a potential effect of the Dissolution of the Monasteries. The decrease in the proportion of monks in England isn’t relevant to the question of Henry VIII’s motivations. Choice B is incorrect. Details about how the monks were treated during the dissolution don’t address Henry VIII’s motivations for enacting the policy in the first place. That the monks could keep the content of their cells (their rooms) and sell off the timber they’d harvested has no impact on the value of the monasteries’ estates—their land holdings. Choice D is incorrect. This choice mentions one impact that the Dissolution of the Monasteries contributed to, two years after it happened. But it doesn’t help explain why Henry VIII might have wanted to enact the policy in the first place.

Question Difficulty: Hard

Question ID 2fdfe002

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 2fdfe002

The following text is adapted from Countee Cullen’s 1926 poem “Thoughts in a Zoo.”

They in their cruel traps, and we in ours, Survey each other’s rage, and pass the hours
Commiserating each the other’s woe, To mitigate his own pain’s fiery glow. Man could but little proffer in exchange
Save that his cages have a larger range. That lion with his lordly, untamed heart
Has in some man his human counterpart, Some lofty soul in dreams and visions wrapped,
But in the stifling flesh securely trapped.

Based on the text, what challenge do humans sometimes experience?

- A. They cannot effectively tame certain wild animals because of a lack of compassion.
- B. They cannot focus on setting attainable goals because of a lack of motivation.
- C. They quickly become frustrated when faced with difficult tasks because of a lack of self-control.
- D. They have aspirations that cannot be fulfilled because of certain limitations.

ID: 2fdfe002 Answer

Correct Answer: D

Rationale

Choice D is the best answer. The text metaphorically likens humans to animals in a zoo, suggesting that humans have dreams that they cannot fulfill because they are trapped.

Choice A is incorrect. The speaker says that the lion has an “untamed heart,” but the speaker doesn’t actually mention anything about humans taming wild animals or a lack of compassion. Choice B is incorrect. The speaker doesn’t suggest that humans lack motivation. Rather, the speaker thinks that humans are “trapped” and prevented from achieving their dreams. Choice C is incorrect. The speaker doesn’t mention anything about humans becoming frustrated or lacking self-control. Rather, the speaker thinks that humans are “trapped” and prevented from achieving their dreams.

Question Difficulty: Hard

Question ID 6a6bbac3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 6a6bbac3

Number and Origin of Clamshell Tools Found at Different Levels Below the Surface in Neanderthal Cave

Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
3–4	99	33
6–7	1	0
4–5	2	0
2–3	7	0
5–6	18	7

Studying tools unearthed at a cave site on the western coast of Italy, archaeologist Paola Villa and colleagues have determined that prehistoric Neanderthal groups fashioned them from shells of clams that they harvested from the seafloor while wading or diving or that washed up on the beach. Clamshells become thin and eroded as they wash up on the beach, while those on the seafloor are smooth and sturdy, so the research team suspects that Neanderthals prized the tools made with seafloor shells. However, the team also concluded that those tools were likely more challenging to obtain, noting that _____

Which choice most effectively uses data from the table to support the research team’s conclusion?

- at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were
- A. easier to collect from the beach than to harvest from the seafloor.
- the highest number of tools were at a depth of 3–4 meters below the surface, which suggests that the Neanderthal
- B. population at the site was highest during the related period of time.
- at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that
- C. Neanderthals preferred to use clamshells from the beach because of their durability.
- the higher number of tools at depths of 5–6 meters below the surface in the cave than at depths of 4–5 meters below
- D. the surface suggests that the size of clam populations changed over time.

ID: 6a6bbac3 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to support the researchers’ conclusion about the harvesting of clamshells by Neanderthals for use as tools. The text explains that Neanderthals used clamshells to make tools and that the sturdiest, and therefore the most desirable, shells for this purpose are found on the seafloor, not on the beach. However, the researchers also concluded that the clamshell tools made from shells from the seafloor are rarer than those made from shells from the beach. Meanwhile the table shows that at each depth, the number of tools made from shells from the beach exceeds the number made from the more desirable shells from the seafloor. The fact that the more desirable shells are less common suggests that it was significantly more difficult to harvest shells from the seafloor than from the beach.

Choice B is incorrect because knowing which depth represents the period of time with the highest Neanderthal population does not help answer the question of why the Neanderthals consistently made more tools from the less desirable shells from the beach than they made from the more desirable shells from the seafloor. Choice C is incorrect because it claims that the beach shells are more durable than the seafloor shells, which contradicts the text’s description of shells from the seafloor as smoother and sturdier than shells from the beach. Choice D is incorrect because knowing which depth has the most artifacts or whether the clam population fluctuated does not help explain why tools made from the less desirable shells from the beach outnumber tools made from the more desirable shells from the seafloor.

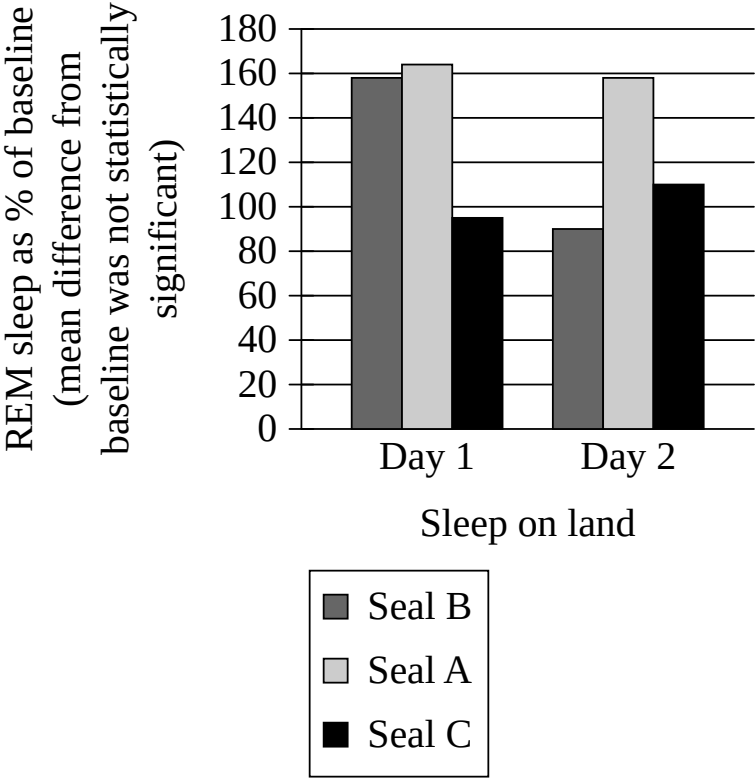
Question Difficulty: Hard

Question ID ccb1ab92

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: ccb1ab92

Fur Seal REM Sleep on Land
after an Extended Period
in Water



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they're in the water, they get almost none. In a study of fur seals' sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study _____

Which choice most effectively uses data from the graph to complete the text?

- A. didn't show significantly less REM sleep during the second day after returning to land than they did during the first day.
- B. showed no significant differences from one another in baseline levels of REM sleep.
- C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.
- D. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.

Correct Answer: C**Rationale**

Choice C is the best answer. If REM sleep were homeostatically regulated in fur seals, then all the seals would compensate with REM levels significantly over baseline after going weeks without REM. We'd also expect the seals to maintain those elevated REM levels for some time. Since seals B and C return very quickly to baseline REM levels, this suggests that REM sleep in fur seals may not be regulated homeostatically.

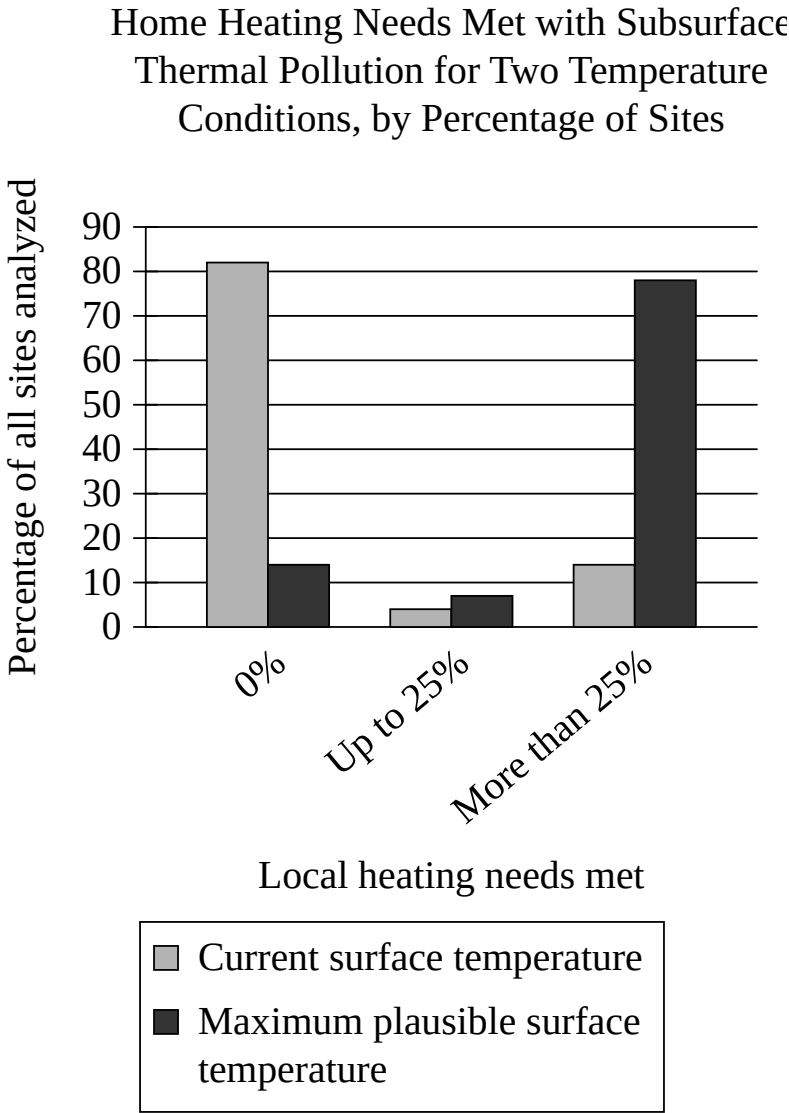
Choice A is incorrect. This doesn't support the conclusion. If REM sleep were homeostatically regulated in fur seals, then we'd suspect the seals to sustain REM levels well above baseline for a prolonged period in order to compensate for weeks of REM deprivation while in the water. Whether or not there's a reduction in REM sleep from day 1 to day 2 doesn't tell us how REM sleep on those days relates to baseline, which is where our focus should be. Choice B is incorrect. The y-axis of this graph doesn't depict baseline levels of REM sleep, but rather shows REM sleep as a percent of baseline. Choice D is incorrect. The graph doesn't depict REM sleep while in the water for the seals in the study. Additionally, we're told fur seals get no REM sleep while in the water, which is significantly different to the values shown in the graph for after they return to land.

Question Difficulty: Hard

Question ID be19faa1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: be19faa1



Urbanization, industrialization, and the warming climate create thermal pollution (excess heat) in the shallow subsurface soil. Susanne A. Benz and colleagues analyzed thousands of sites on three continents under one scenario in which surface temperature remains at the current level and under another in which the surface reaches the maximum plausible temperature. They then categorized each site according to the percentage of local home heating needs that could be met using this excess subsurface heat. The team concluded that if surface temperature approaches the maximum plausible level, the percentage of sites where thermal pollution could feasibly contribute to meeting home heating needs will increase.

Which choice best describes data in the graph that support Benz and colleagues’ conclusion?

Under both temperature conditions, less than 10% of sites were in the up-to-25% group, but at the maximum plausible surface temperature, almost 80% of sites could have all their local heating needs met by thermal pollution.

At current surface temperatures, more than 80% of the sites have no need for supplemental local home heating from subsurface thermal pollution, but at the maximum plausible surface temperature, more than 70% of sites exhibit significantly greater home heating needs.

At current surface temperatures, more than 80% of sites can meet, at most, 25% of local home heating needs with subsurface thermal pollution, but at the maximum plausible surface temperature, more than 80% of sites can meet greater than 25% of local home heating needs.

At current surface temperatures, more than 80% of the sites cannot use subsurface thermal pollution to meet any portion of local home heating needs, but at the maximum plausible surface temperature, that percentage drops below 20%.

ID: be19faa1 Answer

Correct Answer: D

Rationale

Choice D is the best answer. The researchers concluded that as we approach maximum plausible surface temperatures, there will be a larger percentage of sites where thermal pollution could contribute to meeting home heating needs. By showing that only a small percentage of homes can currently use thermal pollution for home heating, and that this percentage would grow much larger at maximum plausible surface temperatures, this choice supports the researchers' conclusion.

Choice A is incorrect. We do not know how many sites could have all (i.e., 100%) of their local heating needs met by thermal pollution, as the graph only classifies sites by whether "0%," "Up to 25%," and "More than 25%" of heating needs could be met. Choice B is incorrect. The graph is not depicting need for supplemental heating from thermal pollution, but rather potential to use thermal pollution for supplemental heating. Choice C is incorrect. The graph indicates that, at current surface temperatures, less than 10% of sites can meet 25% of local home heating needs and that more than 80% of sites cannot meet any local home heating needs.

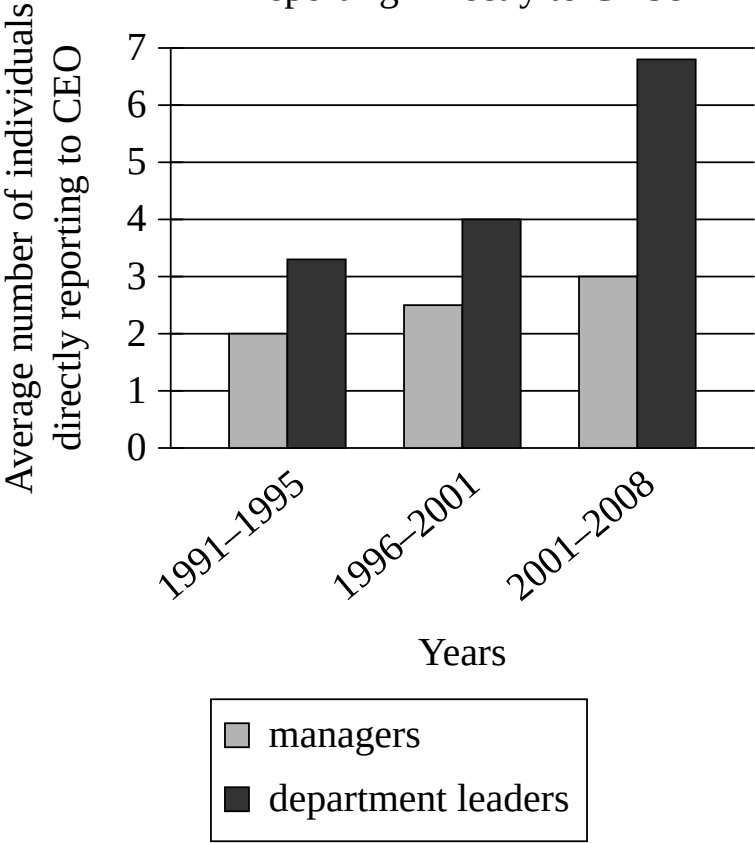
Question Difficulty: Hard

Question ID 645fd11a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 645fd11a

Average Number of Individuals Reporting Directly to CEOs



Considering a large sample of companies, economics experts Maria Guadalupe, Julie Wulf, and Raghuram Rajan assessed the number of managers and leaders from different departments who reported directly to a chief executive officer (CEO). According to the researchers, the findings suggest that across the years analyzed, there was a growing interest among CEOs in connecting with more departments in their companies.

Which choice best describes data from the graph that support the researchers’ conclusion?

The average numbers of managers and department leaders reporting directly to their CEO didn’t fluctuate from the A. 1991–1995 period to the 2001–2008 period.

B. The average number of managers reporting directly to their CEO was highest in the 1996–2001 period.

The average number of department leaders reporting directly to their CEO was greater than the average number of C. managers reporting directly to their CEO in each of the three periods studied.

D. The average number of department leaders reporting directly to their CEO rose over the three periods studied.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data from the graph that support the researchers' conclusion that there is a growing interest among CEOs in connecting with more departments in their companies. The graph shows the average number of individuals reporting directly to CEOs during three different time periods: the individuals are divided into managers and department leaders. The average number of department leaders directly reporting to their CEO during the 1991–1995 period was slightly more than three, during the 1996–2001 period it was four, and during the 2001–2008 period it was almost seven. Thus, the average number of department leaders reporting directly to their CEO rose over the three periods studied, which suggests that CEOs were connecting with more departments.

Choice A is incorrect because the average number of managers and department leaders reporting directly to their CEO rose for both categories between the 1991–1995 and 2001–2008 periods; thus, it isn't true that the average numbers didn't fluctuate. Choice B is incorrect because the average number of managers reporting directly to their CEO was highest in the 2001–2008 period, not in the 1996–2001 period. Choice C is incorrect. Although it correctly describes a feature of the graph, the observation that more department leaders than managers are reporting to CEOs does not by itself address the question of whether CEOs are connecting with more departments over time—to address that question, one needs to know whether the number of department leaders reporting to CEOs is increasing over time.

Question Difficulty: Hard

Question ID 701126bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 701126bc

In superfluorescence, electrical charges known as dipoles emit light in synchronized bursts so intense that they are visible to the eye. Until recently, this phenomenon has only been observed at extremely cold temperatures because dipoles cannot synchronize at higher temperatures. But in a study, Melike Biliroglu and colleagues observed superfluorescence at room temperature in thin films made of perovskite and other similarly crystalline materials; the researchers propose that the formation of shock-absorbing quasiparticles called polarons in the material protects dipoles from thermal interference.

Based on the text, how are polarons believed to be involved in the superfluorescence observed in Biliroglu and colleagues’ study?

- A. Polarons enable superfluorescent bursts to cross from one crystalline material to another.
- B. Polarons allow for the dipoles to synchronize despite higher temperatures.
- C. Polarons accelerate the dipoles’ release of superfluorescent bursts.
- D. Polarons decrease the intensity of the superfluorescent burst.

ID: 701126bc Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately describes Biliroglu and colleagues’ claim about how the polarons function in relation to superfluorescence. The text indicates that “until recently,” superfluorescence (intense, synchronized bursts of light emitted by dipoles) has solely been observed at very cold temperatures. However, it also states that, recently, Biliroglu and colleagues report observing the phenomenon at room temperature. They achieved this using “thin films made of perovskite and other similarly crystalline materials,” which the researchers claim allows for the formation of polarons. They also suggest that these polarons might absorb the thermal shocks that typically disrupt dipole synchronization at warmer temperatures. Thus, based on the text, Biliroglu and colleagues believe that polarons help dipoles synchronize at temperatures well above those at which superfluorescence had previously been observed.

Choice A is incorrect because the text doesn’t address the prospect of a superfluorescent burst moving between crystalline materials or any other mediums. Choice C is incorrect because the text’s discussion of polarons is about how they might enable superfluorescence at higher temperatures than those at which it had previously been observed. Rather than suggesting that polarons speed up superfluorescent bursts, the text suggests that no superfluorescence can occur at room temperature in the absence of polarons. Thus, the text indicates that polarons make superfluorescent bursts more likely to occur at higher temperatures than those at which it had previously been observed, not that polarons accelerate the bursts. Choice D is incorrect because the text’s discussion of polarons is about how they might enable superfluorescence at higher temperatures than those at which it had previously been observed. In the absence of polarons, the text suggests there would be no superfluorescence at room temperature. Thus, rather than decrease the intensity of superfluorescent bursts, polarons make them more likely to occur under certain circumstances.

Question Difficulty: Hard

Question ID c83e0b43

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: c83e0b43

O Pioneers! is a 1913 novel by Willa Cather. In the novel, Cather depicts Alexandra Bergson as a person who takes comfort in understanding the world around her: _____

Which quotation from *O Pioneers!* most effectively illustrates the claim?

- “She looked fixedly up the bleak street as if she were gathering her strength to face something, as if she were trying with all her might to grasp a situation which, no matter how painful, must be met and dealt with somehow.”
- “She had never known before how much the country meant to her. The chirping of the insects down in the long grass had been like the sweetest music. She had felt as if her heart were hiding down there, somewhere, with the quail and the plover and all the little wild things that crooned or buzzed in the sun. Under the long shaggy ridges, she felt the future stirring.”
- “Alexandra drove off alone. The rattle of her wagon was lost in the howling of the wind, but her lantern, held firmly between her feet, made a moving point of light along the highway, going deeper and deeper into the dark country.”
- “Alexandra drew her shawl closer about her and stood leaning against the frame of the mill, looking at the stars which glittered so keenly through the frosty autumn air. She always loved to watch them, to think of their vastness and distance, and of their ordered march. It fortified her to reflect upon the great operations of nature, and when she thought of the law that lay behind them, she felt a sense of personal security.”

ID: c83e0b43 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses a quotation from *O Pioneers!* to illustrate the claim that Alexandra Bergson takes comfort in understanding the world around her. In the quotation, Alexandra is described as enjoying looking at the stars and feeling a “sense of personal security” when she contemplates nature’s order and its governing laws. This suggests that Alexandra takes comfort in understanding the world around her.

Choice A is incorrect because the quotation expresses how Alexandra Bergson attempts to meet difficult situations with determination, not how she takes comfort in understanding the world around her. Choice B is incorrect because the quotation expresses “how much the country meant to” Alexandra Bergson, not how she takes comfort in understanding the world around her. In detailing some of the wildlife surrounding Alexandra, the quotation conveys that nature is important to her but not necessarily that it gives her comfort. Choice C is incorrect because the quotation describes Alexandra driving her wagon down a highway at night; it doesn’t describe how she takes comfort in understanding the world around her or address how she’s feeling as she drives off.

Question Difficulty: Hard

Question ID ac285054

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: ac285054

The domestic sweet potato (*Ipomoea batatas*) descends from a wild plant native to South America. It also populates the Polynesian Islands, where evidence confirms that Native Hawaiians and other Indigenous peoples were cultivating the plant centuries before seafaring first occurred over the thousands of miles of ocean separating them from South America. To explain how the sweet potato was first introduced in Polynesia, botanist Pablo Muñoz-Rodríguez and colleagues analyzed the DNA of numerous varieties of the plant, concluding that Polynesian varieties diverged from South American ones over 100,000 years ago. Given that Polynesia was peopled only in the last three thousand years, the team concluded that _____

Which choice most logically completes the text?

- A. the cultivation of the sweet potato in Polynesia likely predates its cultivation in South America.
- Polynesian peoples likely acquired the sweet potato from South American peoples only within the last three thousand years.
- B. years.
- C. human activity likely played no role in the introduction of the sweet potato in Polynesia.
- D. Polynesian sweet potato varieties likely descend from a single South American variety that was domesticated, not wild.

ID: ac285054 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text's discussion of the sweet potato in Polynesia. The text indicates that the sweet potato is found in Polynesia but originated in South America, and that the sweet potato was being cultivated by Native Hawaiians and other Indigenous peoples in Polynesia long before sea voyages between South America and Polynesia began. The text goes on to note that research by Muñoz-Rodríguez and colleagues has established that the Polynesian varieties of sweet potato split from South American varieties more than 100,000 years ago, which is thousands of years before humans settled in Polynesia. If Polynesian peoples were cultivating the sweet potato before sea voyages between Polynesia and South America began, and if Polynesian varieties of sweet potato diverged from South American varieties well before people were in Polynesia, it can reasonably be concluded that humans didn't play a role in bringing the sweet potato to Polynesia.

Choice A is incorrect. The text doesn't provide any information about when the sweet potato began to be cultivated in South America, so there's no support for the conclusion that cultivation began in Polynesia before it began in South America. Choice B is incorrect because the text indicates that the sweet potato was being cultivated in Polynesia long before sea journeys between Polynesia and South America began. Therefore, it wouldn't be reasonable to conclude that Polynesian peoples acquired the sweet potato from South American peoples. Additionally, the text indicates that the Polynesian varieties of sweet potato diverged from the South American varieties thousands of years before people settled in Polynesia, which suggests that the sweet potato was already present in Polynesia when people arrived. Choice D is incorrect because the text states that the domestic sweet potato, which is found in Polynesia, descends from a wild South American plant, not from a domesticated South American plant. The only people that the text describes as cultivating the sweet potato are Native Hawaiians and other Indigenous peoples of Polynesia.

Question Difficulty: Hard

Question ID dd1757fd

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: dd1757fd

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.

Which finding, if true, would most directly support the claim in the underlined sentence?

The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.

The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.

Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.

Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.

ID: dd1757fd Answer

Correct Answer: A

Rationale

Choice A is the best answer. While many networks can perform navigation tasks, or even mimic grid cells, it doesn't mean they're actually behaving like biological brains—this finding suggests that the rules that govern neural network behavior are completely unlike the way real brains work.

Choice B is incorrect. Although it mentions the rules that are programmed into the networks, this finding wouldn't clarify whether or not these rules have anything to do with the function of biological brains. Choice C is incorrect. This choice suggests that neural networks are modeled after multiple types of brain cells, which sidesteps the question of whether these rule-based networks are genuinely similar to biological brains. Choice D is incorrect. This choice doesn't address the key point of the claim, which is that the apparent similarity between neural networks and biological brains is only due to the rules programmed into the networks. It focuses on training tasks, not the originally programmed rules.

Question Difficulty: Hard

Question ID e185a21f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: e185a21f

One theory behind human bipedalism speculates that it originated in a mostly ground-based ancestor that practiced four-legged “knuckle-walking,” like chimpanzees and gorillas do today, and eventually evolved into moving upright on two legs. But recently, researchers observed orangutans, another relative of humans, standing on two legs on tree branches and using their arms for balance while they reached for fruits. These observations may suggest that _____

Which choice most logically completes the text?

- A. bipedalism evolved because it was advantageous to a tree-dwelling ancestor of humans.
- B. bipedalism must have evolved simultaneously with knuckle-walking and tree-climbing.
- C. moving between the ground and the trees would have been difficult without bipedalism.
- D. a knuckle-walking human ancestor could have easily moved bipedally in trees.

ID: e185a21f Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of the evolution of bipedalism in humans. According to the text, one potential explanation for humans walking upright on two legs is that the behavior evolved from an ancestor that mostly stayed on the ground and walked on four limbs, as modern chimpanzees and gorillas do. However, the finding that orangutans, also a relative of humans, sometimes stand on two legs in trees while using their arms to balance and reach for fruits suggests another possible explanation: perhaps a tree-dwelling ancestor of humans began moving on two legs because it offered an advantage, such as access to certain foods.

Choice B is incorrect because the finding that modern orangutans (a relative of humans) sometimes stand on two legs in trees doesn’t offer any insight into when either bipedalism or tree-climbing behavior emerged in human ancestors. Additionally, the text indicates that one theory is that bipedalism evolved from a mostly ground-based ancestor that was already practicing knuckle-walking, not that bipedalism and knuckle-walking developed at the same time. Choice C is incorrect because the finding that orangutans (a relative of humans) sometimes stand on two legs in trees doesn’t offer any insight into how difficult it would’ve been to move between the ground and the trees without bipedalism; there’s no suggestion that climbing or moving in trees depends on the ability to walk on two legs rather than four, even if that ability might be helpful in certain circumstances. Choice D is incorrect because the finding that orangutans (a relative of humans) sometimes stand on two legs in trees doesn’t suggest that a knuckle-walking human ancestor could’ve easily moved on two legs in trees. Although the text indicates that bipedalism may have evolved from a human ancestor that mostly stayed on the ground and walked on four limbs, it gives no indication of how easy it would’ve been for such an ancestor to move bipedally in trees.

Question Difficulty: Hard

Question ID 64cd7321

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 64cd7321

As media consumption has become increasingly multiplatform and socially mediated, active news acquisition has diminished in favor of an attitude known as “news finds me” (NFM), in which people passively rely on their social networks and ambient media environments for information about current events. Homero Gil de Zúñiga and Trevor Diehl examined data on a representative group of adults in the United States to determine participants’ strength of NFM attitude, political knowledge, and political interest. Although no major election took place sufficiently near the study for Gil de Zúñiga and Diehl to identify causality between NFM and voting behavior, they did posit that NFM may reduce voting probability through an indirect effect.

Which finding, if true, would most directly support the idea advanced by Gil de Zúñiga and Diehl?

NFM attitude tends to increase in strength as major elections approach, and people are significantly more likely to vote in A. major elections than in minor elections.

NFM attitude has a strong negative effect on political knowledge and interest, and there is known to be a strong positive B. correlation between political knowledge and interest and the likelihood of voting.

Political interest is known to have a strong positive effect on likelihood of voting but shows only a weak positive effect on C. political knowledge, and NFM attitude shows little correlation with either political knowledge or political interest.

The likelihood of voting increases as political knowledge increases, and the relationship between NFM attitude and D. political knowledge tends to strengthen as the size of people’s social networks increases.

ID: 64cd7321 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the idea advanced by Homero Gil de Zúñiga and Trevor Diehl that NFM ("news finds me") attitude may reduce voting probability through an indirect effect. The text describes NFM as an attitude that has lowered people's interest in actively acquiring news and introduces Gil de Zúñiga and Diehl's study on the effects of NFM on people in the United States' political knowledge and interest. The text goes on to say that despite the fact that the study didn't occur near a major election, Gil de Zúñiga and Diehl still conclude that NFA may reduce voting probability. If the likelihood, or probability, that a person will vote is linked to that person's level of political knowledge and interest, that would suggest that negatively affecting a person's level of political knowledge and interest would also negatively affect how likely that person is to vote. Thus, if NFM attitude has a negative effect on political knowledge and interest, then it would also likely reduce voting probability.

Choice A is incorrect because the finding that NFM attitude increases as major elections approach wouldn't address the effect of NFM on the likelihood of voting, which is the idea advanced by Gil de Zúñiga and Diehl. Moreover, although the text mentions that there were no major elections that occurred near the time of the study, it doesn't discern between major and minor elections when discussing voting probability. Choice C is incorrect because finding that NFM attitude shows little correlation with either political knowledge or political interest would undermine, not support, Gil de Zúñiga and Diehl's idea that NFM may reduce voting probability because it suggests that NFM has no effect on political knowledge or political interest. Choice D is incorrect. Although Gil de Zúñiga and Diehl's idea would be supported by the finding that the likelihood of voting increases as political knowledge increases, nothing in the text suggests that the researchers' idea hinges on the size of people's social networks.

Question Difficulty: Hard

Question ID 9731a22b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 9731a22b

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

- A. It is the earliest known example of a titanosaur nesting and breeding site.
- B. It was very difficult to excavate given that it was discovered in a mine.
- C. It may have been occupied by other sauropods in addition to titanosaurs.
- D. It is farther north than any other nesting site discovered in South America.

ID: 9731a22b Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a statement about the site discovered by the researchers that is supported by the text. The text discusses Fiorelli and colleagues’ discovery of egg clutches, single eggs, and eggshells in a Brazilian mine. According to the text, the presence of these eggs, which are from the Late Cretaceous period, led the researchers to conclude that the location was once a nesting and breeding site for titanosaurs. The text then explains that the finding is important because of the “previous lack of known nesting sites in northern regions of South America.” If there haven’t been any other discoveries of a nesting site in South America’s northern regions and the site in the Brazilian mine is the first, then the text strongly suggests that the site is farther north than other nesting sites that have been discovered in South America.

Choice A is incorrect because the text doesn’t suggest that the site discovered by Fiorelli and colleagues is the earliest titanosaur nesting and breeding site known to paleontologists but rather that it’s the first nesting site found in northern regions of South America. Moreover, the text doesn’t suggest how the timeline of the newly discovered site compares with that of other titanosaur nesting and breeding sites. Choice B is incorrect because there is no mention in the text about any difficulties that Fiorelli and colleagues faced when they were excavating the nesting and breeding site in the Brazilian mine. Choice C is incorrect because the text doesn’t support the idea that the nesting and breeding site in the Brazilian mine was occupied by sauropods other than titanosaurs. The text simply mentions that titanosaurs are sauropod dinosaurs and presents the researchers’ conclusion that the site they discovered was for titanosaurs.

Question Difficulty: Hard

Question ID 09f9edb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 09f9edb0

In the 1980s, many musicians and journalists in the English-speaking world began to draw attention to music from around the globe—such as mbaqanga from South Africa and quan họ from Vietnam—that can’t be easily categorized according to British or North American popular music genres, typically referring to such music as “world music.” While some scholars have welcomed this development for bringing diverse musical forms to prominence in countries where they’d previously been overlooked, musicologist Su Zheng claims that the concept of world music homogenizes highly distinct traditions by reducing them all to a single category.

Which finding about mbaqanga and quan họ, if true, would most directly support Zheng’s claim?

- A. Mbaqanga and quan họ developed independently of each other and have little in common musically.
- B. Mbaqanga is significantly more popular in the English-speaking world than quan họ is.

Mbaqanga and quan họ are now performed by a diverse array of musicians with no direct connections to South Africa or C. Vietnam.

Mbaqanga and quan họ are highly distinct from British and North American popular music genres but similar to each D. other.

ID: 09f9edb0 Answer

Correct Answer: A

Rationale

Choice A is the best answer. Zheng’s claim is that the idea of world music “homogenizes” (meaning makes similar) distinct kinds of music by reducing them to one category. In other words, Zheng thinks the concept of world music is a harmful oversimplification of diverse musical forms. To support this claim, we need evidence that these musical traditions are so different from one another that they should not fall into the same category. If it’s true that mbaqanga and quan họ developed separately and have little in common musically, then it wouldn’t make sense to lump them into the same category.

Choice B is incorrect. If true, this wouldn’t affect the claim. To support the claim, we need evidence that these musical traditions are so different from one another that they should not fall into the same category. A difference in popularity doesn’t necessarily mean that the two musical traditions shouldn’t be categorized together: instead, we need to know if the music itself is similar or different. Choice C is incorrect. If true, this wouldn’t affect the claim. To support the claim, we need evidence that these musical traditions are so different from each other that they should not fall into the same category. This choice doesn’t do that. Choice D is incorrect. If true, this would actually weaken the claim. Zheng thinks it’s reductive or oversimplifying to put distinct musical traditions into a single category. But if mbaqanga and quan họ are similar to each other, then it would make sense to put them in the same category.

Question Difficulty: Hard

Question ID f39507a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: f39507a3

One challenge when researching whether holding elected office changes a person’s behavior is the problem of ensuring that the experiment has an appropriate control group. To reveal the effect of holding office, researchers must compare people who hold elected office with people who do not hold office but who are otherwise similar to the office-holders. Since researchers are unable to control which politicians win elections, they therefore _____

Which choice most logically completes the text?

- A. struggle to find valid data about the behavior of politicians who do not currently hold office.
- B. can only conduct valid studies with people who have previously held office rather than people who presently hold office.
- C. should select a control group of people who differ from office holders in several significant ways.
- D. will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

ID: f39507a3 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the challenge researchers face when studying the effects of holding elected office on a person’s behavior. The text explains that it’s hard for researchers to test for the effects that elected office has on people because finding people to serve as a control group is difficult. The text indicates that a control group needs to be made up of people who share characteristics of the group being tested but don’t have the variable being tested (in this case, holding elected office). Because researchers aren’t able to influence who wins elections, they’re also unable to determine who would serve as an appropriately similar member of a control group. Thus, it logically follows that researchers will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

Choice A is incorrect because the text focuses on the struggle to put together a control group for experiments; it doesn’t suggest that finding information about politicians’ behavior is difficult. Choice B is incorrect because the experiments mentioned in the text are testing the effects of holding elected office on a person’s behavior. Studying people who have already held elected office wouldn’t provide an opportunity to note any behavioral changes that the position might cause. Choice C is incorrect because the text defines people in a control group as those “who are otherwise similar to the office-holders”; selecting people who differ from the office-holders wouldn’t fit the criteria for an appropriate control group.

Question Difficulty: Hard

Question ID 70aacc03

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 70aacc03

Elizabeth Asiedu has identified a negative correlation between the share of developing countries’ economies derived from natural-resource extraction and those countries’ receipts of foreign investment. This may appear counterintuitive—resource extraction requires initial investments (in extractive technology, for instance) at scales best met by multinational corporations—but Asiedu notes that natural-resource industries’ boom-bust cycle can destabilize local currencies and increase developing countries’ vulnerability to external shocks, creating levels of uncertainty to which foreign investors are typically averse.

Which choice best states the main idea of the text?

- Although it may seem surprising that foreign investment declines in developing countries as natural-resource extraction makes up a larger share of those countries’ economies, that decline happens because resource extraction requires initial

A. investments too large for foreign investors to supply.
- Although developing countries tend to become less dependent on foreign investment as natural-resource industries make up a larger share of their economies, this change may not occur if the boom-bust cycle of those industries

B. destabilizes local currencies or increases countries’ vulnerability to external shocks.
- Although one might expect that foreign investment would increase as natural-resource extraction makes up a larger share of developing countries’ economies, the opposite happens because heavy reliance on natural resources can lead to

C. unattractive conditions for investors.
- Although foreign investors tend to avoid initial investments in natural-resource industries in developing countries, foreign

D. investment may increase significantly as those industries stabilize and the risks associated with them decline.

ID: 70aacc03 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it accurately states the main idea of the text. According to the text, contrary to what some might expect, foreign investment is typically lower in developing countries whose economies are more dependent on natural-resource extraction. The text explains that high reliance on natural-resource extraction can subject a developing country to economic shocks that can destabilize the local currency and introduce economic uncertainty that tends to keep investors away. In other words, although we may think otherwise, foreign investors are less willing to invest in projects in developing countries whose economies are heavily dependent on natural-resource extraction because those economies tend to exhibit instability that investors want to avoid.

Choice A is incorrect. The text does indicate that foreign investment is typically lower in developing countries whose economies are more dependent on natural-resource extraction; the text further indicates that natural-resource extraction requires substantial initial investments (to acquire things like required technologies) for which there are fewer investors willing to participate at this stage than one might think. But the text does not implicate the cost of these initial investments as a reason why foreign investment is less widely available than some might think. Choice B is incorrect. The text indicates that greater dependence on natural-resource extraction makes a developing country less appealing to foreign investors because of associated economic instability. Rather than arguing that the goal of developing countries is to become less dependent on foreign investment, as the phrasing of choice B suggests, the text focuses only on why foreign investors become less involved with such countries, which suggests that more investment would be preferable. Choice D is incorrect. Although the text indicates that natural-resource extraction requires substantial initial investments (to acquire things like required technologies) and that there are fewer likely investors willing to participate at this stage than one might think, the text does not address what investors are likely to do over time as the industry stabilizes itself.

Question Difficulty: Hard

Question ID f9bd4e61

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: f9bd4e61

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht’s influence can be seen in English playwright Caryl Churchill’s 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____

Which choice most logically completes the text?

- A. focus on the characters’ beliefs about social and political issues as revealed by the characters’ actions.
- B. reflect on social and political phenomena not directly related to patriarchy and colonialism.
- C. recognize pertinent social and political parallels between Germany during Brecht’s time and England at the time when Churchill was writing *Cloud 9*.
- D. be dispassionate as they think critically about the social and political questions raised by the play.

ID: f9bd4e61 Answer

Correct Answer: D

Rationale

Choice D is the best answer. Churchill’s play was influenced by Brecht’s belief that theater should elicit an intellectual rather than an emotional response from audiences, making them think about social and political issues that also exist outside of the play. Therefore, it makes sense that Churchill would strive to have audiences think dispassionately (i.e., without emotion) and critically about the social and political questions raised by the play.

Choice A is incorrect. The passage mentions that Churchill was influenced by Brecht’s belief that plays should provoke audience members “to consider social and political realities that extend beyond the characters and events depicted onstage,” so there’s no basis for an inference about audience members deeply engaging with characters’ beliefs and actions. Choice B is incorrect. Reflecting on things that aren’t related to patriarchy and colonialism wouldn’t “engage audiences in an interrogation of patriarchy and colonialism.” That creates a confusing contradiction. Choice C is incorrect. No mention is made of social and political issues in Germany during Brecht’s time, so there’s no basis for this inference.

Question Difficulty: Hard

Question ID f7bd14de

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: f7bd14de

Several scholars have argued that conditions in England in the late ninth through early eleventh centuries—namely, burgeoning literacy amid running conflicts between England’s Anglo-Saxon kingdoms and Danish invaders—were especially conducive to the production of the Old English epic poem *Beowulf*, and they have dated the poem’s composition accordingly. It is not inconceivable that *Beowulf* emerged from such a context, but privileging contextual fit over the linguistic evidence of an eighth- or even seventh-century composition requires a level of justification that thus far has not been presented.

Which choice best states the main idea of the text?

- Although there are some grounds for believing that *Beowulf* was composed between the late ninth and early eleventh centuries, advocates for that view tend to rely on evidence that has been called into question by advocates for an earlier date.
- A. date.
- Although several scholars have dated *Beowulf* to the late ninth through early eleventh centuries, others have argued that doing so privileges a controversial interpretation of the social conditions of the period.
- B. doing so privileges a controversial interpretation of the social conditions of the period.
- Although *Beowulf* fits well with the historical context of England in the late ninth through early eleventh centuries, it fits equally well with the historical context of England in the seventh and eighth centuries.
- C. equally well with the historical context of England in the seventh and eighth centuries.
- Although the claim of a late ninth- through early eleventh-century composition date for *Beowulf* has some plausibility, advocates for the claim have not compellingly addressed evidence suggesting an earlier date.
- D. advocates for the claim have not compellingly addressed evidence suggesting an earlier date.

ID: f7bd14de Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most accurately states the main idea of the text. The text states that some scholars have dated the composition of *Beowulf* to the late ninth through early eleventh centuries due to the poem's fit with that period's historical context. The text goes on to say that while it is "not inconceivable that *Beowulf* emerged from such a context"—that is, it is possible that *Beowulf* was composed during the late ninth through eleventh centuries—there is linguistic evidence that the poem was composed earlier, in the seventh or eighth century. According to the text, favoring the historical context over the linguistic evidence requires justification that scholars have not yet supplied. In other words, the text suggests that scholars who favor the later composition date need to explain why the poem's fit with historical context should take precedence over the linguistic evidence, but they have not yet done so. Thus, the main idea of the text is that while there is some plausibility to the later composition date, advocates for the later date have not compellingly addressed evidence suggestive of an earlier date.

Choice A is incorrect because the text says that scholars who date the poem to the late ninth through early eleventh centuries have failed to account for the linguistic evidence that the poem may have been composed earlier, not that the evidence those scholars cite in favor of their view is unreliable or that anyone has cast doubt on that evidence. In other words, the text does not suggest that there are problems with the evidence cited by advocates of the later composition date, only that there is other evidence of an earlier composition date that those advocates need to consider. Choice B is incorrect because nothing in the text suggests that those scholars who date the poem to the late ninth through early eleventh centuries are giving priority to a controversial view of the social conditions at that time. The text makes no reference to any controversy about how scholars interpret that historical period. Instead, the text suggests that scholars who date the poem on the basis of its fit with the historical context of England in the late ninth through early eleventh centuries have failed to account for linguistic evidence that the poem may have been composed earlier. Choice C is incorrect because the text says nothing about how well the poem fits the historical context of England in the seventh and eighth centuries, let alone that it fits that historical context as well as it fits the historical context of the late ninth through early eleventh centuries. Rather, the text says that there is linguistic evidence that the poem may have been composed in the seventh or eighth century.

Question Difficulty: Hard

Question ID 3fc06a91

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 3fc06a91

Employment by Sector in France and the United States, 1800–2012 (% of total employment)

Year	Agriculture in France	Manufacturing in France	Services in France	Agriculture in US	Manufacturing in US	Services in US
1800	64	22	14	68	18	13
1900	43	29	28	41	28	31
1950	32	33	35	14	33	53
2012	3	21	76	2	18	80

Rows in table may not add up to 100 due to rounding.

Over the past two hundred years, the percentage of the population employed in the agricultural sector has declined in both France and the United States, while employment in the service sector (which includes jobs in retail, consulting, real estate, etc.) has risen. However, this transition happened at very different rates in the two countries. This can be seen most clearly by comparing the employment by sector in both countries in _____

Which choice most effectively uses data from the table to complete the statement?

- A. 1900 with the employment by sector in 1950.
- B. 1800 with the employment by sector in 2012.
- C. 1900 with the employment by sector in 2012.
- D. 1800 with the employment by sector in 1900.

ID: 3fc06a91 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents data from the table that most effectively complete the statement about the rates at which employment shifted in France and the United States. The text states that over the last two hundred years employment in the agricultural sector has declined while employment in the service sector has risen in both France and the US, and the data from the table reflect these trends. The text asserts, however, that the transition from agriculture to services “happened at very different rates in the two countries.” This assertion is best supported by a comparison of data from 1900 and 1950: the table shows that in those years, employment in agriculture went from 43% to 32% in France (a decline of 11 percentage points) and from 41% to 14% in the US (a decline of 27 percentage points), and that employment in services went from 28% to 35% in France (an increase of 7 percentage points) and from 31% to 53% in the US (an increase of 22 percentage points). In other words, the rate of change was greater in the US than in France for both sectors.

Choice B is incorrect because comparing the data for 1800 and 2012 would suggest a similar rate of change in the two countries, not very different rates: employment in agriculture went from 64% in 1800 to 3% in 2012 in France, which is close to the change from 68% in 1800 to 2% in 2012 in the US, while employment in services went from 14% in 1800 to 76% in 2012 in France, which is close to the change from 13% in 1800 to 80% in 2012 in the US. Choice C is incorrect because comparing the data for 1900 and 2012 would suggest a similar rate of change in the two countries rather than very different rates: employment in agriculture went from 43% in 1900 to 3% in 2012 in France, which is close to the change from 41% in 1900 to 2% in 2012 in the US, while employment in services went from 28% in 1900 to 76% in 2012 in France, which is close to the change from 31% in 1900 to 80% in 2012 in the US. Choice D is incorrect because comparing the data for 1800 and 1900 would suggest a similar rate of change in the two countries, not very different rates: employment in agriculture went from 64% in 1800 to 43% in 1900 in France, which is fairly close to the change from 68% in 1800 to 41% in 1900 in the US, while employment in services went from 14% in 1800 to 28% in 1900 in France, which is close to the change from 13% in 1800 to 31% in 1900 in the US.

Question Difficulty: Hard

Question ID 96802cc0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 96802cc0

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

- The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum’s long-suspected role
A. in motor coordination.
- Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway
B. connecting the VTA and the cerebellum.
- The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers
C. have been uncovering additional functions.
- Technological limitations have historically hindered the study of the cerebellum, but the recent development of new
D. technologies has led to greater insights into its functions.

ID: 96802cc0 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The text says that the cerebellum has long been thought to regulate motor functioning, but new research shows that it may also have other functions—including regulating emotion and social behavior and some connection to motivation and rewards processing.

Choice A is incorrect. The VTA is described as “a center for motivation and reward processing,” and the discovery of the pathway between the VTA and the cerebellum supports the theory that the cerebellum is involved in functions other than motor coordination. Choice B is incorrect. The text says that recent research has identified this pathway, but it doesn’t discuss any previous “widely accepted claims” about the pathway’s function. The “widespread acknowledgement” mentioned early in the passage is about the cerebellum alone, not its connection to the VTA. Choice D is incorrect. The text never discusses any technological limitations or any new technologies.

Question Difficulty: Hard

Question ID ec93e52c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: ec93e52c

Archaeologist Petra Vaiglova, anthropologist Xinyi Liu, and their colleagues investigated the domestication of farm animals in China during the Bronze Age (approximately 2000 to 1000 BCE). By analyzing the chemical composition of the bones of sheep, goats, and cattle from this era, the team determined that wild plants made up the bulk of sheep’s and goats’ diets, while the cattle’s diet consisted largely of millet, a crop cultivated by humans. The team concluded that cattle were likely raised closer to human settlements, whereas sheep and goats were allowed to roam farther away.

Which finding, if true, would most strongly support the team’s conclusion?

- Analysis of the animal bones showed that the cattle’s diet also consisted of wheat, which humans widely cultivated in
- A. China during the Bronze Age.
 - B. Further investigation of sheep and goat bones revealed that their diets consisted of small portions of millet as well.
 - C. Cattle’s diets generally require larger amounts of food and a greater variety of nutrients than do sheep’s and goats’ diets.
 - D. The diets of sheep, goats, and cattle were found to vary based on what the farmers in each Bronze Age settlement could grow.

ID: ec93e52c Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents a finding that, if true, would most strongly support the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were. The text explains that Vaiglova, Liu, and their colleagues analyzed the chemical composition of sheep, goat, and cattle bones from the Bronze Age in China in order to investigate the animals’ domestication, or their adaptation from a wild state to a state in which they existed in close connection with humans. According to the text, the team’s analysis showed that sheep and goats of the era fed largely on wild plants, whereas cattle fed on millet—importantly, a crop cultivated by humans. If analysis of the animal bones shows that the cattle’s diet also consisted of wheat, another crop cultivated by humans in China during the Bronze Age, the finding would support the team’s conclusion by offering additional evidence that cattle during this era fed on human-grown crops—and, by extension, that humans raised cattle relatively close to the settlements where they grew these crops, leaving goats and sheep to roam farther away in areas with wild vegetation, uncultivated by humans.

Choice B is incorrect because if it were true that sheep’s and goats’ diets consisted of small portions of millet, which the text states was a crop cultivated by humans, the finding would suggest that sheep and goats were raised relatively close to human settlements, weakening the team’s conclusion that cattle were likely raised closer to those settlements than sheep and goats were. Choice C is incorrect because the finding that cattle generally require more food and nutrients than do sheep and goats wouldn’t support the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were. Nothing in the text suggests that cattle were incapable of obtaining sufficient food and nutrients without access to human-grown crops. Hence, even if cattle’s diets are found to have different requirements than the diets of sheep and goats, the cattle could have met those requirements from food located far from human settlements. Choice D is incorrect because if it were true that the diets of sheep, goats, and cattle varied based on what the farmers in each Bronze Age settlement could grow, the finding would weaken the team’s conclusion that cattle were likely raised closer to human settlements than sheep and goats were, suggesting instead that all three types of animals were raised close enough to human settlements to feed on those settlements’ crops.

Question Difficulty: Hard

Question ID 4889580c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 4889580c

Archaeologists and historians used to believe that the Maya civilization during its Classic period (roughly 250–900) lacked agricultural marketplaces. One reason for this belief was that these scholars misunderstood the ecology of the regions the Maya inhabited. Marketplaces typically emerge because different individuals or groups want to trade resources they control for resources they don't control. Scholars seriously underestimated the ecological diversity of the Maya landscape and thus assumed that _____

Which choice most logically completes the text?

- A. marketplaces likely would not have attracted many traders from outside the regions controlled by the Maya.
- B. farming practices would have been largely the same throughout Maya lands even if the crops people produced varied significantly.
- C. marketplaces would not have enabled Maya people to acquire many products different from those they already produced.
- D. farmers would trade agricultural products only if they had already produced enough to meet their own needs.

ID: 4889580c Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents the conclusion that most logically follows from the text's discussion of scholars' understanding of Maya ecology and agricultural marketplaces. The text indicates that scholars used to believe that during the Classic period, the Maya civilization didn't have agricultural marketplaces. According to the text, scholars held this view because they misunderstood the ecology of areas where the Maya lived. The text points out that people tend to create marketplaces in order to acquire resources they don't otherwise control. Agricultural marketplaces would have allowed farmers who produced one type of crop to trade that crop for other types of crops that they didn't produce. The text goes on to say, however, that scholars underestimated the ecological diversity of the Maya areas, meaning that scholars thought that the Maya landscape produced a smaller range of resources than it actually produced. Taken together, then, this information suggests that scholars assumed that marketplaces wouldn't have allowed Maya people to acquire products different from the products they already produced: that is, if everyone produced the same array of crops, as scholars mistakenly believed, then there wouldn't have been any need for marketplaces where people could trade those crops.

Choice A is incorrect because the text doesn't say anything about trade between the Maya and people from outside the regions controlled by the Maya. Although scholars' mistaken belief that the Maya lands weren't very ecologically diverse would give those scholars a reason to think that the Maya didn't have marketplaces, it wouldn't lead scholars to assume that traders from outside Maya lands were uninterested in acquiring resources produced by the Maya. Even if the Maya actually did produce only a small array of resources throughout their lands, there is no reason to believe from the text that people outside Maya lands also produced these same resources and thus would have no need to trade with the Maya people. Choice B is incorrect because the text indicates that scholars underestimated the ecological diversity of the Maya lands, which suggests that they mistakenly believed that the Maya produced a relatively small array of resources throughout their territory, not that the crops the Maya produced varied significantly throughout the Maya lands. Although the scholars might have assumed that a lack of ecological diversity suggests that Maya farming practices were largely the same everywhere, the text does not support that they also assumed there was a lot of variation in the crops that Maya people produced. In fact, the text states that marketplaces emerge when people want to obtain resources they don't already control. If it were the case that scholars assumed that the crops Maya people produced varied significantly, this would have led them to conclude that Maya people likely established marketplaces so they could trade for resources they didn't already possess, not that the Maya civilization lacked marketplaces. Choice D is incorrect because nothing in the text suggests that scholars assumed that farmers wouldn't trade their agricultural products unless they had already met their own needs with those products. Instead, the text says that scholars thought that the Maya lands produced a smaller array of resources than they actually did, which the text suggests led scholars to assume that the Maya didn't have any need for marketplaces. The scholars' mistaken belief has no bearing on the issue of whether farmers met their own needs before trading their products.

Question Difficulty: Hard

Question ID d0f51067

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: d0f51067

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

- A. Dog breeds would not exist without many years of human intervention in dogs’ reproduction.
- B. Research fails to confirm a commonly held belief about dog breeds and behavior.
- C. The dog breeds most popular among owners have often changed over the past 160 years.
- D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.

ID: d0f51067 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately states the main idea of the text. The text indicates that dog owners typically claim that some dog breeds are “more likely than others to have particular personality traits.” In other words, the text points out that a commonly held belief about dog breeds is that their personality traits are heritable. The text then states that Kathleen Morrill and colleagues undertook research about dog trait heritability and found that “behavior varies widely among dogs of the same breed.” Because Morrill and colleagues found evidence for variability rather than consistency in the behavior of dogs of the same breed, the statement that research fails to uphold a commonly held belief about dog breeds and behavior accurately reflects the main idea of the text.

Choice A is incorrect. Although the text mentions that humans have long intervened in dogs’ reproduction by intentionally crossbreeding certain dogs, it doesn’t argue that such intervention is essential to the existence of dog breeds. Choice C is incorrect because the text doesn’t discuss the popularity of any dog breeds; breeds are mentioned as having certain traits, but the text says nothing about the popularity of these breeds or traits. Choice D is incorrect. Although the text briefly mentions that Morrill and colleagues conducted a study about dog traits using both surveys and DNA sequencing, this is not the main focus of the text. The text concerns the study’s results about the heritability of dog traits, not the particular methodology used by Morrill and colleagues.

Question Difficulty: Hard

Question ID aaddd60f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: aaddd60f

Scientists studying Mars long thought the history of its crust was relatively simple. One reason for this is that geologic and climate data collected by a spacecraft showed that the crust was largely composed of basalt, likely as a result of intense volcanic activity that brought about a magma ocean, which then cooled to form the planet’s surface. A study led by Valerie Payré focused on additional information—further analysis of data collected by the spacecraft and infrared wavelengths detected from Mars’s surface—that revealed the presence of surprisingly high concentrations of silica in certain regions on Mars. Since a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica, Payré concluded that _____

Which choice most logically completes the text?

- the information about silica concentrations collected by the spacecraft is likely more reliable than the silica information gleaned from infrared wavelengths detected from Mars’s surface.
- high silica concentrations on Mars likely formed from a different process than that which formed the crusts of other planets.
- having a clearer understanding of the composition of Mars’s crust and the processes by which it formed will provide more insight into how Earth’s crust formed.
- Mars’s crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.

ID: aaddd60f Answer

Correct Answer: D

Rationale

Choice D is the best answer. Cooling magma would create basalt, but “a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica.” Since Mars’s crust does contain large amounts of silica, it is unlikely that Mars’s crust was formed exclusively by cooling magma. Therefore, there were likely other major geological events that created the high silica concentrations.

Choice A is incorrect. Although the passage discusses these two methods of collecting data about Mars’s crust, it never compares their reliability, so there’s no basis for this inference. Choice B is incorrect. The passage never mentions anything about the crusts of other planets, so there’s no basis for this inference. Choice C is incorrect. The passage never mentions Earth’s crust, so there’s no basis for this inference.

Question Difficulty: Hard

Question ID 1c69ff20

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 1c69ff20

For many years, the only existing fossil evidence of mixopterid eurypterids—an extinct family of large aquatic arthropods known as sea scorpions and related to modern arachnids and horseshoe crabs—came from four species living on the paleocontinent of Laurussia. In a discovery that expands our understanding of the geographical distribution of mixopterids, paleontologist Bo Wang and others have identified fossilized remains of a new mixopterid species, *Terropterus xiushanensis*, that lived over 400 million years ago on the paleocontinent of Gondwana.

According to the text, why was Wang and his team’s discovery of the *Terropterus xiushanensis* fossil significant?

A. The fossil constitutes the first evidence found by scientists that mixopterids lived more than 400 million years ago.

The fossil helps establish that mixopterids are more closely related to modern arachnids and horseshoe crabs than
B. previously thought.

The fossil helps establish a more accurate timeline of the evolution of mixopterids on the paleocontinents of Laurussia
C. and Gondwana.

The fossil constitutes the first evidence found by scientists that mixopterids existed outside the paleocontinent of
D. Laurussia.

ID: 1c69ff20 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it states why Wang and his team’s discovery of the *Terropterus xiushanensis* fossil was significant. The text explains that up until Wang and his team’s discovery, the only fossil evidence of mixopterids came from the paleocontinent of Laurussia. Wang and his team, however, identified fossil remains of a mixopterid species from the paleocontinent Gondwana. Therefore, the team’s discovery was significant because the fossil remains of a mixopterid species were outside of the paleocontinent Laurussia.

Choice A is incorrect. Although the text states that Wang and his team identified fossilized remains of a mixopterid species that lived more than 400 million years ago, it doesn’t indicate that mixopterid fossils previously found by scientists dated to a more recent period than that. Choice B is incorrect. Although the text states that mixopterids are related to modern arachnids and horseshoe crabs, it doesn’t suggest that the fossil discovered by Wang and his team confirmed that this relationship is closer than scientists had previously thought. Choice C is incorrect because the team’s fossil established the presence of mixopterids on Gondwana, not on Laurussia. Moreover, the text only discusses the fossil in relation to the geographical distribution of mixopterids, not in relation to their evolution.

Question Difficulty: Hard

Question ID 39e440e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 39e440e4

Archaeologists have held that the Casarabe culture, which emerged in the southwestern Amazon basin in the first millennium CE, was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness. Recently, however, archaeologist Heiko Prümers and colleagues conducted a study of the region using remote-sensing technology that enabled them to create three-dimensional images of the jungle-covered landscape from above, and the researchers concluded that the Casarabe people developed a form of urbanism in the Amazon basin.

- Which finding about the remote-sensing images, if true, would most directly support Prümers and colleagues’ conclusion?
- A. They show shapes consistent with widely separated settlements of roughly equal small size surrounded by uncultivated jungle.
 - B. They show shapes consistent with long-distance footpaths running from Casarabe territories to large cities outside the region inhabited by the Casarabe people.
 - C. They show shapes consistent with scattered small farms created by clearing jungle areas near sources of fresh water.
 - D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.

ID: 39e440e4 Answer

Correct Answer: D

Rationale

Choice D is the best answer. This finding, if true, would support the archaeologists’ conclusion. Dense central settlements linked to smaller ones would provide evidence of cities and suburbs—in other words, “a form of urbanism.”

Choice A is incorrect. This finding, if true, would weaken the archaeologists’ conclusion. Widely separated, small settlements with jungle in between would support the long-held belief that the Casarabe culture “was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness.” Choice B is incorrect. This choice wouldn’t support the researchers’ conclusion. These large cities are located outside Casarabe territory, which doesn’t show evidence of Casarabe urbanism. Choice C is incorrect. This finding wouldn’t support the archaeologists’ conclusion. Scattered small farms in jungle clearings are not good evidence to support the existence of cities (“a form of urbanism”).

Question Difficulty: Hard

Question ID d0fbf1ae

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: d0fbf1ae

Algae living within the tissues of corals play a critical role in keeping corals, and the marine ecosystems they are part of, thriving. Some coral species appear brown in color when healthy due to the algae colonies living in their tissues. In the event of an environmental stressor, the algae can die or be expelled, causing the corals to appear white. To recover the algae, the bleached corals then begin to produce bright colors, which block intense sunlight, encouraging the light-sensitive algae to recolonize the corals.

What does the text most strongly suggest about corals that produce bright colors?

- A. These corals have likely been subjected to stressful environmental conditions.
- B. These corals are likely more vulnerable to exposure from intense sunlight than white corals are.
- C. These corals have likely recovered from an environmental event without the assistance of algae colonies.
- D. These corals are more likely to survive without algae colonies than brown corals are.

ID: d0fbf1ae Answer

Correct Answer: A

Rationale

Choice A is the best answer. The text says that corals produce bright colors to block sunlight and encourage algae to recolonize after “an environmental stressor.” From this, we can infer that corals that produce bright colors have probably been subjected to an environmental stressor.

Choice B is incorrect. The text says that corals produce bright colors to block intense sunlight, which protects the light-sensitive algae that keep the coral healthy. In other words, bright colors make the coral’s health less vulnerable to intense sunlight. Choice C is incorrect. The text says that corals produce bright colors to encourage algae to recolonize, not that they have recovered without the assistance of algae colonies. Choice D is incorrect. The text never compares the likelihood of differently colored corals surviving without algae colonies.

Question Difficulty: Hard

Question ID 3f4ab688

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 3f4ab688

In a research paper, a student criticizes some historians of modern African politics, claiming that they have evaluated Patrice Lumumba, the first prime minister of what is now the Democratic Republic of the Congo, primarily as a symbol rather than in terms of his actions.

Which quotation from a work by a historian would best illustrate the student’s claim?

- “Lumumba is a difficult figure to evaluate due to the starkly conflicting opinions he inspired during his life and continues to inspire today.”
A.
- “The available information makes it clear that Lumumba’s political beliefs and values were largely consistent throughout his career.”
B.
- “Lumumba’s practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention.”
C.
- “Many questions remain about Lumumba’s ultimate vision for an independent Congo; without new evidence coming to light, these questions are likely to remain unanswered.”
D.

ID: 3f4ab688 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it illustrates the student’s claim about some historians viewing Lumumba primarily as a symbol. This quotation argues that Lumumba “warrants” (or deserves) “scholarly attention” as a symbol and not for his “practical accomplishments”—that is, his actions as prime minister—which “can be passed over quickly,” or dismissed as being of comparatively little importance. Thus, the quotation expresses the view that the student criticizes some historians for holding.

Choice A is incorrect. Although this quotation touches on the difficulty of evaluating Lumumba’s legacy, it doesn’t address how historians of modern African politics view him as a symbol. Choice B is incorrect. While this quotation mentions Lumumba’s political beliefs, it doesn’t discuss historians viewing him as a symbol. Choice D is incorrect. This quotation touches on Lumumba’s vision for his country, but it doesn’t discuss historians viewing him as a symbol.

Question Difficulty: Hard

Question ID 08395130

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 08395130

The Hubble Space Telescope (HST) is projected to maintain operation until at least 2030, but it has already revolutionized high-resolution imaging of solar-system bodies in visible and ultraviolet (UV) light wavelengths, notwithstanding that only about 6% of the bodies imaged by the HST are within the solar system. NASA researcher Cindy L. Young and colleagues assert that a new space telescope dedicated exclusively to solar-system observations would permit an extensive survey of minor solar-system bodies and long-term UV observation to discern how solar-system bodies change over time. Young and colleagues’ recommendation therefore implies that the HST _____

Which choice most logically completes the text?

- A. will likely continue to be used primarily to observe objects outside the solar system.
- B. will no longer be used to observe solar system objects if the telescope recommended by Young and colleagues is deployed.
- C. can be modified to observe the features of solar system objects that are of interest to Young and colleagues.
- D. lacks the sensors to observe the wavelengths of light needed to discern how solar system bodies change over time.

ID: 08395130 Answer

Correct Answer: A

Rationale

Choice A is the best answer. The HST will operate until at least 2030, but it’s only observing stuff inside our solar system 6% of the time. If we could get a different telescope to observe stuff inside our solar system 100% of the time and take more extensive images of certain things, then the HST could continue to be used mainly for observing stuff outside the solar system.

Choice B is incorrect. This inference is too strong to be supported by the text. Even if the new telescope is deployed, the HST might still be used as it’s being used now. Based on the text, the new telescope would just be used for more extensive and long-term imaging of solar system bodies, which doesn’t necessarily overlap with the HST. Choice C is incorrect. This inference isn’t supported. The text never mentions the possibility of modifying the HST, so there is no basis to make this inference. Rather, the researchers suggest using a different telescope to more closely observe certain objects. Choice D is incorrect. This inference is too strong to be supported. The text doesn’t give us enough info to assume that the HST lacks any particular sensors.

Question Difficulty: Hard

Question ID 26f5c4ba

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 26f5c4ba

In 2022, Crystal Reeck and colleagues studied whether the decision-making modes that guide consumers influence their choice between nonenvironmentally friendly standard electricity plans and environmentally friendly green plans that cap electricity usage. Study participants who self-reported using either an Affect Mode or Role Mode—which prioritize choices that have a stronger positive emotional or social impact, respectively—were more likely to select a green plan. Conversely, participants using a Calculation Mode—which aims to minimize both financial cost and personal inconvenience—were more likely to select a standard plan, even when the green option was cheaper. This finding suggests that participants using a Calculation Mode _____

Which choice most logically completes the text?

- _____ were equally unlikely to factor the financial savings of the green plan into their decision-making as were participants using either the Affect or Role Modes.
- _____ may have been less strongly motivated to appear socially responsible with their choice of plan than they realized.
- _____ may have determined that the green plan imposed additional burdens on them that were not sufficiently offset by the potential financial savings.
- _____ were less likely to believe that the green plan was truly cost-effective than were participants using either the Affect or Role Modes.

ID: 26f5c4ba Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text's discussion of the influence of decision-making modes on consumers' choices of different electricity plans. The Affect Mode describes choices based on the decision-maker's preferred emotional outcome, the Role Mode describes choices based on the decision-maker's social motivations, and the Calculation Mode describes choices based on minimizing financial costs and maximizing the decision-maker's convenience. Thus, if those using a Calculation Mode disfavor the green plan, it would be due to some financial or convenience burden the green plan imposes that the other doesn't. The text indicates that the green plan had a lower financial cost but was nonetheless rejected by participants using the Calculation Mode. It therefore follows that the green plan likely imposed a convenience burden that outweighed potential financial savings for these participants.

Choice A is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, unlike the Affect or Role Modes, the Calculation Mode would almost certainly involve taking price differences into account. Choice B is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, there is no reason to think that participants using the Calculation Mode at any point took social responsibility into account, let alone think that they discovered a disparity between their expectations and reality. Choice D is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, the text doesn't indicate that cost-effectiveness is a factor for participants using the Affect or Role Modes, and there is no basis to compare their beliefs in the truthfulness of the green-plan description with those of the Calculation Mode users.

Question Difficulty: Hard

Question ID 14189fbb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 14189fbb

Having written the impassioned call to arms “Letter to the Spanish Americans” in 1791, Peruvian intellectual Juan Pablo Viscardo y Guzmán is often considered a forerunner for the independence movements in Latin America. But Viscardo’s role in history would have remained insignificant were it not for Venezuelan revolutionary Francisco de Miranda, who was handed the unpublished letter after Viscardo’s death. Miranda not only helped circulate the letter, but his edits and footnotes to the text position Miranda as a central figure in the text’s creation.

Which choice best states the main idea of the text?

- A. The original authorship of “Letter to the Spanish Americans” is disputed by contemporary historians.
- B. The majority of the most eloquently stated arguments in “Letter to the Spanish Americans” were written by Miranda.
- C. Miranda played a crucial role in influencing the content and distribution of “Letter to the Spanish Americans.”
- D. “Letter to the Spanish Americans” persuaded many people in Latin America to pursue national independence.

ID: 14189fbb Answer

Correct Answer: C

Rationale

Choice C is the best answer. The text describes how Miranda circulated, edited, and added footnotes to “Letter to the Spanish Americans,” and it claims that the letter and its author would have “remained insignificant” if it weren’t for Miranda’s efforts.

Choice A is incorrect. The text only says that Viscardo wrote the letter and that Miranda edited it: it never suggests that the original authorship of the letter is up for debate. Choice B is incorrect. This contradicts the text. The text says that Miranda edited and footnoted the letter, but it identifies Viscardo as the letter’s author. It also never identifies certain arguments as more eloquent than others. Choice D is incorrect. This is outside the scope of the text. The paragraph describes Miranda’s role in the creation and distribution of the letter, but it doesn’t get into the effects of the letter on other people.

Question Difficulty: Hard

Question ID cef77aa7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: cef77aa7

Geoglyphs are large-scale designs of lines or shapes created in a natural landscape. The Nazca Lines were created in the Nazca Desert in Peru by several Indigenous civilizations over a period of many centuries. Peruvian archaeologist Johny Isla specializes in these geoglyphs. At a German exhibit about the Nazca Lines, he saw an old photograph of a large geoglyph of a whalelike figure and was surprised that he didn’t recognize it. Isla returned to Peru and used a drone to search a wide area, looking for the figure from the air. This approach suggests that Isla thought that if he hadn’t already seen it, the whalelike geoglyph _____

Which choice most logically completes the text?

- A. must represent a species of whale that went extinct before there were any people in Peru.
- B. is actually located in Germany, not Peru, and isn’t part of the Nazca Lines at all.
- C. is probably in a location Isla hadn’t ever come across while on the ground.
- D. was almost certainly created a long time after the other Nazca Lines geoglyphs were created.

ID: cef77aa7 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of Johny Isla and the whalelike geoglyph. The text indicates that the German exhibit about the Nazca Lines included a photograph showing a whalelike geoglyph that Isla hadn’t known about before attending the exhibit, even though Isla “specializes in” Nazca Lines geoglyphs. Given his expertise, and his surprise at being unfamiliar with the whale glyph, the text strongly suggests that Isla believed he would have noticed the glyph if he had been to its location. Thus, the text implies that the whalelike geoglyph is likely in a location Isla had not previously been to.

Choice A is incorrect because the text doesn’t address either the species of whale that the geoglyph is meant to represent or its relationship to the earliest humans in the area that is now Peru. Choice B is incorrect. Although the text indicates that the photograph of the whalelike geoglyph was on display at a “German exhibit,” that exhibit was specifically “about the Nazca Lines,” which the text indicates are located in Peru. Choice D is incorrect. Although the text does indicate that the glyphs were created “over a period of many centuries,” the text doesn’t address when in that period of time any particular glyphs were created.

Question Difficulty: Hard

Question ID 9c591ff7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 9c591ff7

Some *Astyanax mexicanus*, a river-dwelling fish found in northeast Mexico, have colonized caves in the region. Although there is little genetic difference between river and cave *A. mexicanus* and all members of the species can emit the same sounds, biologist Carole Hyacinthe and colleagues found that the context and significance of those sounds vary by location—e.g., the click that river-dwelling *A. mexicanus* use to signal aggression is used by cave dwellers when foraging—and the acoustic properties of cave fish sounds show some cave-specific variations as well. Hyacinthe and colleagues note that differences in sonic communication could accumulate to the point of inhibiting interbreeding among fish from different locations, suggesting that _____

Which choice most logically completes the text?

- although *A. mexicanus* living in rivers are genetically similar to those living in caves, river fish rely on sonic communication less than cave fish do.
- although *A. mexicanus* is a single species at present, it could be in the process of splitting into distinct populations with different characteristics.
- although all *A. mexicanus* emit sounds, the fish living in rivers produce some sounds that the fish living in caves do not, and vice versa.
- although *A. mexicanus* from different locations can interbreed currently, river fish and cave fish are sufficiently genetically distinct that they can be considered separate species.

ID: 9c591ff7 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of *A. mexicanus*. According to the text, *A. mexicanus*, a river-dwelling fish species, has colonized caves. The fish that live in rivers and those that live in caves show no significant genetic differences and are all capable of making the same sounds. The text indicates, however, that Hyacinthe and colleagues found that sounds that the river fish use in a particular context and for a particular purpose are used in a different context and for a different purpose by the cave fish. Additionally, the sounds made by cave fish show some differences depending on the specific cave the fish inhabit. The text goes on to state that Hyacinthe and colleagues have noted that differences in how the fish use sound to communicate could eventually become so great that they prevent fish from different locations from interbreeding. In other words, the river fish might eventually only breed with other river fish (with whom they share characteristics regarding sonic communication that they do not share with cave fish), while the cave fish might only breed with other cave fish for a similar reason. In context, this observation suggests that even though the fish are a single species right now, they could be in the process of splitting into distinct populations with different characteristics.

Choice A is incorrect because there is no information in the text suggesting that the river fish are less reliant on sonic communication than the cave fish are. Although the text does indicate that the river fish and cave fish are genetically similar, the text describes both groups as using sonic communication and says nothing to indicate that one group depends on that communication more than the other group does. Choice C is incorrect. The text states that all members of the species can emit the same sounds but that the function and context of sounds differ depending on whether the fish live in rivers or caves, but it does not indicate that river fish produce sounds that cave fish do not or vice versa. Choice D is incorrect because it contradicts the text. The text says that there is little genetic difference between the river fish and the cave fish, not that the river fish and cave fish are so genetically distinct that they can be considered separate species.

Question Difficulty: Hard

Question ID 5d122d45

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 5d122d45

Psychologists Dacher Keltner and Jonathan Haidt have argued that experiencing awe—a sensation of reverence and wonder typically brought on by perceiving something grand or powerful—can enable us to feel more connected to others and thereby inspire us to act more altruistically. Keltner, along with Paul K. Piff, Pia Dietze, and colleagues, claims to have found evidence for this effect in a recent study where participants were asked to either gaze up at exceptionally tall trees in a nearby grove (reported to be a universally awe-inspiring experience) or stare at the exterior of a nearby, nondescript building. After one minute, an experimenter deliberately spilled a box of pens nearby.

Which finding from the researchers’ study, if true, would most strongly support their claim?

- Participants who had been looking at the trees helped the experimenter pick up significantly more pens than did

A. participants who had been looking at the building.
- Participants who helped the experimenter pick up the pens used a greater number of positive words to describe the trees

B. and the building in a postexperiment survey than did participants who did not help the experimenter.
- Participants who did not help the experimenter pick up the pens were significantly more likely to report having

C. experienced a feeling of awe, regardless of whether they looked at the building or the trees.
- Participants who had been looking at the building were significantly more likely to notice that the experimenter had

D. dropped the pens than were participants who had been looking at the trees.

ID: 5d122d45 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents a finding that, if true, would most strongly support the researchers' claim that they found evidence that experiencing awe can make people feel more connected to others and thus more likely to behave altruistically (with beneficial and unselfish concern for others). According to the text, the researchers tested for this effect by first having participants look at either something known to be awe-inspiring (very tall trees) or something ordinary (a plain building) and then purposely spilling pens near the participants. The finding that participants who had looked at the trees helped pick up significantly more pens than did participants who had looked at the building would support the researchers' claim by demonstrating that the people who had experienced awe behaved more altruistically when the experimenter needed help than the other participants did.

Choice B is incorrect because a finding about helpful participants using positive words to describe the trees and the building after the experiment was over wouldn't have any bearing on the researchers' claim that experiencing awe increases altruistic behavior. The text doesn't address the use of positive words to describe things or suggest any connection between using such words and having experienced awe, so that behavior wouldn't serve as evidence that experiencing awe played a role in promoting helpful behavior. Choice C is incorrect because a finding that participants who didn't help the experimenter were significantly more likely than others to report having experienced awe whether they had looked at the building or the trees would weaken the researchers' claim that experiencing awe increases altruistic behavior by suggesting that the opposite might be true—that experiencing awe is in fact linked to choosing not to act in a way that benefits someone else. Choice D is incorrect because a finding about participants noticing that the experimenter had dropped the pens wouldn't have any bearing on the researchers' claim about people behaving altruistically. Being aware of a challenge or problem isn't necessarily beneficial on its own and isn't the same as offering help, so the finding wouldn't support the idea that experiencing awe increases altruistic behavior.

Question Difficulty: Hard

Question ID 156ff681

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 156ff681

Many governments that regularly transfer money to individuals—to provide supplemental incomes for senior citizens, for example—have long done so electronically, but other countries typically have distributed physical money and have only recently developed electronic transfer infrastructure. Researchers studied the introduction of an electronic transfer system in one such location and found that recipients of electronic transfers consumed a different array of foods than recipients of physical transfers of the same amount did. One potential explanation for this result is that individuals conceive of and allocate funds in physical money differently than they conceive of and allocate funds in electronic form.

Which finding from the study, if true, would most directly weaken the potential explanation?

- A. Recipients of electronic transfers typically spent their funds at a slower rate than recipients of physical transfers did.
Nearly every recipient of an electronic transfer withdrew the entire amount in physical money shortly after receiving the B. transfer.
- C. Recipients of physical transfers tended to purchase food about as frequently as recipients of electronic transfers did.
Some recipients of physical transfers received small amounts of money relatively frequently, while others received large D. amounts relatively infrequently.

ID: 156ff681 Answer

Correct Answer: B

Rationale

Choice B is the best answer. This would weaken the explanation. If the recipients of electronic money immediately withdrew it all as physical money, then both kinds of recipients ended up spending physical money on food. So there must be some other explanation why those who initially received electronic money ate different kinds of food.

Choice A is incorrect. This wouldn't weaken the explanation. If anything, it actually supports it: it demonstrates that recipients of electronic money and recipients of physical money have different spending habits. Choice C is incorrect. This wouldn't weaken the explanation. The explanation we're testing this choice against is about the way that people might "conceive of and allocate" physical and electronic funds differently. This choice only focuses on the timing, not what they spend the money on. Choice D is incorrect. This would have no impact on the explanation. It doesn't tell us anything about possible differences between the spending habits of those who spend physical money and those who spend money electronically.

Question Difficulty: Hard

Question ID 8545ccfe

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 8545ccfe

Icebergs generally appear to be mostly white or blue, depending on how the ice reflects sunlight. Ice with air bubbles trapped in it looks white because much of the light reflects off the bubbles. Ice without air bubbles usually looks blue because the light travels deep into the ice and only a little of it is reflected. However, some icebergs in the sea around Antarctica appear to be green. One team of scientists hypothesized that this phenomenon is the result of yellow-tinted dissolved organic carbon in Antarctic waters mixing with blue ice to produce the color green.

Which finding, if true, would most directly weaken the team’s hypothesis?

- A. White ice doesn’t change color when mixed with dissolved organic carbon due to the air bubbles in the ice.
- B. Dissolved organic carbon has a stronger yellow color in Antarctic waters than it does in other places.
- C. Blue icebergs and green icebergs are rarely found near each other.
- D. Blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon.

ID: 8545ccfe Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would weaken the scientists’ hypothesis about icebergs that appear to be green. The text indicates that most icebergs are either mostly white or blue in color but that some icebergs in Antarctica appear to be green. The text goes on to say that the scientists hypothesized that this green color occurs when yellow-tinted dissolved organic carbon in ocean waters mixes with blue ice. A finding that both blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon would suggest that something other than yellow-tinted organic carbon causes some icebergs’ green color, since the blue icebergs that contain yellow-tinted organic carbon remained blue instead of turning green.

Choice A is incorrect because, according to the text, the scientists’ hypothesis was that blue icebergs, not white ones, change color when their ice mixes with yellow-tinted dissolved organic carbon. A finding that white ice, because of its air bubbles, doesn’t change color when it’s mixed with dissolved organic carbon would therefore have no bearing on the scientists’ hypothesis. Choice B is incorrect because the text focuses only on Antarctic icebergs that appear to be green. It doesn’t indicate that icebergs in locations other than Antarctica have been found to have a green hue. A finding that dissolved organic carbon has a stronger yellow color in Antarctic waters than in other places would therefore have no bearing on the scientists’ hypothesis that green color in icebergs in Antarctica is caused by yellow-tinted dissolved organic carbon mixing with blue ice. Choice C is incorrect because, according to the text, the scientists’ hypothesis was that blue icebergs turn green when their ice mixes with yellow-tinted dissolved organic carbon in the sea around them. If that’s correct, one would expect blue icebergs and green icebergs to be located at a distance from each other since all blue icebergs in an area where the waters contain yellow-tinted dissolved organic carbon would take on a green hue. A finding that blue icebergs and green icebergs are rarely found near each other would therefore strengthen, not weaken, the researchers’ hypothesis.

Question Difficulty: Hard

Question ID e946a32e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: e946a32e

Boldly mixing elements of poetry, fiction, drama, philosophy, and manifesto, Puerto Rican writer Giannina Braschi creates cross-genre literature that explores themes such as immigration and independence. Her works have inspired responses from individuals across different fields and in a wide range of formats, from musical compositions and a comic book to architecture and furniture design. In an essay, a student asserts that the production of these diverse creations by others is reflective of Braschi’s own approach to crafting literature.

Which quotation from a scholarly review of Braschi’s work best supports the student’s claim?

- “Braschi is the focus of a 2020 collection of essays in which fifteen scholars from seven different countries delved into
- A. the linguistic and structural patterns of her writings.”
- “Braschi’s eagerness to push boundaries and blend genres within literature invites us to consider how other art forms
- B. might also engage with literature.”
- “Before settling in New York City, where she would go on to become a college professor, Braschi studied both literature
- C. and philosophy in several cities around the world.”
- “In addition to her creative literary works, Braschi has produced academic pieces analyzing writings by Miguel de
- D. Cervantes, Federico García Lorca, and other authors.”

ID: e946a32e Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the quotation that best supports the student’s claim about Braschi. By describing how Braschi’s blending of genres invites her audience to think about how other art forms could also engage with literature, the quotation supports the idea that the diversity of responses to Braschi’s work reflects Braschi’s own approach to creating literature.

Choice A is incorrect because the quotation describes scholars from different countries writing essays about Braschi’s use of language in her writings; it doesn’t address how Braschi’s creation of cross-genre literature inspires diverse types of responses, which is the claim the student makes. Choice C is incorrect because the quotation focuses on the fact that Braschi studied in several different cities, which doesn’t address the student’s claim that Braschi’s creation of cross-genre literature inspires diverse types of responses. Choice D is incorrect because the quotation lists some of the authors who Braschi has written academic works about, which is irrelevant to the student’s claim that Braschi’s creation of cross-genre literature inspires diverse types of responses.

Question Difficulty: Hard

Question ID dc87adf4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: dc87adf4

Barchester Towers is an 1857 novel by Anthony Trollope. In the novel, Trollope’s portrayal of Dr. Proudie underscores the character’s exaggerated sense of his own abilities: _____

Which quotation from *Barchester Towers* most effectively illustrates the claim?

- “It must not...be taken as proved that Dr. Proudie was a man of great mental powers, or even of much capacity for
- A. business, for such qualities had not been required in him.”
- “[Dr. Proudie] was comparatively young, and had, as he fondly flattered himself, been selected as possessing such gifts,
- B. natural and acquired, as must be sure to recommend him to a yet higher notice.”
- “[Dr. Proudie’s] residence in the metropolis, rendered necessary by duties thus entrusted to him, his high connexions, and
- C. the peculiar talents and nature of the man, recommended him to persons in power.”
- “[Dr. Proudie] was certainly possessed of sufficient tact to answer the purpose for which he was required without making
- D. himself troublesome.”

ID: dc87adf4 Answer

Correct Answer: B

Rationale

Choice B is the best answer. In this quotation, Dr. Proudie is described as "fondly flatter[ing] himself" that he has gifts that "must be sure to recommend him to a yet higher notice." In other words, he expects his skills to push him to greater fame and success. This implies an exaggerated sense of his own abilities, which matches the claim we’re trying to support.

Choice A is incorrect. This quotation doesn’t describe Proudie’s view of himself, nor does it paint him in an especially flattering light. Instead, by saying his mental powers and business skill are not "proved," it implies that he is actually dim-witted and bad at business. Choice C is incorrect. This choice describes Proudie’s closeness to power and importance, but it doesn’t show what Proudie thinks of himself. Proudie is not describing himself or his abilities here. The narrator is. Choice D is incorrect. While this quotation offers Proudie very mild praise, it doesn’t show what Proudie thinks of himself or his own abilities, which is what the claim focuses on.

Question Difficulty: Hard

Question ID 3bfc73b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 3bfc73b

An archaeological team led by Piotr Bieliński and Sultan al-Bakri found remnants of a 4,000-year-old Bronze Age board game at a site in Oman. Little is left of the game except a stone board, which is carved with a grid and has places to hold game pieces. Some scholars claim that the game was largely played by traders.

Which finding, if true, would most directly support the scholars’ claim?

- Other examples of the game dating to the same period have been found in the remains of several homes in the region,
- A. including in one home that may have belonged to a trader.
 - Similar games have been found in other sites dating to the same period that were connected to the site in Oman via trade routes.
 - B. The other known examples of the game dating to the same period have been found along routes that seem to have been used primarily by traders at the time.
 - C. Remnants of other goods have been found at the site in Oman that probably also reached the location through trade.
 - D.

ID: 3bfc73b Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most directly support the scholars’ claim about the board game. The text explains that the remains of a 4,000-year-old board game were found in Oman. The text then states that scholars claim this board game was played mostly by traders. If the other known examples of this board game from the same time period were discovered along routes that seem to have been used primarily by traders, this finding would directly support the scholars’ claim because it suggests that the game was largely played by traders who brought it with them for entertainment as they traveled.

Choice A is incorrect because this finding would suggest only that a single trader may have possessed examples of the board game, perhaps for the purpose of trading or selling the game to residents of Oman. For this reason, the finding wouldn’t directly support the scholars’ claim that the majority of the game’s players were traders. Choice B is incorrect because this finding doesn’t mention the board game at all, referring only to similar games found at other sites, and would therefore provide no direct support for the scholars’ claim about the board game. Choice D is incorrect because this finding doesn’t mention the board game at all, referring only to the remains of other goods found at the site in Oman, and would therefore provide no direct support for the scholars’ claim about the board game.

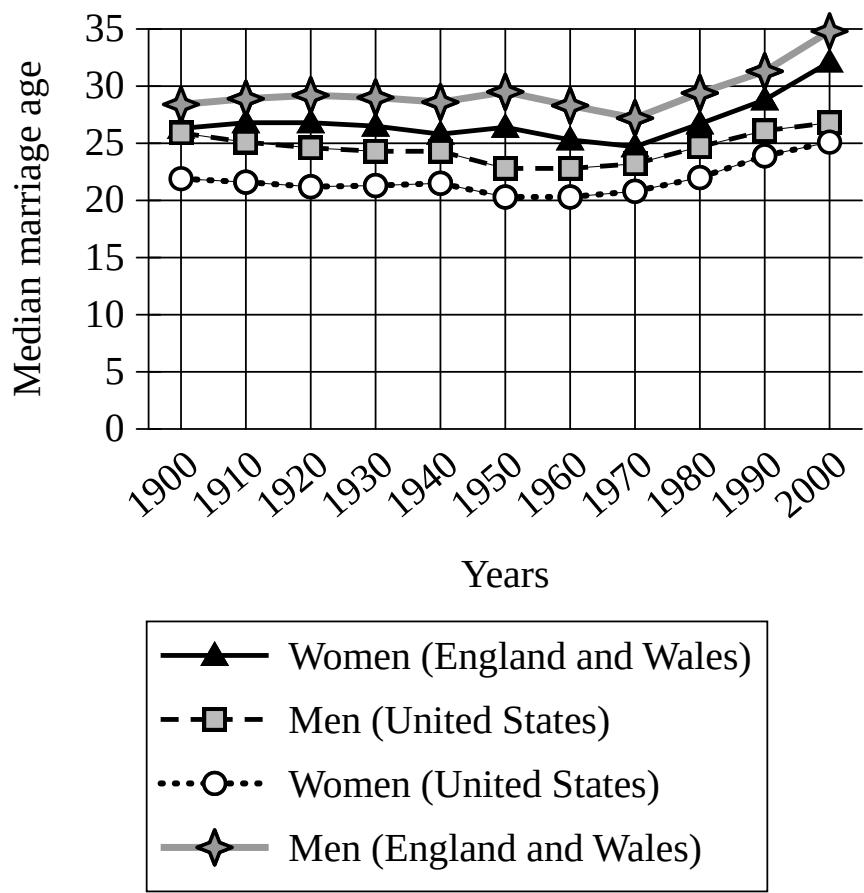
Question Difficulty: Hard

Question ID 53c6c179

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 53c6c179

Median Ages of First Marriage for Men and Women in the United States and in England and Wales, 1900–2000



A sociology student is reading an essay on the median age of first marriage in Western countries throughout the twentieth century. The author of the essay cites factors common to these countries that the author believes caused an increase in the median age of first marriage, such as new technologies that shortened the time needed for domestic chores, making two-person households less necessary and living alone more viable. The student asserts that beyond these factors there must be additional ones specific to particular Western countries that influenced the increase of age at first marriage.

Which choice most effectively uses data from the graph that support the student’s assertion?

Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for A. men in the United States.

- In England and Wales, the median age of first marriage was consistently higher for men than for women between 1900 and 2000, but this was not always the case in the United States.
- C. The median age of first marriage for men in England and Wales was lower in 1970 than in 1950 or 1990.
- Between 1900 and 2000, the median age of first marriage for women in England and Wales was consistently higher than for women in the United States, as was the case for men.

ID: 53c6c179 Answer

Correct Answer: A

Rationale

Choice A is the best answer. The student concluded that country-specific factors influence changes to median age of first marriage. This choice describes a time period when the rate of change differed between countries, suggesting that country-specific factors may have played a role in these changes.

Choice B is incorrect. This choice misreads the graph. The median age of first marriage was consistently higher for men than for women in the United States during the time period depicted. Choice C is incorrect. The student concluded that country-specific factors influence changes to median age of first marriage. However, this choice doesn't provide any contrasts between countries and thus doesn't support the idea of country-specific factors influencing median age of first marriage. Choice D is incorrect. The students' conclusion is about changes that occurred during the 20th century. This choice provides broad information about the century as a whole, so it doesn't give insight into how median age at first marriage changed over time.

Question Difficulty: Hard

Question ID 1bf2173e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: 1bf2173e

In a three-year study of parasitic infections by *Anomotaenia brevis* tapeworms in *Temnothorax nylander* ants, entomologist Susanne Foitzik and colleagues found something unexpected: rather than reducing its host’s fitness, as is typical of parasites, *A. brevis* greatly extends the lifespan of a *T. nylander* worker ant and seems to halt the effects of aging. Furthermore, those infected receive special treatment, ceasing their share of labor to sustain the colony and remaining in the nest as uninfected workers feed, groom, and transport them. By contrast, the researchers observed that uninfected workers in parasitized colonies have shortened lifespans, most likely because the _____

Which choice most logically completes the text?

- uninfected workers are at high risk for direct exposure to *A. brevis* in the course of providing social care to the infected

A. workers in the nest.
- need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to

B. the uninfected workers.
- high level of activity maintained by the uninfected workers makes them better able than infected workers to quickly

C. disperse when the nest is attacked by a predator.
- average lifespan of *T. nylander* worker ants in colonies without parasitic activity typically falls well below three years, the

D. range covered by the study.

ID: 1bf2173e Answer

Correct Answer: B

Rationale

Choice B is the best answer. The text describes a study examining the relationship between a species of parasitic tapeworm, *A. brevis*, and its host insect, the *T. nylander* ant. According to the text, researchers were surprised to find that the tapeworm extends the life of its ant host, rather than reducing it. The text goes on to state that the infected ants end up doing less work to sustain the colony and that as a result, the uninfected ants take on the infected ants' share of labor in addition to caring for them in their infected state. The study's researchers also observed that the uninfected ants have shorter lifespans than expected. If the infected ants, who are doing less work in the colony, have longer lifespans, it can be inferred that the less an ant works, the longer it will live. The opposite of this statement can also be inferred: the more an ant works, the shorter its life. So, since the workload within the colony is being redistributed so that the infected ants work less while the uninfected ants work more (as they take on the neglected duties of the infected ants and also care for those ants), then it can be inferred that the lifespans of the uninfected ants are shortened because the need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.

Choice A is incorrect because the text does not indicate how *A. brevis* is transmitted to the ants or assert that uninfected ants are more likely to be directly exposed to *A. brevis* while caring for infected ants. Choice C is incorrect because the text makes no mention of the relative abilities of infected and uninfected ants to escape predators: in fact, predators are not mentioned in the text at all. Choice D is incorrect because the text does not supply any information about the average lifespans of the ants in colonies without parasitic activity; the text only indicates factors that lengthen and shorten the lifespans of ants in parasitized colonies.

Question Difficulty: Hard

Question ID df91532e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: df91532e

In the “language nest” model of education, Indigenous children learn the language of their people by using it as the medium of instruction and socialization at pre-K or elementary levels. In their 2016 study of a school in an Anishinaabe community in Ontario, Canada, scholars Lindsay Morcom and Stephanie Roy (who are Anishinaabe themselves) found that the model not only imparted fluency in the Anishinaabe language but also enhanced students’ pride in Anishinaabe culture overall. Given these positive effects, Morcom and Roy predict that the model increases the probability that as adults, former students of the school will transmit the language to younger generations in their community.

Which finding, if true, would most strongly support the researchers’ prediction?

As Anishinaabe adults who didn’t attend the school feel roughly the same degree of cultural pride as the former students of A. the school feel.

B. After transferring to the school, new students experience an increase in both fluency and academic performance overall.

As adults, former students of the school are just as likely to continue living in their community as individuals who didn’t C. attend the school.

As they complete secondary and higher education, former students of the school experience no loss of fluency or D. cultural pride.

ID: df91532e Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support the researchers' prediction about the language nest model of education. The text states that Morcom and Roy studied the effects of the language nest model of education on students at an Anishinaabe school, and they found that the model—which is used with students during pre-K or elementary school—increased students' fluency in the Anishinaabe language and pride in Anishinaabe culture. The researchers predicted that the students' positive early experiences with the Anishinaabe language would lead them to be more likely to later share the language with younger generations. If former students maintain full fluency and cultural pride after finishing secondary and higher education, it follows that they would be both able and motivated to share what they know with others; this would likely result in a higher probability of transmitting the language to younger generations, as the researchers predict.

Choice A is incorrect because finding that Anishinaabe adults who didn't attend the school feel approximately the same degree of cultural pride as those adults who did attend wouldn't support the researchers' prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people. Choice B is incorrect because finding that new students experience increased performance in language fluency and academics would suggest that the school has a positive effect on students when they attended but wouldn't reveal anything about those students' later actions as adults (such as their likelihood of sharing their knowledge with younger generations). Choice C is incorrect because finding that Anishinaabe adults who attended the school are equally likely to stay in the community as adults who didn't attend the school wouldn't support the researchers' prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people.

Question Difficulty: Hard

Question ID 8584f3ce

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 8584f3ce

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada’s coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant’s rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadow’s health overall.

Which finding, if true, would most directly undermine Foster’s hypothesis?

- At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.
- At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.
- At several sites not included in the study, eelgrass meadows’ health correlates negatively with the length of residence and size of otter populations.
- At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.

ID: 8584f3ce Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would weaken Foster's hypothesis that damage to eelgrass roots improves the health of eelgrass meadows by boosting genetic diversity. The text indicates that sea otters damage eelgrass roots but that eelgrass meadows near Vancouver Island, where there's a large otter population, are comparatively healthy. When Foster and her colleagues compared the Vancouver Island eelgrass meadows to those that don't have established otter populations, the researchers found that the Vancouver Island meadows are more genetically diverse than the other meadows are. This finding led Foster to hypothesize that damage to the eelgrass roots encourages eelgrass reproduction, thereby improving genetic diversity and the health of the meadows. If, however, other meadows not included in the study are less healthy the larger the local otter population is and the longer the otters have been in residence, that would suggest that damage to the eelgrass roots, which would be expected to increase with the size and residential duration of the otter population, isn't leading meadows to be healthier. Such a finding would therefore weaken Foster's hypothesis.

Choice A is incorrect because finding that small, recently introduced otter populations are near other eelgrass meadows in the study wouldn't weaken Foster's hypothesis. If otter populations were small and only recently established, they wouldn't be expected to have caused much damage to eelgrass roots, so even if those eelgrass meadows were less healthy than the Vancouver Island meadows, that wouldn't undermine Foster's hypothesis. In fact, it would be consistent with Foster's hypothesis since it would suggest that the greater damage caused by larger, more established otter populations is associated with healthier meadows. Choice B is incorrect because the existence of areas with otters but without eelgrass meadows wouldn't reveal anything about whether the damage that otters cause to eelgrass roots ultimately benefits eelgrass meadows. Choice D is incorrect because the health of plants other than eelgrass would have no bearing on Foster's hypothesis that damage to eelgrass roots leads to greater genetic diversity and meadow health. It would be possible for otters to have a negative effect on other plants while nevertheless improving the health of eelgrass meadows by damaging eelgrass roots.

Question Difficulty: Hard

Question ID 9391b7cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 9391b7cc

If some artifacts recovered from excavations of the settlement of Kuulo Kataa, in modern Ghana, date from the thirteenth century CE, that may lend credence to claims that the settlement was founded before or around that time. There is other evidence, however, strongly supporting a fourteenth century CE founding date for Kuulo Kataa. If both the artifact dates and the fourteenth century CE founding date are correct, that would imply that _____

Which choice most logically completes the text?

- A. artifacts from the fourteenth century CE are more commonly recovered than are artifacts from the thirteenth century CE.
- B. the artifacts originated elsewhere and eventually reached Kuulo Kataa through trade or migration.
- C. Kuulo Kataa was founded by people from a different region than had previously been assumed.
- D. excavations at Kuulo Kataa may have inadvertently damaged some artifacts dating to the fourteenth century CE.

ID: 9391b7cc Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of artifacts and Kuulo Kataa’s founding date. If it were true both that Kuulo Kataa was founded in the fourteenth century CE and that artifacts found in excavations of the settlement are from the thirteenth century CE, it would be reasonable to conclude that the artifacts weren’t created in the Kuulo Kataa settlement. That would suggest, then, that the artifacts originated somewhere else and eventually reached the settlement through trading or as people migrated.

Choice A is incorrect because the existence of thirteenth-century CE artifacts recovered during excavations of a settlement founded in the fourteenth century CE isn’t logically connected to artifacts from one century being more commonly recovered than artifacts from another century. Rather than suggesting anything about how frequently artifacts from different times are found, the existence of artifacts confirmed as predating the settlement’s founding suggests that those items arrived in Kuulo Kataa during or after its establishment. Choice C is incorrect because the text focuses on time periods and says nothing about which region the founders of Kuulo Kataa have been thought to come from; similarly, the text doesn’t suggest anything about where the thirteenth-century CE artifacts originated other than not from Kuulo Kataa. Therefore, it isn’t logical to conclude that the mere existence of artifacts confirmed as predating the Kuulo Kataa settlement suggests that the founders of the settlement came from a particular region other than one previously assumed. Choice D is incorrect because the existence of artifacts from the thirteenth century CE at a site dated to the fourteenth century CE doesn’t imply that fourteenth-century objects were damaged during excavations. There’s nothing in the text to suggest that any objects were damaged; rather, the existence of artifacts confirmed as predating the settlement’s founding suggests that those items were brought to Kuulo Kataa during or after its establishment.

Question Difficulty: Hard

Question ID 9452092c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 9452092c

Effects of Mycorrhizal Fungi on 3 Plant Species

Plant species	Mycorrhizal host	Average mass of plants grown in soil containing mycorrhizal fungi (in grams)	Average mass of plants grown in soil treated to kill fungi (in grams)
Corn	yes	15.1	3.8
Marigold	yes	10.2	2.4
Broccoli	no	7.5	7

Mycorrhizal fungi in soil benefits many plants, substantially increasing the mass of some. A student conducted an experiment to illustrate this effect. The student chose three plant species for the experiment, including two that are mycorrhizal hosts (species known to benefit from mycorrhizal fungi) and one nonmycorrhizal species (a species that doesn't benefit from and may even be harmed by mycorrhizal fungi). The student then grew several plants from each species both in soil containing mycorrhizal fungi and in soil that had been treated to kill mycorrhizal and other fungi. After several weeks, the student measured the plants' average mass and was surprised to discover that _____

Which choice most effectively uses data from the table to complete the statement?

- broccoli grown in soil containing mycorrhizal fungi had a slightly higher average mass than broccoli grown in soil that
A. had been treated to kill fungi.
- corn grown in soil containing mycorrhizal fungi had a higher average mass than broccoli grown in soil containing
B. mycorrhizal fungi.
- marigolds grown in soil containing mycorrhizal fungi had a much higher average mass than marigolds grown in soil that
C. had been treated to kill fungi.
- corn had the highest average mass of all three species grown in soil that had been treated to kill fungi, while marigolds
D. had the lowest.

ID: 9452092c Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to complete the statement. The text explains that mycorrhizal hosts are plants that benefit from the presence of mycorrhizal fungi in the soil and that some such plants produce more mass when grown in the presence of these fungi, while for nonmycorrhizal species the fungi either have no effect or may be harmful. The experiment included two mycorrhizal hosts (corn and marigold) and one nonmycorrhizal species (broccoli). Given the claim in the text that nonmycorrhizal species will see either no difference or a decrease in mass when exposed to mycorrhizal fungi, the student would likely have been surprised by the higher average mass for broccoli grown in the presence of the fungi than the broccoli grown in the soil treated to kill fungi.

Choice B is incorrect. Although this choice accurately describes the corn data from the table, the fact that the mycorrhizal host corn is more massive in the presence of the fungi likely fits with what the student expected and would therefore not be surprising. Choice C is incorrect. Although this choice accurately describes the marigold data from the table, the fact that the mycorrhizal host marigold is more massive in the presence of the fungi is likely what the student expected and thus would not be surprising. Choice D is incorrect because it does not accurately represent the data in the table—when grown in soil treated to kill fungi, corn had an average mass of 3.8 g while broccoli had an average mass of 7g—and because making comparisons among the plants in the no-fungi condition, by itself, does not provide a basis to compare the average mass of mycorrhizal hosts and nonmycorrhizal species grown in the presence of the fungi with those grown in the soil treated to kill fungi.

Question Difficulty: Hard

Question ID 2a075bd1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 2a075bd1

Indigenous cultures possess unique knowledge of the medicinal uses of plants. According to a 2021 study, 73 percent of the medicinal uses of plants native to North America are reflected in the vocabulary of a single Indigenous language. However, as more and more Indigenous people exclusively speak a globally dominant language, such as English, their ancestral languages faded from daily use. These facts lend added importance to tribal nations’ efforts to preserve their languages. By ensuring the continued use of Cherokee, Ojibwe, and the hundreds of other Indigenous languages in what is now the United States, tribal nations are also _____

Which choice most logically completes the text?

- A. increasing the number of medicinal plants represented in the vocabularies of Indigenous languages.
- B. transmitting terms for medicinal plants from Indigenous languages to globally dominant languages.
- C. preserving knowledge about the medicinal value of plants native to the tribal nations’ lands.
- D. ensuring that citizens of tribal nations have physical access to medicinal plants.

ID: 2a075bd1 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of the relationship between Indigenous languages and knowledge of the medicinal uses of plants. The text states that Indigenous cultures possess special knowledge of the medicinal uses of plants, which is reflected in their vocabulary. The text then discusses how tribal nations are working to preserve their languages, whose daily use is declining as globally dominant languages become increasingly dominant in Indigenous communities. Given that the languages of tribal nations in what is now the United States function as repositories of knowledge about plants’ medicinal uses, it logically follows that continued use of those languages will assist with passing on knowledge about the medicinal value of plants native to the tribal nations’ lands.

Choice A is incorrect because the text states that preserving Indigenous languages will increase the knowledge, not the number, of medicinal plants. Choice B is incorrect because the text is concerned with how vocabulary about the medicinal value of plants can be preserved through the continued daily use of Indigenous languages, not with how such vocabulary can be incorporated into globally dominant, non-Indigenous languages. Moreover, the text explains that the exclusive use of globally dominant languages in Indigenous communities comes at an expense to the continued daily use of those communities’ languages. Given this relationship, it is unlikely globally dominant languages would borrow Indigenous vocabulary pertaining to plants’ medicinal uses. Choice D is incorrect because the text doesn’t discuss physical access to medicinal plants, instead focusing on Indigenous knowledge and language surrounding the medicinal uses of plants.

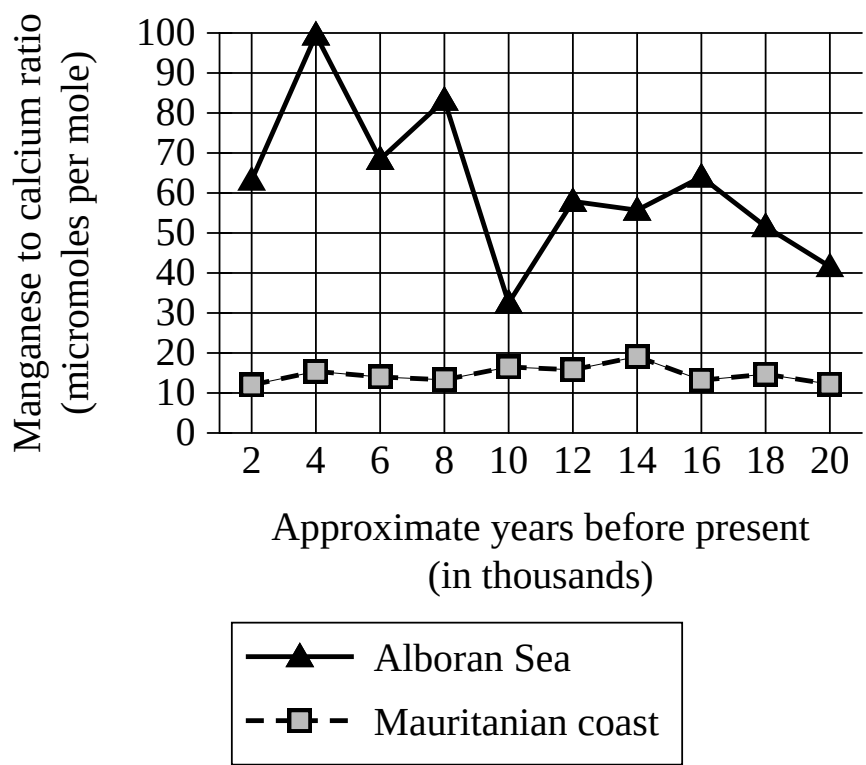
Question Difficulty: Hard

Question ID 3233c162

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 3233c162

Ratio of Manganese to Calcium in Samples from Alboran Sea and Mauritanian Coast



The population of the coral *Lophelia pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Using the ratio of manganese to calcium, which inversely correlates with ocean oxygenation levels, marine scientist Rodrigo da Costa Portilho-Ramos and colleagues evaluated whether oxygenation played a role in the declines of *L. pertusa*. The researchers concluded that oxygenation may have been important in the Alboran Sea but not near the Mauritanian coast, since _____

Which choice most effectively uses data from the graph to complete the statement?

- a substantial increase in oxygenation in the Alboran Sea corresponded with the local decline in *L. pertusa*, but the opposite relationship between oxygenation and *L. pertusa* was found near the Mauritanian coast.
- L. pertusa* declined in the Alboran Sea during a period of substantial local decline in oxygenation, but *L. pertusa* declined near the Mauritanian coast during a period of little local change in oxygenation.
- oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after the decline, whereas oxygenation near the Mauritanian coast was relatively low both before and after the decline in *L. pertusa*.
- oxygenation in the Alboran Sea tended to be substantially higher than oxygenation near the Mauritanian coast during the period studied.

Correct Answer: B

Rationale

Choice B is the best answer because it effectively uses data from the graph to complete the statement about Rodrigo da Costa Portilho-Ramos and colleagues' conclusion. The graph shows the ratio of manganese to calcium in *L. pertusa* coral samples from the Alboran Sea and the Mauritanian coast. The graph reflects time in approximate years before present: in other words, the greater the number in years noted on the graph's horizontal axis, the farther that moment is in the past. The text indicates that the researchers tested the samples to determine whether oxygenation played a role in the decline of *L. pertusa*. The text goes on to note that a change in the ratio of manganese to calcium would signal an inverse, or opposite, change in oxygenation. According to the graph, the ratio of manganese to calcium in samples from the Alboran Sea increased from about 30 micromoles per mole 10,000 years ago to about 80 micromoles per mole 8,000 years ago, which means that oxygenation decreased between 10,000 and 8,000 years ago. Meanwhile, there was almost no discernible change in the ratio of manganese to calcium in samples from the Mauritanian coast between 12,000 and 10,000 years ago. According to the text, the population of *L. pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Thus, the increase in the ratio of manganese to calcium around 9,000 years ago in the Alboran Sea coincides with the decline in the *L. pertusa* population, suggesting an association between the decrease in oxygenation and the decline in population of the coral. No such relationship is suggested around 11,000 years ago near the Mauritanian coast. So, oxygenation likely played a role in the *L. pertusa* decline in the Alboran Sea but not in the coral's decline near the Mauritanian coast.

Choice A is incorrect because it asserts the opposite of what the graph indicates regarding oxygenation in the Alboran Sea, and it misrepresents what the graph indicates about oxygenation near the Mauritanian coast. The graph indicates that at the time of the decline in *L. pertusa* (approximately 9,000 years ago), the samples from the Alboran Sea contained a ratio of manganese to calcium that was increasing. According to the text, this ratio inversely correlates with ocean oxygenation levels, so if the ratio was increasing, oxygenation was decreasing, not substantially increasing. Furthermore, the graph shows that the ratio of manganese to calcium remained relatively stable in coral samples from the Mauritanian coast during the period studied, which suggests that there was no discernible relationship between oxygenation and the coral's population decline in that location, not that there was a substantial decrease in oxygenation corresponding to the coral's decline. Choice C is incorrect. Although the graph suggests that the level of oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after—because the ratio of manganese to calcium inversely correlates with ocean oxygenation levels and this ratio was lower before the decline than after—the graph doesn't support the claim that oxygenation near the Mauritanian coast was consistently low before and after the coral's decline there. Rather, the graph indicates that relative to coral samples from the Alboran Sea, the ratio of manganese to calcium in samples from near the Mauritanian coast was consistently low, which suggests that oxygenation levels were relatively high both before and after the decline of *L. pertusa*. Choice D is incorrect because it states the opposite of what the graph indicates: the graph shows that throughout the period studied, the ratio of manganese to calcium was higher in coral samples from the Alboran Sea than it was in samples from near the Mauritanian coast. Since the text indicates that the ratio of manganese to calcium inversely correlates with ocean oxygenation levels, oxygenation in the Alboran Sea was therefore lower than, not higher than, oxygenation near the Mauritanian coast during the period studied. Moreover, even if choice D did accurately represent the graph, it wouldn't effectively complete the statement since a comparison of the ocean oxygenation levels at the two locations is not relevant to the claim that a decline in oxygenation levels was associated with the decline of *L. pertusa* in the Alboran Sea but not near the Mauritanian coast.

Question Difficulty: Hard

Question ID 7f0be746

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 7f0be746

The following text is from Milan Kundera’s 1984 novel *The Unbearable Lightness of Being* (translated by Michael Henry Heim in 1984). Karenin is a dog that belongs to Tomas and Tereza.

Karenin was not overjoyed by the move to Switzerland [from Prague]. Karenin hated change. Dog time cannot be plotted along a straight line; it does not move on and on, from one thing to the next. It moves in a circle like the hands of a clock, which—they, too, unwilling to dash madly ahead—turn round and round the face, day in and day out following the same path. In Prague, when Tomas and Tereza bought a new chair or moved a flower pot, Karenin would look on in displeasure. It disturbed his sense of time. It was as though they were trying to dupe the hands of the clock by changing the numbers on its face.

©1984 by Milan Kundera. Translation ©1984 by Harper & Row, Publishers, Inc.

Which choice best states the main idea of the text?

- As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to
- A. disruption.
- B. After he’s moved to a new home, Karenin’s negative response to changes has become more pronounced.
- C. Similar to Tomas and Tereza, Karenin comprehends time as circular rather than as a straightforward progression.
- As is the case for other dogs, Karenin’s sense of time seems to accelerate depending on the objects and places that
- D. surround him.

ID: 7f0be746 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. The text begins by stating that Tomas and Tereza's dog Karenin felt disrupted by a recent move because of his dislike of change. The text then goes on to suggest that this is because the way a dog experiences time differs from the way humans experience time: time for a dog doesn't move linearly, going "on and on, from one thing to the next," but instead moves circularly, "like the hands of a clock." That is, time for a dog is experienced as a cyclical pattern characterized by routine and predictability, with each day "following the same path." The text then concludes by providing examples of seemingly insignificant changes in routine that profoundly "disturbed [Karenin's] sense of time," causing him to feel displeasure. Thus, the main idea of the text is that Karenin's sense of time as a dog involves a strong preference for predictability and an aversion to disruption.

Choice B is incorrect. Although the text emphasizes Karenin's displeasure with the recent move to a new home, it doesn't suggest that the move has made his negative responses more pronounced than they once were. Rather, in accounting for Karenin's displeasure with the move to Switzerland, the text explains that Karenin generally has a negative response to any kind of change. Choice C is incorrect because the text doesn't suggest that Karenin comprehends time similarly to how Tomas and Tereza comprehend it. On the contrary, the text strongly implies a contrast between dogs' circular experience of time with the way humans experience time as a straightforward progression that can be "plotted on a straight line." Choice D is incorrect because the text provides no indication that a change in the places and objects surrounding Karenin causes him to feel as though time is accelerating. Although the text does use the language of "dash[ing] madly ahead" in relation to time, the phrase appears in the context of a comparison illustrating how dogs experience time: time for a dog moves just as the hands of a clock do, in a circle and "unwilling to dash madly ahead"—that is, always in a regular and predictable way.

Question Difficulty: Hard

Question ID a44bbd6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: a44bbd6b

Several studies of sediment (e.g., dirt, pieces of rock, etc.) in streams have shown an inverse correlation between sediment grain size and downstream distance from the primary sediment source, suggesting that stream length has a sorting effect on sediment. In a study of sediment sampled at more than a dozen sites in Alpine streams, however, geologists Camille Litty and Fritz Schlunegger found that cross-site variations in grain size were not associated with differences in downstream distance, though they did not conclude that downstream distance is irrelevant to grain size. Rather, they concluded that sediment influx in these streams may have been sufficiently spatially diffuse to prevent the typical sorting effect from being observed.

- Which finding about the streams in the study, if true, would most directly support Litty and Schlunegger’s conclusion?
- The streams regularly experience portions of their banks collapsing into the water at multiple points upstream of the A. sampling sites.
 - The streams contain several types of sediment that are not typically found in streams where the sorting effect has been B. demonstrated.
 - The streams mostly originate from the same source, but their lengths vary considerably due to the different courses they C. take.
 - The streams are fed by multiple tributaries that carry significant volumes of sediment and that enter the streams D. downstream of the sampling sites.

ID: a44bbd6b Answer

Correct Answer: A

Rationale

Choice A is the best answer. This finding would support the conclusion. If stream banks are collapsing into the water at multiple points, then sediment is getting into the water at those various points. This supports the conclusion that the inflow of sediment is very spread out.

Choice B is incorrect. This finding wouldn’t support the conclusion. The conclusion is about the influx of sediment being “spatially diffuse,” meaning spread out over a large area. The type of sediment wouldn’t have an impact on the conclusions.

Choice C is incorrect. This finding wouldn’t support the conclusion. It doesn’t say anything about the influx of sediment being “spatially diffuse” (spread out). Choice D is incorrect. This finding wouldn’t support the conclusion. Any sediment that enters downstream of the sampling sites wouldn’t end up in the samples, so it wouldn’t affect the findings or the conclusion.

Question Difficulty: Hard

Question ID 3f236877

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 3f236877

Ratified by more than 90 countries, the Nagoya Protocol is an international agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge of wild plants and animals are utilized by agricultural corporations. However, the protocol has shortcomings. For example, it allows corporations to insist that their agreements with communities to conduct research on the commercial uses of the communities’ resources and knowledge remain confidential. Therefore, some Indigenous advocates express concern that the protocol may have the unintended effect of _____

Which choice most logically completes the text?

- A. diminishing the monetary reward that corporations might derive from their agreements with Indigenous communities.
- limiting the research that corporations conduct on the resources of the Indigenous communities with which they have signed agreements.
- preventing independent observers from determining whether the agreements guarantee equitable compensation for Indigenous communities.
- discouraging Indigenous communities from learning new methods for harvesting plants and animals from their corporate partners.

ID: 3f236877 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the argument about an unintended effect of the Nagoya Protocol. The text explains that the Nagoya Protocol is an agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge are used by corporations. The text then states that the protocol allows corporations to keep their agreements with Indigenous communities confidential, about which some Indigenous advocates express concern. Choice C, when inserted into the blank, gives a good justification for the advocates’ concern: such secrecy could mean that the public is unable to determine whether participating Indigenous communities were properly compensated under these agreements.

Choice A is incorrect. The text suggests that because corporations can keep their agreements with Indigenous communities confidential, Indigenous communities, not corporations, might not be compensated fairly. Choice B is incorrect because the text doesn’t suggest that the ability of corporations to keep their agreements with Indigenous communities confidential would place limits on how much research corporations can undertake. Choice D is incorrect because the text doesn’t indicate that Indigenous communities aim to learn new harvesting methods from their corporate partners. Rather, the text suggests that corporations use the knowledge of Indigenous communities for their research.

Question Difficulty: Hard

Question ID 5b74feb9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 5b74feb9

Political scientists who favor the traditional view of voter behavior claim that voting in an election does not change a voter’s attitude toward the candidates in that election. Focusing on each US presidential election from 1976 to 1996, Ebonya Washington and Sendhil Mullainathan tested this claim by distinguishing between subjects who had just become old enough to vote (around half of whom actually voted) and otherwise similar subjects who were slightly too young to vote (and thus none of whom voted). Washington and Mullainathan compared the attitudes of the groups of subjects toward the winning candidate two years after each election.

Which finding from Washington and Mullainathan’s study, if true, would most directly weaken the claim made by people who favor the traditional view of voter behavior?

- Subjects’ attitudes toward the winning candidate two years after a given election were strongly predicted by subjects’
- A. general political orientation, regardless of whether subjects were old enough to vote at the time of the election.
 - B. winning candidate two years later than they held at the time of the election.
 - C. later than did subjects who were not old enough to vote in that election.
 - D. more likely to express negative attitudes than positive attitudes toward the winning candidate in that election.

ID: 5b74feb9 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would weaken the claim made by people who favor the traditional view of voter behavior. According to the text, people who favor that view believe that voting in an election doesn't change a voter's attitude toward the candidates in that election. If Washington and Mullainathan found that two years after an election, attitudes toward the winning candidate were significantly more polarized among subjects who had voted than among subjects who had been too young to vote, that would suggest that the act of voting did have an effect on the voters' attitudes toward the candidates, which would undermine the claim that voting doesn't change voters' attitudes.

Choice A is incorrect because a finding about links between subjects' attitudes and general political orientation, regardless of age and ability to vote, wouldn't address the presence or absence of changes in attitudes among those subjects who did actually vote. Therefore, the finding wouldn't have any bearing on the claim that voting in an election doesn't change a voter's attitude toward the candidates in that election. Choice B is incorrect because a finding that positive attitudes toward a winning candidate significantly increased in the two years after the election among subjects who had been too young to vote would involve only people who didn't vote; therefore, the finding wouldn't have any bearing on the claim that when people do vote, the act of voting doesn't change their attitudes toward the candidates. Choice D is incorrect because the finding that subjects in both groups were more likely to have negative attitudes than positive attitudes toward the winning candidate two years after an election would reflect all subjects' attitudes at one particular time whether they voted or not, rather than the presence or absence of a change in voters' attitudes after voting. Therefore, the finding would neither weaken nor strengthen the claim that voting in an election doesn't change a voter's attitude toward the candidates.

Question Difficulty: Hard

Question ID 0198240a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 0198240a

Under normal atmospheric pressure at Earth’s surface, water molecules form a tetrahedral network stabilized by hydrogen bonds between adjacent molecules. Extreme high pressure, such as can be found in deep ocean waters, destabilizes these bonds and compresses water’s structure, allowing water molecules within organisms to permeate proteins and impede crucial biological functions; yet deep-sea organisms known as piezophiles have adapted to extreme pressure. Studies have found a positive correlation between the depths that various piezophiles inhabit and concentrations of a compound called trimethylamine N-oxide (TMAO) in their muscle tissues, which has led a team of researchers to hypothesize that TMAO reduces water’s compressibility.

Which finding, if true, would most directly support the researchers’ hypothesis?

- Water molecules are found to be impervious to TMAO even when the water molecules’ tetrahedral configuration has been
- A. distorted by high pressure.
- B. Examination of TMAO’s molecular structure shows that TMAO molecules retain their shape even as pressure increases.
- A positive correlation is found between concentrations of TMAO and the rate at which water’s molecular structure
- C. compresses as pressure increases.
- Analysis of water’s molecular structure under high pressure reveals that hydrogen bonds are more stable when TMAO is
- D. present than when it is not.

ID: 0198240a Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support the researchers' hypothesis that TMAO reduces water's compressibility. The text explains that at great depths in the ocean, extreme pressure compresses the molecular structure of water by destabilizing the hydrogen bonds between adjacent molecules, thereby allowing water to penetrate proteins and harm the associated organisms. However, deep-sea organisms called piezophiles have adapted to live at these depths and previous studies show a positive correlation between the depth at which a piezophile species lives and the species' level of the compound TMAO. Because this hypothesis links TMAO levels with reduced compressibility of water's tetrahedral molecular structure, a finding that TMAO helps maintain the hydrogen bonds between water molecules under high pressure would strongly support that hypothesis.

Choice A is incorrect. Although the researchers' hypothesis suggests a relationship between TMAO and water molecules' tetrahedral molecular structure, that relationship involves TMAO helping maintain water's tetrahedral molecular structure under high pressure; as presented in the text, the hypothesis doesn't contend that water molecules are impervious to, or incapable of being penetrated by, TMAO. Choice B is incorrect because the text discusses how the molecular structure of water, not TMAO, is compressed under extreme pressure and never addresses how TMAO might be affected by such pressure. Choice C is incorrect because the researchers' hypothesis holds that water under extreme pressure is more resistant, not less, to being compressed when TMAO concentrations are higher. Moreover, the positive correlation mentioned in the text is between TMAO concentrations and the depths at which piezophiles live, not between concentrations of TMAO and the rate at which water's molecular structure compresses as pressure increases.

Question Difficulty: Hard

Question ID c2c61e7d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: c2c61e7d

Researchers hypothesized that a decline in the population of dusky sharks near the mid-Atlantic coast of North America led to a decline in the population of eastern oysters in the region. Dusky sharks do not typically consume eastern oysters but do consume cownose rays, which are the main predators of the oysters.

Which finding, if true, would most directly support the researchers’ hypothesis?

- Declines in the regional abundance of dusky sharks’ prey other than cownose rays are associated with regional declines
- A. in dusky shark abundance.
- Eastern oyster abundance tends to be greater in areas with both dusky sharks and cownose rays than in areas with only
- B. dusky sharks.
- Consumption of eastern oysters by cownose rays in the region substantially increased before the regional decline in
- C. dusky shark abundance began.
- D. Cownose rays have increased in regional abundance as dusky sharks have decreased in regional abundance.

ID: c2c61e7d Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers' hypothesis about the connection between the dusky shark population decline and the eastern oyster population decline. The text indicates that although dusky sharks don't usually eat eastern oysters, they do consume cownose rays, which are the main predators of eastern oysters. An increase in the abundance of cownose rays in the region in response to a decline in the abundance of dusky sharks would directly support the researchers' hypothesis: a higher number of cownose rays would consume more eastern oysters, driving down the oyster population.

Choice A is incorrect because a finding that there's an association between a decline in the regional abundance of some of dusky sharks' prey and the regional abundance of dusky sharks wouldn't directly support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. Although such a finding might help explain why shark abundance has declined, it would reveal nothing about whether the shark decline is related to the oyster decline. Choice B is incorrect because a finding that eastern oyster abundance tends to be greater when dusky sharks and cownose rays are present than when only dusky sharks are present wouldn't support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. The text indicates that the sharks prey on the rays, which are the main predators of the oysters; if oyster abundance is found to be greater when rays are present than when rays are absent, that would suggest that rays aren't keeping oyster abundance down, and thus that a decline in rays' predators, which would be expected to lead to an increase in the abundance of rays, wouldn't bring about a decline in oyster abundance as the researchers hypothesize. Choice C is incorrect because a finding that consumption of eastern oysters by cownose rays increased substantially before dusky sharks declined in regional abundance wouldn't support the researchers' hypothesis that the decline in dusky sharks has led to a decline in eastern oysters in the region. Such a finding would suggest that some factor other than shark abundance led to an increase in rays' consumption of oysters and thus to a decrease in oyster abundance, thereby weakening the researchers' hypothesis.

Question Difficulty: Hard

Question ID 825dc766

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 825dc766

King Lear is a circa 1606 play by William Shakespeare. In the play, the character of King Lear attempts to test his three daughters’ devotion to him. He later expresses regret for his actions, as is evident when he _____

Which choice most effectively uses a quotation from *King Lear* to illustrate the claim?

- A. says of himself, “I am a man / more sinned against than sinning.”
- B. says during a growing storm, “This tempest will not give me leave to ponder / On things would hurt me more.”
- C. says to himself while striking his head, “Beat at this gate that let thy folly in / And thy dear judgement out!”
- D. says of himself, “I will do such things— / What they are yet, I know not; but they shall be / The terrors of the earth!”

ID: 825dc766 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively uses a quotation from *King Lear* to illustrate the claim that King Lear expresses regret for his actions. In the quotation, Lear describes striking himself on the head—the same act he’s engaged in as he speaks, and one that suggests he’s deeply upset with himself. Referring to himself in the second person (with “thy”), the character exclaims “Beat at this gate that let thy folly in / And thy dear judgement out!” Lear refers metaphorically to his own mind as a gate that has allowed folly, or poor judgement, to enter and good judgement to escape. This suggests that Lear regrets his attempts to test his three daughters’ devotion to him, regarding those attempts as examples of the folly that has entered the gate of his mind.

Choice A is incorrect because this quotation doesn’t express King Lear’s sense of regret over his own actions; instead, it expresses his belief that the harm that others have done to him (or the extent to which they have “sinned against” him) outweighs whatever harm he himself has caused by “sinning.” Choice B is incorrect because this quotation doesn’t express King Lear’s sense of regret over his own actions; instead, it expresses his thoughts about an approaching storm (“this tempest”), which he believes “will not give [him] leave to ponder,” or time to consider, the harm that he will continue to experience (“things” that “would hurt [him] more”). Choice D is incorrect because this quotation expresses King Lear’s vow to commit terrible actions (or “things” that “shall be / The terrors of the earth”) in the future, not his regret over actions that he’s already taken.

Question Difficulty: Hard

Question ID 0240d11c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 0240d11c

In the twentieth century, ethnographers made a concerted effort to collect Mexican American folklore, but they did not always agree about that folklore’s origins. Scholars such as Aurelio Espinosa claimed that Mexican American folklore derived largely from the folklore of Spain, which ruled Mexico and what is now the southwestern United States from the sixteenth to early nineteenth centuries. Scholars such as Américo Paredes, by contrast, argued that while some Spanish influence is undeniable, Mexican American folklore is mainly the product of the ongoing interactions of various cultures in Mexico and the United States.

Which finding, if true, would most directly support Paredes’s argument?

- The folklore that the ethnographers collected included several songs written in the form of a *décima*, a type of poem
- A. originating in late sixteenth-century Spain.
 - B. Much of the folklore that the ethnographers collected had similar elements from region to region.
 - C. Most of the folklore that the ethnographers collected was previously unknown to scholars.
 - D. clearly recent origin.
- Most of the folklore that the ethnographers collected consisted of *corridos*—ballads about history and social life—of a

ID: 0240d11c Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support Paredes's argument that Mexican-American folklore is mostly the result of cultural interactions in Mexico and the United States rather than an adaptation of Spanish folklore. The text describes a disagreement among scholars about whether Mexican-American folklore mostly derived from the folklore of Spain or originated in Mexico and the United States as cultures there have interacted. The latter view is the argument that Paredes puts forward. If Mexican-American folklore collected in the twentieth century mostly consists of ballads about history and social life that originated recently, then that would support Paredes's argument, since it would suggest that the folklore mostly arose after Spanish rule ended in the early nineteenth century and that the folklore reflects cultural interactions in Mexico and the United States rather than traditions from Spain.

Choice A is incorrect because the inclusion of songs influenced by sixteenth-century Spanish poetry among Mexican-American folklore collected in the twentieth century would not support Paredes's view that the folklore was the result of cultural interactions in Mexico and the United States rather than an offshoot of Spanish folklore. If anything, the presence of such songs among the folklore collected in the twentieth century would weaken Paredes's argument, since it would reflect the influence of Spanish culture on the folklore. Choice B is incorrect because the mere presence of similarities in Mexican-American folklore across regions would not be sufficient to draw a conclusion about where the folklore originated, let alone to support Paredes's argument that the folklore reflects cultural interactions in Mexico and the United States. In fact, since Paredes argued that Mexican-American folklore is the product of various cultures interacting in Mexico and the United States, he would likely expect there to be regional variations in folklore as different cultures have interacted in different places. Choice C is incorrect because scholars' previous ignorance of the folklore would have no bearing on Paredes's argument that Mexican-American folklore mostly reflects cultural interactions in Mexico and the United States. The folklore's origins are independent of scholars' knowledge of the folklore.

Question Difficulty: Hard

Question ID 4d3e3c52

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 4d3e3c52

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell’s electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as “highly reproducible, concise, and practical.”

What does the text most strongly suggest about conventional ETL fabrication?

- It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.

It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.

It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.

It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

ID: 4d3e3c52 Answer

Correct Answer: A

Rationale

Choice A is the best answer. Conventional solar cell fabrication increases “the cost of manufacturing at scale,” but spray coating might get rid of that problem.

Choice B is incorrect. This is not completely supported by the text. While it’s true that conventional ETL fabrication is expensive at scale, there’s nothing in the text that mentions other perovskite solar cell architectures. Choice C is incorrect. This choice does not match the text. Only one conventional method of ETL fabrication is described, so we can’t compare the solution loss in this method to that of other conventional methods. Choice D is incorrect. This choice isn’t supported by the text. The text never suggests that the effectiveness of solar cells changes based on their method of fabrication.

Question Difficulty: Hard

Question ID d1b8a9ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: d1b8a9ad

Disco remains one of the most ridiculed popular music genres of the late twentieth century. But as scholars have argued, the genre is far less superficial than many people believe. Take the case of disco icon Donna Summer: she may have been associated with popular songs about love and heartbreak (subjects hardly unique to disco, by the way), but like many Black women singers before her, much of her music also reflects concerns about community and identity. These concerns are present in many of the genre’s greatest songs, and they generally don’t require much digging to reveal.

What does the text most strongly suggest about the disco genre?

- It has been unjustly ignored by most scholars despite the importance of the themes addressed by many of the genre’s

A. songs.
- B. It evolved over time from a superficial genre focused on romance to a genre focused on more serious concerns.
- C. It has been unfairly dismissed for the inclusion of subject matter that is also found in other musical genres.
- It gave rise to a Black women’s musical tradition that has endured even though the genre itself faded in the late twentieth

D. century.

ID: d1b8a9ad Answer

Correct Answer: C

Rationale

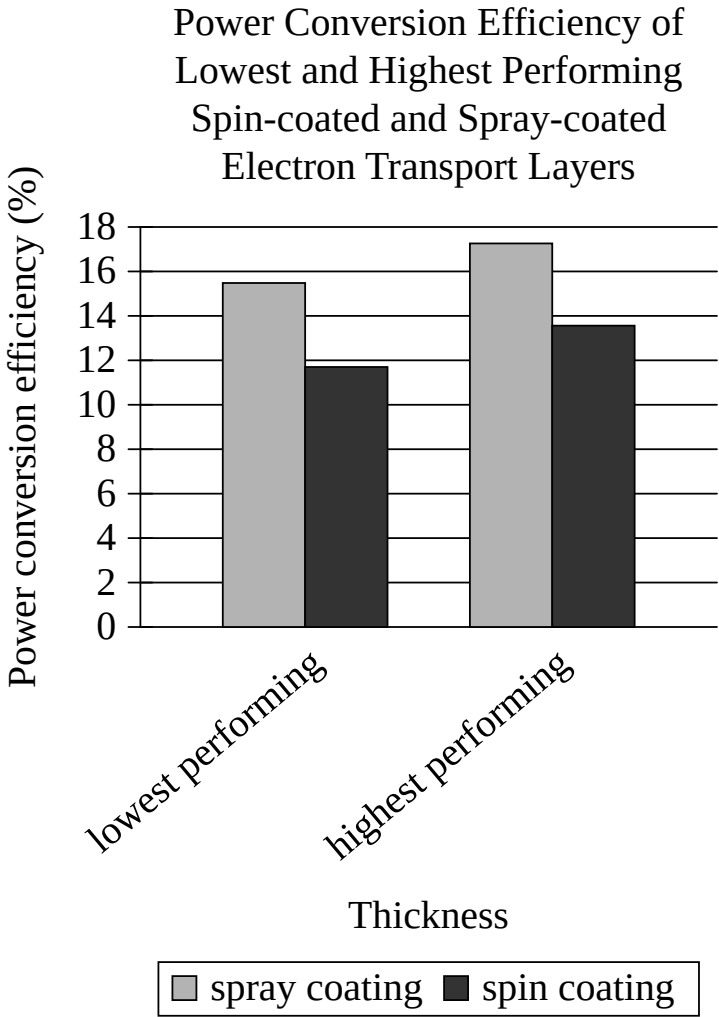
Choice C is the best answer. The text argues that disco is "far less superficial" than its popular perception might indicate, and that love and heartbreak are "subjects hardly unique to disco."

Choice A is incorrect. This choice conflicts with the text, which says that scholars argue that disco "is far less superficial than many people believe." Choice B is incorrect. This choice says the opposite of what the text suggests. The writer argues that the genre is not as superficial as commonly believed, but that it always reflected "concerns about community and identity." Choice D is incorrect. The text doesn’t support this choice. There’s nothing in the text about disco giving rise to an enduring Black women’s musical tradition.

Question Difficulty: Hard

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 02848335



Perovskite solar cells convert light into electricity more efficiently than earlier kinds of solar cells, and manufacturing advances have recently made them commercially attractive. One limitation of the cells, however, has to do with their electron transport layer (ETL), through which absorbed electrons must pass. Often the ETL is applied through a process called spin coating, but such ETLs are fairly inefficient at converting input power to output power. André Taylor and colleagues tested a novel spray coating method for applying the ETL. The team produced ETLs of various thicknesses and concluded that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells.

Which choice best describes data from the graph that support Taylor and colleagues’ conclusion?

Both the ETL applied through spin coating and the ETL applied through spray coating showed a power conversion efficiency greater than 10% at their lowest performing thickness.

The lowest performing ETL applied through spray coating had a higher power conversion efficiency than the highest performing ETL applied through spin coating.

The highest performing ETL applied through spray coating showed a power conversion efficiency of approximately 13%, while the highest performing ETL applied through spin coating showed a power conversion efficiency of approximately C. 11%.

There was a substantial difference in power conversion efficiency between the lowest and highest performing ETLs D. applied through spray coating.

ID: 02848335 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the graph that support Taylor and colleagues' conclusion that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells. The text explains that perovskite solar cells' efficiency at converting light into electricity is diminished by their electron transport layer (ETL), which is applied through spin coating, but that Taylor's team devised a new spray coating method for applying the ETL that improves its power conversion efficiency. The graph displays data on the power conversion efficiency of solar cells in tests conducted by Taylor's team, with bars for both the highest- and lowest-performing ETLs in two data categories: spray coating and spin coating. According to the graph, the lowest-performing ETL applied through spray coating had a power conversion efficiency of between 14% and 16%, while the highest-performing ETL applied through spin coating had a power conversion efficiency of less than 14%. These data confirm that ETLs applied through novel spray coating are more efficient than those applied through traditional spin coating. Thus, the data support Taylor and colleagues' conclusion about spray coating's potential value.

Choice A is incorrect. Although this claim correctly describes the data in the graph by stating that both the lowest-performing ETL applied through spin coating and the lowest-performing ETL applied through spray coating had a power conversion efficiency greater than 10%, this relationship in the data doesn't support or relate to Taylor and colleagues' conclusion that spray coating promises greater efficiency for solar cells than traditional spin coating does. Choice C is incorrect. This claim does address the greater power conversion efficiency of the highest-performing ETL applied through spray coating, compared with the highest-performing ETL applied through spin coating. However, it also incorrectly cites the value for the efficiency of the highest-performing ETL applied through spray coating as approximately 13%, instead of a value between 14% and 16%, and the value for the efficiency of the highest-performing ETL applied through spin coating as approximately 11%, instead of a value between 12% and 14%, as shown in the graph. Choice D is incorrect because Taylor and colleagues' conclusion is based on the difference in the power conversion efficiency of ETLs applied through spray coating and that of ETLs applied through spin coating, not on the difference between the highest- and lowest-performing ETLs applied through just spray coating.

Question Difficulty: Hard

Question ID 95dbdf51

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 95dbdf51

Laura Mulvey has theorized that in narrative film, shots issuing from a protagonist’s point of view compel viewers to identify with the character. Such identification is heightened by “invisible editing,” or editing so inconspicuous that it renders cuts between shots almost unnoticeable. Conversely, Mulvey proposes that conspicuous editing or an absence of point-of-view shots would induce a more critical stance toward a protagonist. Consider, for example, the attic scene in Alfred Hitchcock’s *The Birds*, a conspicuously edited sequence of tens of shots, few of which correspond to the protagonist’s point of view. According to Mulvey’s logic, this scene should affect viewers by _____

Which choice most logically completes the text?

- A. obscuring their awareness of the high degree of artifice involved in constructing the montage.
- B. lessening their identification with the protagonist, if not alienating them from the character altogether.
- C. compelling them to identify with the film’s director, whose proxy is the camera, and not with the protagonist.
- D. diverting their attention away from the film’s content and toward its stylistic attributes.

ID: 95dbdf51 Answer

Correct Answer: B

Rationale

Choice B is the best answer. We’re told that point-of-view shots and “invisible editing” make audiences identify with a character. We’re also told that obvious editing and a lack of point-of-view shots have the opposite effect. Since the sequence in *The Birds* falls into this second category, it should have the effect of reducing the audience’s connection with the protagonist.

Choice A is incorrect. The passage doesn’t mention viewers’ awareness of artifice (i.e., camera trickery) in films, so there’s no basis for this inference. Choice C is incorrect. The passage doesn’t mention the director at all, so there’s no basis for this inference. Choice D is incorrect. The passage doesn’t discuss whether a film’s “stylistic attributes” may distract viewers from the film’s story, so there’s no basis for this inference.

Question Difficulty: Hard

Question ID 55df0275

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 55df0275

Ablation Rates for Three Elements in Cosmic Dust, by Dust Source

Element	SPC	AST	HTC	OCC
iron	20%	28%	90%	98%
potassium	44%	74%	97%	100%
sodium	45%	75%	99%	100%

Earth’s atmosphere is bombarded by cosmic dust originating from several sources: short-period comets (SPCs), particles from the asteroid belt (ASTs), Halley-type comets (HTCs), and Oort cloud comets (OCCs). Some of the dust’s material vaporizes in the atmosphere in a process called ablation, and the faster the particles move, the higher the rate of ablation. Astrophysicist Juan Diego Carrillo-Sánchez led a team that calculated average ablation rates for elements in the dust (such as iron and potassium) and showed that material in slower-moving SPC or AST dust has a lower rate than the same material in faster-moving HTC or OCC dust. For example, whereas the average ablation rate for iron from AST dust is 28%, the average rate for _____

Which choice most effectively uses data from the table to complete the example?

- A. iron from SPC dust is 20%.
- B. sodium from OCC dust is 100%.
- C. iron from HTC dust is 90%.
- D. sodium from AST dust is 75%.

ID: 55df0275 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively completes the example regarding the ablation rate of iron. The table shows the ablation rates for three elements—iron, potassium, and sodium—found in cosmic dust that comes from one of four sources. The text says that the ablation rate for a given element in slower-moving SPC and AST dust was lower than the ablation rate for that same element in faster-moving HTC or OCC dust. The text then presents the first part of an example of this pattern, describing an ablation rate of 28% for iron in AST dust. The information that iron from HTC dust had an ablation rate of 90% is therefore the most effective way to complete this example—the comparison of a relatively low ablation rate for iron in slower-moving AST dust with a relatively high ablation rate for iron in faster-moving HTC dust illustrates the tendency of ablation rates for a given element to be lower in slower-moving dust than in faster-moving dust.

Choice A is incorrect because the text indicates that SPC dust, like AST dust, moves relatively slowly; a comparison of the ablation rates of iron from two slower-moving dust sources could not be an example of the difference between ablation rates in slower-moving dust and faster-moving dust, which is the pattern that the example is supposed to illustrate. Choice B is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Choice D is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Additionally, any ablation rate from AST dust would be ineffective in this example since AST dust is referenced in the first part of the example and thus additional data focused on AST dust would not illustrate a variation across dust types.

Question Difficulty: Hard

Question ID 80fd9454

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 80fd9454

Percentage of Available Eggs Eaten by Cane Toad Tadpoles

Amphibian species (common name)	Percentage of eggs eaten	Native to Australia	Produces bufadienolide
Little red tree frog	1%	yes	no
Cane toad	90%	no	yes
Short-footed frog	7%	yes	no
Striped burrowing frog	10%	yes	no
Dainty green tree frog	1%	yes	no

Native to Latin America, the cane toad was introduced to Australia in the 1930s. In recent decades, tadpoles in the Australian population have been shown to consume eggs of their own species. A 2022 study showed that when presented with cane toad eggs as well as eggs of native Australian amphibians, cane toad tadpoles disproportionately consumed eggs of their own species. This behavior results from their attraction to bufadienolide, a chemical produced by the eggs of cane toads but not by the eggs of native amphibians. However, using data from this study, a student wishes to argue that the presence of bufadienolide doesn’t entirely explain the cane toad tadpoles’ preference for certain eggs over others.

Which choice best describes data from the table that support the student’s argument?

- The tadpoles consumed a higher percentage of the striped burrowing frog eggs than they did of the eggs of the dainty green tree frog.

The tadpoles left a certain percentage of the eggs of each of the five species unharmed, thus ultimately allowing them to hatch.

The tadpoles consumed a lower percentage of the short-footed frog eggs than they did of the eggs of their own species.

The tadpoles consumed the same percentage of the dainty green tree frog eggs as they did of the little red tree frog eggs.

ID: 80fd9454 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to support the student’s argument about the role of bufadienolide in the egg preferences of cane toad tadpoles. For each of five amphibian species included in the 2022 study, the table gives the percentage of available eggs that the cane toad tadpoles ate. According to the table, the tadpoles ate 10% of striped burrowing frog eggs and 1% of dainty green tree frog eggs, which suggests a preference for striped burrowing frog eggs over dainty green tree frog eggs. The table also indicates that neither of these species’ eggs produces bufadienolide. Thus, these data suggest that something other than the presence or absence of bufadienolide is needed to adequately explain the tadpoles’ egg preferences.

Choice B is incorrect. Although the table shows that for each of the five amphibian species, the cane toad tadpoles ate less than 100% of that species’ eggs, which demonstrates that the tadpoles did indeed leave some eggs for each species unharmed, this fact alone is irrelevant to the tadpoles’ preferences for some species’ eggs over other species’ eggs. Choice C is incorrect. Although the table indicates that the cane toad tadpoles ate 90% of the cane toad eggs and 7% of the short-footed frog eggs, which suggests that they prefer cane toad eggs over short-footed frog eggs, the table also indicates that cane toad eggs produce bufadienolide, whereas short-footed frog eggs do not. Therefore, these data are not sufficient to exclude that bufadienolide alone could explain the tadpoles’ preference for some species’ eggs over other species’ eggs. Choice D is incorrect. Although the table shows that for both dainty green tree frog eggs and little red tree frog eggs, the cane toad tadpoles ate 1% of those species’ eggs, it also indicates that neither produces bufadienolide. Thus, these data alone don’t indicate bufadienolide’s role in the tadpoles’ egg preferences.

Question Difficulty: Hard

Question ID 63e7799d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 63e7799d

In vertical inheritance, parents pass genes to their offspring, but in horizontal transfer (HT), one species, often bacteria, passes genetic material to an unrelated species. In a 2022 study, herpetologist Atsushi Kurabayashi and his team investigated HT in multicellular organisms—namely, snakes and frogs in Madagascar. The team detected *BovB*—a gene transmitted vertically in snakes—in many frog species. The apparent direction of gene transfer seems counterintuitive because frogs usually don’t survive encounters with snakes and so wouldn’t be able to transmit the newly acquired gene to offspring, but the team concluded that *BovB* is indeed transmitted from snakes to frogs, either directly or indirectly, via HT.

Which finding, if true, would most directly support the team’s conclusion?

- A. *BovB* can be transmitted across frog species through HT.
- B. Parasites known to feed on species of snakes and frogs in which the *BovB* gene occurs also carry *BovB*.
- C. *BovB* cannot be reliably transmitted from a snake species to bacteria that are usually encountered by frog species.
- D. Frog species with *BovB* show few discernible advantages as compared with frog species that do not carry *BovB*.

ID: 63e7799d Answer

Correct Answer: B

Rationale

Choice B is the best answer. If there are parasites that feed on both snakes and frogs, they could carry *BovB* from snakes to frogs. HT could occur “indirectly” through those encounters, which frogs are more likely to survive than snake encounters.

Choice A is incorrect. The team’s conclusion specifically argues that *BovB* is transmitted from snakes to frogs via HT, and this choice doesn’t mention snakes. Choice C is incorrect. The team argues that *BovB* is transmitted from snakes to frogs via HT, and this answer choice discusses a scenario in which the gene is not transmitted. Choice D is incorrect. Whether or not frog species with *BovB* are advantaged isn’t relevant to the team’s conclusion about how the gene is transmitted.

Question Difficulty: Hard

Question ID 0dba14e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 0dba14e6

The increased integration of digital technologies throughout the process of book creation in the late 20th and early 21st centuries lowered the costs of book production, but those decreased costs have been most significant in the manufacturing and distribution process, which occurs after the authoring, editing, and design of the book are complete. This suggests that in the late 20th and early 21st centuries, _____

Which choice most logically completes the text?

- digital technologies made it easier than it had been previously for authors to write very long works and get them published.
- A. published.
- customers generally expected the cost of books to decline relative to the cost of other consumer goods.
- B. customers generally expected the cost of books to decline relative to the cost of other consumer goods.
- publishers increased the variety of their offerings by printing more unique titles but also printed fewer copies of each title.
- C. title.
- the costs of writing, editing, and designing a book were less affected by the technologies used than were the costs of manufacturing and distributing a book.
- D. manufacturing and distributing a book.

ID: 0dba14e6 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents the conclusion that most logically follows from the text’s discussion of how digital technologies affected the process of book creation. The text explains that in the late 20th and early 21st centuries digital technologies lowered book production costs most significantly in manufacturing and distribution. The text goes on to point out that authoring, editing, and book design are distinct steps in the process that occur before manufacturing and distribution. Because the savings connected to digital technologies have been most significant in manufacturing and distribution, it’s reasonable to infer that those technologies had less of an effect on writing, editing, and designing books.

Choice A is incorrect because the text focuses on lowered book production costs that occur after authoring has taken place; there’s no indication in the text whether digital technologies made writing and publishing lengthy books easier. Choice B is incorrect. Although it’s logical to conclude that customers would expect the cost of books to decline if production costs have declined, the text doesn’t address customer expectations for the cost of books or any other consumer goods. Choice C is incorrect because the text focuses broadly on how digital technologies have affected the cost of the publishing process; it doesn’t address the kinds of books being published or how many copies are printed.

Question Difficulty: Hard

Question ID 860803dd

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 860803dd

Biologist Valentina Gómez-Bahamón and her team have investigated two subspecies of the fork-tailed flycatcher bird that live in the same region in Colombia, but one subspecies migrates south for part of the year, and the other doesn't. The researchers found that, due to slight differences in feather shape, the feathers of migratory forked-tailed flycatcher males make a sound during flight that is higher pitched than that made by the feathers of nonmigratory males. The researchers hypothesize that fork-tailed flycatcher females are attracted to the specific sound made by the males of their own subspecies, and that over time the females' preference will drive further genetic and anatomical divergence between the subspecies.

Which finding, if true, would most directly support Gómez-Bahamón and her team's hypothesis?

- The feathers located on the wings of the migratory fork-tailed flycatchers have a narrower shape than those of the
- A. nonmigratory birds, which allows them to fly long distances.
- Over several generations, the sound made by the feathers of migratory male fork-tailed flycatchers grows progressively
- B. higher pitched relative to that made by the feathers of nonmigratory males.
- Fork-tailed flycatchers communicate different messages to each other depending on whether their feathers create high-
- C. pitched or low-pitched sounds.
- The breeding habits of the migratory and nonmigratory fork-tailed flycatchers remained generally the same over several
- D. generations.

ID: 860803dd Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support Gómez-Bahamón and her team's hypothesis about fork-tailed flycatchers. The text indicates that although two subspecies of the birds live in the same region, the tail feathers of the migrating males make a higher-pitched sound than the tail feathers of the nonmigrating males do. Gómez-Bahamón and her team hypothesize that female fork-tailed flycatchers are attracted to the particular sound made by the tail feathers of males of their own subspecies, which will bring about additional "genetic and anatomical divergence" between the two subspecies. If it were found that the pitch generated by the tail feathers of migrating males is getting higher over successive generations, it would indicate that the shape of the migrating subspecies' tail feathers is diverging further from that of the nonmigrating subspecies. And if females continue to prefer the sounds of the males of their own subspecies, the females of the migrating subspecies will become acclimated to increasingly higher pitches over subsequent generations, causing further divergence between the subspecies. Thus, if it were found that migrating males' tail feathers were producing higher pitches over time, that would support the researchers' hypothesis.

Choice A is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding is about the shape of wing feathers and how that affects long-distance flight, whereas the hypothesis is about the shape of tail feathers and how that relates to female mate preference. Choice C is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding focuses on how the tail feather sounds communicate different messages, which doesn't address differences between the subspecies or female preferences. Choice D is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. The finding that breeding habits haven't changed for either subspecies does not, by itself, suggest anything about female preferences or divergence between the two subspecies.

Question Difficulty: Hard

Question ID a44c7bd4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: a44c7bd4

Some ethicists hold that the moral goodness of an individual’s actions depends solely on whether the actions themselves are good, irrespective of the context in which they are carried out. Philosopher L. Sebastian Purcell has shown that surviving works of Aztec (Nahua) philosophy express a very different view. Purcell reveals that these works posit an ethical system in which an individual’s actions are judged in light of how well they accord with the individual’s role in society and how well they contribute to the community. To the extent that these works are representative of Aztec thought, Purcell’s analysis suggests that _____

Which choice most logically completes the text?

the Aztecs would have disputed the idea that the morality of an individual’s actions can be assessed by appealing to

A. standards of behavior that are independent of the individual’s social circumstances.

the Aztecs would not have accepted the notion that the morality of an individual’s actions can be fairly evaluated by

B. people who do not live in the same society as that individual.

actions by members of Aztec society who contributed a great deal to their community could be judged as morally good

C. even if those actions were inconsistent with behaviors the Aztecs regarded as good in all contexts.

similar actions performed by people in different social roles in Aztec society would have been regarded as morally

D. equivalent unless those actions led to different outcomes for the community.

ID: a44c7bd4 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion about Aztec (Nahua) ethics. The text indicates that, according to Purcell’s interpretation of available Aztec philosophical works, the Aztec ethical system views an individual’s actions in relation to that individual’s societal role and how the actions affect the community. The text contrasts this view with another held by some ethicists, namely that actions are morally good or bad regardless of the context in which they occur. Thus, Purcell’s analysis suggests that the Aztecs would have asserted that the morality of an individual’s actions are rooted in that person’s position in the community and the actions’ effects and therefore cannot be determined in the absence of that context.

Choice B is incorrect. Although the text indicates that morally judging an action according to Aztec ethics requires an understanding of the action’s effects and the individual’s social circumstances, it does not specify that only members of that society can acquire this information. Choice C is incorrect because it implies that the Aztecs considered some actions good or bad regardless of the surrounding context, which contradicts the text’s claim that the Aztecs believed that the morality of an individual’s action is dependent on the action’s effects on the community and the person’s specific circumstances. Choice D is incorrect. Although the text indicates that in Aztec ethics the morality of an action depends in part on how it affects the community, this is only one of the two factors—the other being the person’s societal role—that need to be considered. Therefore, it is possible that two actions with the same effect on the community could be considered morally distinct if they are performed by individuals in different social roles.

Question Difficulty: Hard

Question ID a7c52fa4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: a7c52fa4

Mean Time (in Seconds) Spent per Flower for Four Pollinator Genera

Pollinator genus	Seconds per intact pin flower	Seconds per damaged pin flower	Seconds per intact thrum flower	Seconds per damaged thrum flower
<i>Habropoda</i>	2.7	5.4	4.1	9.5
<i>Osmia</i>	5.2	8.2	7.1	8.3
<i>Pierid</i>	2.6	4.0	2.4	1.9
<i>Xylocopa</i>	2.3	2.8	2.5	2.2

To study how floral damage affects the behavior of pollinators, such as bees, a team of researchers punched holes in the floral tissue of flowers from the vine yellow jessamine (*Gelsemium sempervirens*), a plant that produces flowers that have either a long pistil and a short stamen (pin morphs) or a short pistil and a long stamen (thrum morphs). The researchers then compared the time different insect pollinators spent visiting intact pin and thrum flowers to the time such pollinators spent visiting the artificially damaged pin and thrum flowers. The researchers concluded that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator.

Which choice best describes data from the table that support the researchers’ conclusion?

- For pin flowers, damage led to longer times per flower in all pollinator genera, whereas for thrum flowers, damage led to
- A. longer times per flower only in *Habropoda* and *Osmia*.
- Compared with pollinators belonging to the genus *Osmia*, pollinators belonging to the genus *Xylocopa* spent less time on
- B. damaged pin flowers but more time on damaged thrum flowers.
- Damage led to shorter times per thrum flower in three pollinator genera (*Osmia*, *Pierid*, and *Xylocopa*), whereas it led to
- C. longer times per thrum flower in one pollinator genus (*Habropoda*).
- Pollinators belonging to the genus *Habropoda* spent 2.7 seconds on intact pin flowers and 4.1 seconds on intact thrum
- D. flowers.

ID: a7c52fa4 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it describes data from the table that support the researchers’ conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator. The table presents the average time pollinators spent per floral morph. The data in the table shows that for pin flowers, the average time spent per flower by all pollinator genera was higher for flowers that had been artificially damaged than for intact flowers. By contrast, for thrum flowers, the difference in time spent on intact and damaged flowers is seen for only some pollinator genera. This supports the researchers’ conclusion that both floral morph and the genus of the pollinator are factors involved in the effect that floral damage has on time pollinators spend per flower.

Choice B is incorrect. The table shows that pollinators belonging to the genus *Xylocopa* spent less time on both damaged pin flowers and damaged thrum flowers than pollinators belonging to the genus *Osmia* did. Choice C is incorrect. The table shows that the artificial damage to the thrum flowers led to shorter average times spent by pollinators on those flowers for only two of the four pollinator genera represented in the table, not three. Furthermore, this choice doesn’t address the effect the artificial damage had on pin flowers; thus, even if accurate, this evidence doesn’t support the researchers’ conclusion that both floral morph and pollinator genus affect time spent per flower when a flower has been damaged. Choice D is incorrect. While this statement is true based on the information in the table, it doesn’t provide evidence that supports the researchers’ conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator because this choice provides information only about how much time one pollinator genus spent on intact flowers.

Question Difficulty: Hard

Question ID a13c1c66

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: a13c1c66

Many animals, including humans, must sleep, and sleep is known to have a role in everything from healing injuries to encoding information in long-term memory. But some scientists claim that, from an evolutionary standpoint, deep sleep for hours at a time leaves an animal so vulnerable that the known benefits of sleeping seem insufficient to explain why it became so widespread in the animal kingdom. These scientists therefore imply that _____

Which choice most logically completes the text?

- A. prolonged deep sleep is likely advantageous in ways that have yet to be discovered.
- B. most traits perform functions that are hard to understand from an evolutionary standpoint.
- C. it is more important to understand how widespread prolonged deep sleep is than to understand its function.
- D. many traits that provide significant benefits for an animal also likely pose risks to that animal.

ID: a13c1c66 Answer

Correct Answer: A

Rationale

Choice A is the best answer. It most logically completes the text. The text says that some scientists can’t explain why prolonged deep sleep is so widespread, given that the known benefits of sleep don’t seem to make up for how vulnerable it leaves an animal. This suggests that prolonged deep sleep probably has unknown evolutionary benefits that make up for the vulnerability.

Choice B is incorrect. It doesn’t logically complete the text. The text only discusses the benefits and risks of one trait: sleep. So there’s no basis for an inference about “most traits.” Choice C is incorrect. It doesn’t logically complete the text. The text says that it is already known that prolonged deep sleep is widespread in the animal kingdom. Rather, what some scientists can’t explain is WHY prolonged deep sleep is so widespread, given that the known benefits of sleep don’t seem to make up for how vulnerable it leaves an animal. Choice D is incorrect. It doesn’t logically complete the text. The text only discusses the benefits and risks of one trait: sleep. So there’s no basis for an inference about “many traits.”

Question Difficulty: Hard

Question ID f942646f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: f942646f

Researchers Suchithra Rajendran and Maximilian Popfinger modeled varying levels of passenger redistribution from short-haul flights (flights of 50 to 210 minutes, from takeoff to landing) to high-speed rail trips. Planes travel faster than trains, but air travel typically requires 3 hours of lead time for security, baggage handling, and boarding that rail travel doesn’t, so short-haul routes take similar amounts of time by air and by rail. However, the model suggests that as rail passenger volumes approach current capacity limits, long lead times emerge. Therefore, for rail to remain a viable alternative to short-haul flights, _____

Which choice most logically completes the text?

- A. rail systems should offer fewer long-haul routes and airlines should offer more long-haul routes.
- B. rail systems may need to schedule additional trains for these routes.
- C. security, baggage handling, and boarding procedures used by airlines may need to be implemented for rail systems.
- D. passengers who travel by rail for these routes will need to accept that lead times will be similar to those for air travel.

ID: f942646f Answer

Correct Answer: B

Rationale

Choice B is the best answer. Air travel usually requires much more “lead time” than train travel, so short flights end up taking the same amount of time as a train trip to the same destination. But train travel starts to need more “lead time” when the trains approach their capacity limits. This suggests that train companies should add more trains for these routes if they want to encourage travelers to take a train instead of a plane.

Choice A is incorrect. This inference isn’t supported. The text never discusses “long-haul routes” for either air travel or rail travel, so there is no basis to make this inference. Choice C is incorrect. This inference isn’t supported. The text only mentions these procedures to explain why the “lead time” is so long for air travel. It never suggests that trains need to start implementing these procedures too. Choice D is incorrect. This inference isn’t supported. The goal is to make sure that trains “remain a viable alternative” to short flights, which suggests that anything that makes train travel take longer should be avoided.

Question Difficulty: Hard

Question ID 16025337

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 16025337

The following text is adapted from William Shakespeare’s 1609 poem “Sonnet 27.” The poem is addressed to a close friend as if he were physically present.

Weary with toil, I [hurry] to my bed, The dear repose for limbs with travel tired; But then begins a journey in my head
To work my mind, when body’s work’s expired: For then my thoughts—from far where I abide—
[Begin] a zealous pilgrimage to thee, And keep my drooping eyelids open wide,

What is the main idea of the text?

- A. The speaker is asleep and dreaming about traveling to see the friend.
- B. The speaker is planning an upcoming trip to the friend’s house.
- C. The speaker is too fatigued to continue a discussion with the friend.
- D. The speaker is thinking about the friend instead of immediately falling asleep.

ID: 16025337 Answer

Correct Answer: D

Rationale

Choice D is the best answer it most accurately states the main idea of the text. The speaker describes going to bed for “repose” (rest) but finding that his thoughts are focused on the friend the speaker is addressing, and the thoughts are keeping the speaker awake.

Choice A is incorrect because the speaker isn’t asleep; the thoughts about the friend are keeping the speaker awake. Choice B is incorrect because the speaker isn’t talking about taking a literal trip; rather, the speaker uses the metaphor of a journey to describe internal thoughts. Choice C is incorrect because the speaker isn’t having a discussion with the friend.

Question Difficulty: Hard

Question ID 20583752

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 20583752

“The Poet Walt Whitman” is an 1887 essay by José Martí, a Cuban author and political activist, originally written in Spanish. In the essay, Martí explores the value of literature, arguing that a society’s spiritual well-being depends on the character of its literary culture: _____

Which quotation from a translation of “The Poet Walt Whitman” most effectively illustrates the claim?

- “Poetry, which brings together or separates, which fortifies or brings anguish, which shores up or demolishes souls, which gives or robs men of faith and vigor, is more necessary to a people than industry itself, for industry provides them with a means of subsistence, while literature gives them the desire and strength for life.”
- “Every society brings to literature its own form of expression, and the history of the nations can be told with greater truth by the stages of literature than by chronicles and decades.”
- “Where will a race of men go when they have lost the habit of thinking with faith about the scope and meaning of their actions? The best among them, those who consecrate Nature with their sacred desire for the future, will lose, in a sordid and painful annihilation, all stimulus to alleviate the ugliness of humanity.”
- “Listen to the song of this hardworking and satisfied nation; listen to Walt Whitman. The exercise of himself exalts him to majesty, tolerance exalts him to justice, and order to joy.”

ID: 20583752 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively illustrates the claim that Martí argues that a society’s spiritual well-being depends on the character of its literary culture. In the quotation, Martí asserts that poetry is “more necessary to a people than industry itself” and that it has the power to provide people with “faith and vigor.” He also adds that literature gives people “the desire and strength for life.” Therefore, this quotation shows that Martí believes that literature is a societal necessity because it uplifts people and nourishes their spiritual well-being.

Choice B is incorrect. Although this quotation emphasizes the importance of literature, it focuses on how the nature of a society is reflected in that society’s literature rather than on literature’s value for people’s spiritual well-being. Choice C is incorrect. Although this quotation involves an element of spirituality, it doesn’t discuss literature. The quotation instead focuses on humanity’s actions. Choice D is incorrect because this quotation mainly focuses on the importance of Walt Whitman rather than on the value of literature in general.

Question Difficulty: Hard

Question ID 8f3a200e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 8f3a200e

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH_3 (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH_3 likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH_3 .

- Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH_3 ?
- A. Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.
 - B. Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.
 - C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.
 - D. It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.

ID: 8f3a200e Answer

Correct Answer: C

Rationale

Choice C is the best answer because it states a conclusion the researchers likely agree with, given the details in the text. The text explains that a biosignature gas is a gas that can be used as an indicator that a planet harbors some form of life and some astronomers have proposed that NH_3 could serve as a biosignature gas. The researchers evaluating this claim found that the atmosphere of rocky planets would be unlikely to reach "detectably high levels" of NH_3 without biological activity, which would support the proposal of NH_3 serving as a biosignature gas. However, the text also states that mini-Neptune planets can produce NH_3 in the absence of biological activity. Thus, the text is structured to lead to the conclusion that detectable levels of NH_3 in the atmospheres of rocky planets could constitute a biosignature, but that is not the case for detectable levels of the gas in the atmospheres of mini-Neptune planets.

Choice A is incorrect because the text indicates that biological activity likely accounts for detectable levels of NH_3 in the atmospheres of rocky planets but mini-Neptune planets can have detectable levels of NH_3 in their atmospheres in the absence of biological activity. Therefore, both rocky planets and mini-Neptune planets can have detectable levels of atmospheric NH_3 . Choice B is incorrect because the text states that for NH_3 to reach detectable levels in the atmospheres of rocky planets likely means they harbor biological activity, meaning that rocky planets with detectable NH_3 usually harbor biological activity. However, that does not entail that every rocky planet with biological activity will have detectable levels of NH_3 in their atmospheres. Choice D is incorrect because the text claims only that some astronomers have proposed using NH_3 as a biosignature gas without mentioning a minimum concentration of atmospheric NH_3 that must be met for it to function as a biosignature gas.

Question Difficulty: Hard

Question ID db2da2bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: db2da2bf

In 2019, 20 previously unknown moons were confirmed to be orbiting Saturn. Three of the moons have prograde orbits (orbiting in the direction the planet spins), and the other 17 have retrograde orbits (orbiting in the opposite direction of the planet’s spin). All but one of the 20 moons are thought to be remnants of bodies that orbited Saturn until they broke apart in collisions. Although the one exceptional moon orbits in the same direction as the planet’s spin, its orbit is highly eccentric compared to the rest, which may suggest that it has a different origin than the other 19 moons.

Based on the text, which choice best describes the moon with the eccentric orbit?

- A. It doesn’t have a retrograde orbit, but it likely has the same origin as the moons with retrograde orbits.
- B. Its orbit is so tilted with respect to the other moons’ orbits that it’s neither prograde nor retrograde.
- C. It has a prograde orbit that is likely the result of having collided with another body orbiting Saturn.
- D. It has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

ID: db2da2bf Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most accurately describes the moon with the eccentric orbit. The text indicates that three of the 20 newly discovered moons have prograde orbits, meaning that they orbit Saturn in the same direction as the planet’s spin, while the other 17 moons have retrograde orbits, meaning that they orbit Saturn in the opposite direction of the planet’s spin. The text then states that 19 of the 20 moons appear to be the remains of earlier bodies that orbited Saturn but were broken apart in collisions. The one exception is a moon that orbits Saturn in the same direction as the planet’s spin, meaning that the exceptional moon’s orbit is prograde. The text goes on to state that the exceptional moon’s orbit is so eccentric that the moon may have formed through a different process than the other 19 moons. The moon with the eccentric orbit, therefore, has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

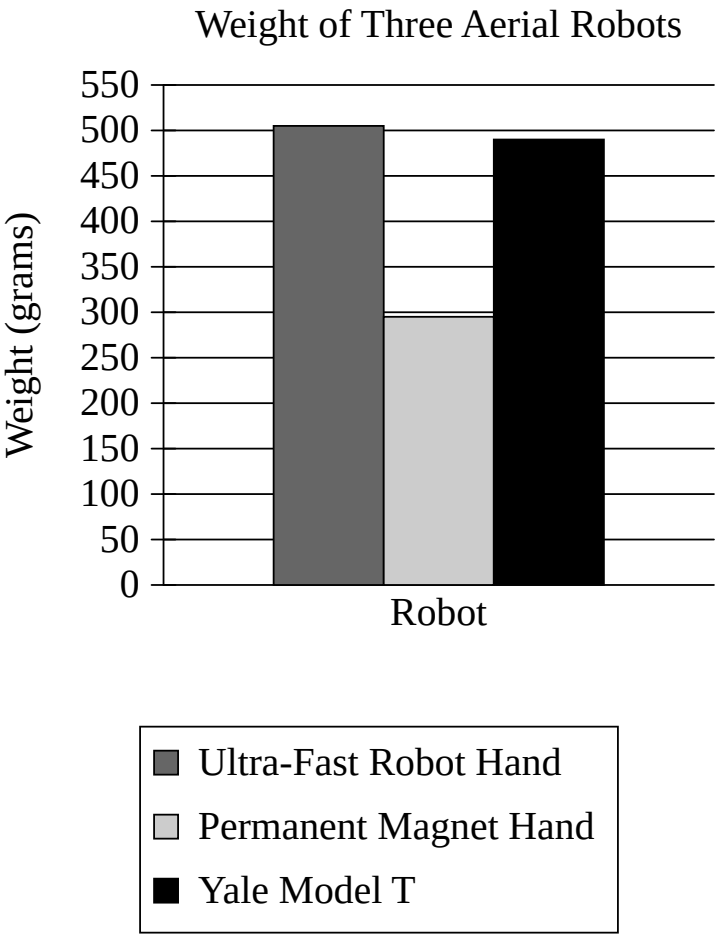
Choice A is incorrect because nothing in the text supports the idea that the moon with the eccentric orbit likely has the same origin as the moons with retrograde orbits. Although it’s true that the moon has a prograde orbit (and thus doesn’t have a retrograde orbit), the only information the text provides about the moon’s origin is that it may be different than the origin of the other 19 moons. Choice B is incorrect because the text states that the moon in question orbits Saturn in the same direction as the planet’s spin, meaning that the moon’s orbit is prograde, not that its orbit is neither prograde nor retrograde. Choice C is incorrect because the text merely notes that the moon in question has a prograde orbit without giving any indication of what likely caused that orbit.

Question Difficulty: Hard

Question ID df37c087

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: df37c087



Aerial robots vary considerably in their holding force; the Ultra-Fast Robot Hand, for example, has a holding force of 56 newtons, more than twice that of the Permanent Magnet Hand and more than four times that of the Yale Model T. Since an aerial robot must lift its own weight along with its cargo, engineer Jiawei Meng and colleagues used a ratio of each robot’s holding force to the robot’s weight to calculate payload capacity, with higher ratios corresponding to greater capacity, concluding that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Which choice best describes data in the graph that support Meng and colleagues’ conclusion?

- A. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than 450 grams.
- B. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than the Permanent Magnet Hand does.
- C. The Yale Model T has a lower holding force than the Permanent Magnet Hand despite weighing more.
- D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

ID: df37c087 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it describes data in the graph that support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. According to the text, payload capacity is calculated by using a ratio of a robot's holding force to the robot's weight, and higher ratios indicate a greater payload capacity. The Ultra-Fast Robot Hand has a holding force of 56 newtons, four times greater than that of the Yale Model T. Additionally, the graph shows that the Ultra-Fast Robot Hand has a weight of approximately 500 grams, slightly more than the Yale Model T's weight of approximately 480 grams. Therefore, the Ultra-Fast Robot Hand has a higher ratio of holding force to weight than the Yale Model T. Since higher ratios correspond to greater payload capacity, the information from the graph indicating that the Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T combined with the information in the text ultimately supports the conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Choice A is incorrect. Although, according to the graph, it's true that both the Ultra-Fast Robot Hand and the Yale Model T weigh more than 450 grams, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the two robots. Choice B is incorrect. Although, according to the graph, it's true that the Ultra-Fast Robot Hand and the Yale Model T both weigh more than the Permanent Magnet Hand does, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the Ultra-Fast Robot Hand and the Yale Model T. Furthermore, the comparison to the Permanent Magnet Hand is irrelevant to the claim about the relative ratios and payload capacities of the Ultra-Fast Robot Hand and the Yale Model T. Choice C is incorrect. Although the text states that the Yale Model T has a lower holding force than the Permanent Magnet Hand, the graph provides no information about holding force. Moreover, information about the Permanent Magnet Hand is irrelevant to the conclusion by Meng and colleagues, which only concerns the Ultra-Fast Robot Hand and the Yale Model T.

Question Difficulty: Hard

Question ID e6e6be2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: e6e6be2d

Overgrazing by purple sea urchins has caused many kelp forests along North America’s west coast to be replaced by urchin barrens—areas stripped of vegetation and covered in purple sea urchins. Urchins in barrens persist in a state of starvation that lessens their nutritional value—and thus their appeal—to many predators. Sarah Gravem and colleagues placed sunflower sea stars, a once-abundant predator species suffering massive population declines in recent years, in aquariums that each contained a nutritionally poor and a nutritionally rich purple sea urchin. The researchers found that the sea stars selected the nutritionally rich urchin in 42.7% of trials and the nutritionally poor urchin in 37.5% of trials, suggesting that

Which choice most logically completes the text?

- sunflower sea stars are willing to hunt sea urchins, but if given a choice, they will prey on other more nutritious marine

A. animals instead.
- sunflower sea stars are reluctant to feed on both nutritionally poor and nutritionally rich sea urchins and are therefore

B. unlikely to thrive in kelp forests.
- sunflower sea stars are less likely to consume sea urchins in barrens than other species of sea stars are, putting

C. sunflower sea stars at a high risk of extinction.
- sunflower sea stars do not always avoid foraging on nutritionally poor sea urchins, making sunflower sea star population

D. recovery a potentially important tool for controlling urchin barrens.

ID: e6e6be2d Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents the conclusion that most logically follows from the text's discussion of the researchers' findings about purple sea urchins and sunflower sea stars. The text explains that urchin barrens are areas that used to be kelp forests but are now covered by purple sea urchins. The text suggests that because there is no more vegetation to be consumed in those areas, the urchins exist in a state of starvation that makes them less nutritional for many predators. The text goes on to explain that in a study with a choice between two purple sea urchins, sunflower sea stars (a predator species that has been substantially declining) consumed a nutritionally rich urchin 42.7% of the time and a nutritionally poor urchin 37.5% of the time. Because the sunflower sea stars didn't always avoid consuming nutritionally poor urchins, even when nutritionally rich ones were available, it follows that helping sunflower sea star populations to grow could help control urchin barrens by increasing the number of sea stars that may consume and thus remove nutritionally poor purple sea urchins from barrens.

Choice A is incorrect because the text indicates only that when presented with purple sea urchins, the sunflower sea stars in the study consumed both nutritionally rich and nutritionally poor ones. It doesn't suggest that sunflower sea stars generally prefer other marine animals that are more nutritious; there's no mention of other marine animals. Choice B is incorrect because the text doesn't suggest that sunflower sea stars are generally reluctant to feed on sea urchins. In fact, the text indicates that the sunflower sea stars in the study did consume sea urchins, feeding on both nutritionally poor and nutritionally rich ones. Choice C is incorrect because the text addresses only the willingness of sunflower sea stars to consume the type of sea urchins found in barrens (nutritionally poor sea urchins), not how likely other species of sea stars are to consume them.

Question Difficulty: Hard

Question ID 0dccbf17

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: 0dccbf17

Henry Ossawa Tanner’s 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore _____

Which choice most logically completes the text?

- A. risk judging Tanner’s painting by standards that may not be historically appropriate.
- B. tend to conflate Tanner’s political views with those of his contemporaries.
- C. forgo analyzing Tanner’s painting in favor of analyzing his political activity.
- D. wrongly assume that Tanner’s painting was intended as a critique of his fellow artists.

ID: 0dccbf17 Answer

Correct Answer: A

Rationale

Choice A is the best answer. It most logically completes the text. The text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. This suggests that scholars who attribute those modern values to Tanner’s painting are risking judging the painting by standards that are not historically accurate.

Choice B is incorrect. It doesn’t logically complete the text. The text argues that Tanner AND his contemporaries may have been unfamiliar with modern views. It never suggests that Tanner’s views were different from his contemporaries’ views.

Choice C is incorrect. It doesn’t logically complete the text. The text never suggests that scholars should analyze Tanner’s political activity instead of his painting. Rather, the text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. Choice D is incorrect. It doesn’t logically complete the text. The text never suggests that Tanner wanted to critique his contemporaries with his painting. Rather, the text argues that Tanner AND his contemporaries may have been unfamiliar with modern beliefs and values.

Question Difficulty: Hard

Question ID 37e15265

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 37e15265

“The Young Girl” is a 1920 short story by Katherine Mansfield. In the story, the narrator takes an unnamed seventeen-year-old girl and her younger brother out for a meal. In describing the teenager, Mansfield frequently contrasts the character’s pleasant appearance with her unpleasant attitude, as when Mansfield writes of the teenager, _____

Which quotation from “The Young Girl” most effectively illustrates the claim?

- “I heard her murmur, ‘I can’t bear flowers on a table.’ They had evidently been giving her intense pain, for she positively

A. closed her eyes as I moved them away.”
- “While we waited she took out a little, gold powder-box with a mirror in the lid, shook the poor little puff as though she

B. loathed it, and dabbed her lovely nose.”
- “I saw, after that, she couldn’t stand this place a moment longer, and, indeed, she jumped up and turned away while I went

C. through the vulgar act of paying for the tea.”
- “She didn’t even take her gloves off. She lowered her eyes and drummed on the table. When a faint violin sounded she

D. winced and bit her lip again. Silence.”

ID: 37e15265 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most effectively illustrates the claim in the text that in describing the teenaged girl, Mansfield contrasts the character’s pleasant appearance with her unpleasant attitude. In the quotation, Mansfield describes the teenager as having a “lovely nose” (a compliment about her appearance) but also as treating her makeup puff “as though she loathed it” (a judgment suggesting her unpleasant attitude).

Choice A is incorrect because the teenager's reaction to the flowers doesn't make it clear that she has an unpleasant attitude, and nothing in the quotation indicates that any part of her appearance is pleasant. Choice C is incorrect because the quotation suggests that the teenager has an unpleasant attitude (being upset with the location and leaving the table before the narrator has paid for the meal) but doesn't give any indication that she has a pleasant appearance. Choice D is incorrect because the quotation suggests that the teenager may have an unpleasant attitude (lowering her eyes, wincing, and sitting in silence) but doesn't give any indication that any part of her appearance is pleasant.

Question Difficulty: Hard

Question ID f9c4bdab

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: f9c4bdab

A student is writing a paper about *One Night in Miami...*, a 2020 film directed by Regina King and written by Kemp Powers. Powers adapted the film’s screenplay from his 2013 play, which he wrote after learning about a 1964 meeting that took place in Miami, Florida, between four prominent figures of the Civil Rights movement: Malcolm X, Muhammad Ali, Jim Brown, and Sam Cooke. The student claims that although Powers was inspired by this meeting, the film is best understood not as a precise retelling of historical events but rather as a largely imagined but informed representation of them.

Which quotation from an article about *One Night in Miami...* would be the most effective evidence for the student to include in support of this claim?

- “When Powers learned of the meeting, he initially planned to write a much longer work about its four famous participants rather than focusing on the meeting itself.”
- “*One Night in Miami...* received numerous awards and nominations, including an Academy Award nomination for Powers for Best Adapted Screenplay.”
- “Powers has described *One Night in Miami...* as the story of four friends encouraging and supporting one another while engaged in a crucial political debate about how best to achieve equality for Black people in the United States.”
- “Powers could find only the most superficial historical details about the meeting, so he read extensively about the four individuals and their thinking at the time in an effort to portray what might have happened between them.”

ID: f9c4bdab Answer

Correct Answer: D

Rationale

Choice D is the best answer because it provides a quotation that effectively supports the student’s claim about the film *One Night in Miami*.... The quotation states that in researching the play on which the film was based, Kemp Powers only found superficial details about what actually happened during the 1964 meeting in Miami between four leading Civil Rights leaders, meaning that there is very little information about the meeting in the historical record. In the absence of greater details, it wouldn’t have been possible for the film to be a precise retelling of the historical events it depicts. The quotation explains that to compensate for this lack of information about the meeting, Powers did extensive research into the four figures and how they thought at the time in order to speculate in an informed way about what they might have said or what might have occurred between them. Therefore, the quotation effectively supports the claim that the film is best understood not as a precise retelling of a historical event but as a deeply informed imaginative rendering of that event.

Choice A is incorrect. Although the quotation discusses how on learning about the 1964 meeting in Miami, Powers was inspired to write a play and, later, to adapt it into a screenplay, it doesn’t discuss Powers’s approach to representing what had occurred in the meeting. Instead, it states that Powers didn’t initially plan to write a story only “focusing on the meeting itself” but rather had considered writing a “much longer” and more expansive work about the meeting’s four participants. Choice B is incorrect because the quotation doesn’t discuss Powers’s approach to representing historical events in his play and in the film; instead, the quotation focuses on the film’s positive critical reception by mentioning that it received numerous awards and nominations. Choice C is incorrect. Although the quotation references historical events that are discussed directly in the play and film by explaining how the four historical figures featured in the story engage in political debates about contemporary issues, it doesn’t specify to what extent Powers’s representation of what occurred during the 1964 meeting in Miami is a factual retelling of what happened and how much is an imaginative rendering of what might have happened. Rather, the quotation focuses on Powers’s description of the film’s basic premise and how the characters engage with the historical context of its setting.

Question Difficulty: Hard

Question ID 5632ffb4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 5632ffb4

In a study of the cognitive abilities of white-faced capuchin monkeys (*Cebus imitator*), researchers neglected to control for the physical difficulty of the tasks they used to evaluate the monkeys. The cognitive abilities of monkeys given problems requiring little dexterity, such as sliding a panel to retrieve food, were judged by the same criteria as were those of monkeys given physically demanding problems, such as unscrewing a bottle and inserting a straw. The results of the study, therefore, _____

Which choice most logically completes the text?

- could suggest that there are differences in cognitive ability among the monkeys even though such differences may not actually exist.
- are useful for identifying tasks that the monkeys lack the cognitive capacity to perform but not for identifying tasks that the monkeys can perform.
- should not be taken as indicative of the cognitive abilities of any monkey species other than *C. imitator*.
- reveal more about the monkeys’ cognitive abilities when solving artificial problems than when solving problems encountered in the wild.

ID: 5632ffb4 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the study of capuchin monkeys’ cognitive abilities. The text explains that the study failed to distinguish between outcomes for the tasks performed by the capuchin monkeys, such that simpler tasks requiring less dexterity, or skill, were judged by the same criteria as tasks that demanded more dexterity. Because the study didn’t account for this discrepancy, the researchers might have assumed that observed differences in performance were due to the abilities of the monkeys rather than the complexity of the tasks. In other words, the results may suggest cognitive differences among the monkeys even though such differences may not really exist.

Choice B is incorrect because the text focuses on the fact that the tasks assigned to the capuchin monkeys in the study varied in difficulty and that the variety wasn’t taken into consideration. The text doesn’t suggest that the capuchin monkeys couldn’t perform certain tasks, just that some tasks were more difficult to do. Choice C is incorrect because the text doesn’t suggest that the study’s results are indicative of the abilities of capuchin monkeys but not of other monkey species; in fact, the text suggests that the results may not even be an accurate reflection of capuchin monkeys’ abilities. Choice D is incorrect because the text doesn’t indicate that the researchers compared results for artificial tasks with those for tasks encountered in the wild, although the tasks described in the text—sliding a panel and putting a straw in a bottle—are presumably artificial.

Question Difficulty: Hard

Question ID b32c4b3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: b32c4b3a

The Intertropical Convergence Zone (ITCZ), a band of clouds that encircles Earth in the tropics and is a major rainfall source, shifts position in response to temperature variations across Earth’s hemispheres. Data from Huagapo Cave in Peru suggest the ITCZ shifted south during the Little Ice Age (circa 1300–1850), but a shift as far into South America as Huagapo should have led to dry conditions in Central America, which is inconsistent with climate models. To resolve the issue, geologist Yemane Asmerom and colleagues collected data from Yok Balum Cave in Central America and compared them with the Huagapo data. They concluded that during the Little Ice Age, the ITCZ may have expanded northward and southward rather than simply shifted.

Which finding from Asmerom and colleagues’ study, if true, would most directly support their conclusion?

- Neither the Yok Balum data nor the Huagapo data show significant local variations in temperature during the Little Ice Age.
- Both the Yok Balum data and the Huagapo data show increased temperatures and prolonged dry conditions during the Little Ice Age.
- The Yok Balum data show prolonged dry conditions during the same portions of the Little Ice Age in which the Huagapo data show heightened levels of rainfall.
- The Yok Balum data and the Huagapo data show strongly correlated patterns of high rainfall during the Little Ice Age.

ID: b32c4b3a Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support Asmerom and colleagues' conclusion that the ITCZ may have expanded northward and southward rather than shifting south during the Little Ice Age. The text indicates that the ITCZ, a band of clouds in the tropics that is a significant rainfall source, can change position. Data from Peru's Huagapo Cave suggest that the ITCZ shifted south during the Little Ice Age. But according to the text, if the ITCZ moved into South America in that way, then Central America should have been drier than climate models suggest it was. In other words, rainfall should have been reduced in Central America because the ITCZ, a significant rainfall source, had shifted into South America, but climate models do not show such a reduction in Central America. The text goes on to say that Asmerom and colleagues tried to resolve this apparent conflict by collecting data from Yok Balum cave in Central America and comparing them with data from Huagapo, which led the researchers to conclude that the ITCZ may have expanded both northward and southward rather than simply shifting south. If it is true that Yok Balum in Central America and Huagapo in South America show strongly correlated patterns of high rainfall during the Little Ice Age, such a finding would support Asmerom and colleagues' conclusion by suggesting that the two areas were affected by the same rainfall source, and thus that the ITCZ may have expanded rather than shifted.

Choice A is incorrect because there is no information in the text about how, if at all, the ITCZ affects temperature in areas where it is located. Rather, the text states that temperature variations across Earth's hemispheres can shift the position of the ITCZ. Finding that neither Yok Balum nor Huagapo data show evidence of significant local variations in temperature during the Little Ice Age would have no clear bearing on Asmerom and colleagues' claim. Choice B is incorrect because finding that both Yok Balum and Huagapo experienced prolonged dry conditions during the Little Ice Age would not support Asmerom and colleagues' conclusion that the ITCZ, a major source of rainfall, may have expanded northward and southward rather than simply shifting south. Dry conditions in both locations would suggest that the ITCZ did not cover either location. Additionally, finding that temperatures were elevated in both locations would have no clear bearing on Asmerom and colleagues' conclusion, since there is no information in the text that indicates how, if at all, the ITCZ affects temperature. Choice C is incorrect because finding that Yok Balum experienced prolonged dry conditions at the same time that Huagapo experienced high rainfall would weaken Asmerom and colleagues' conclusion, not strengthen it. Such a finding would suggest that the ITCZ shifted south and left Central America dry rather than expanding both northward and southward.

Question Difficulty: Hard

Question ID d102706f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: d102706f

Estimates of Tyrannosaurid Bite Force

Study	Year	Estimation method	Approximate bite force (newtons)
Cost et al.	2019	muscular and skeletal modeling	35,000–63,000
Gignac and Erickson	2017	tooth-bone interaction analysis	8,000–34,000
Meers	2002	body-mass scaling	183,000–235,000
Bates and Falkingham	2012	muscular and skeletal modeling	35,000–57,000

The largest tyrannosaurids—the family of carnivorous dinosaurs that includes *Tarbosaurus*, *Albertosaurus*, and, most famously, *Tyrannosaurus rex*—are thought to have had the strongest bites of any land animals in Earth’s history. Determining the bite force of extinct animals can be difficult, however, and paleontologists Paul Barrett and Emily Rayfield have suggested that an estimate of dinosaur bite force may be significantly influenced by the methodology used in generating that estimate.

Which choice best describes data from the table that support Barrett and Rayfield’s suggestion?

The study by Meers used body-mass scaling and produced the lowest estimated maximum bite force, while the study by A. Cost et al. used muscular and skeletal modeling and produced the highest estimated maximum.

In their study, Gignac and Erickson used tooth-bone interaction analysis to produce an estimated bite force range with a B. minimum of 8,000 newtons and a maximum of 34,000 newtons.

The bite force estimates produced by Bates and Falkingham and by Cost et al. were similar to each other, while the C. estimates produced by Meers and by Gignac and Erickson each differed substantially from any other estimate.

The estimated maximum bite force produced by Cost et al. exceeded the estimated maximum produced by Bates and D. Falkingham, even though both groups of researchers used the same method to generate their estimates.

ID: d102706f Answer

Correct Answer: C

Rationale

Choice C is the best answer because it accurately describes data from the table that support Barrett and Rayfield’s suggestion about bite force estimates. According to the text, Barrett and Rayfield believe that estimates of dinosaur bite force may be strongly influenced by the methods used to produce them—that is, that different methods may produce significantly different results. The table shows that the studies by Bates and Falkingham and by Cost et al. used the same estimation method (muscular and skeletal modeling) and produced similar bite force estimates (approximately 35,000–57,000 newtons and 35,000–63,000 newtons, respectively). The study by Meers, however, used body-mass scaling and produced a much higher bite force estimate (183,000–235,000 newtons), while the study by Gignac and Erickson used tooth-bone interaction analysis and produced a much lower bite force estimate (8,000–34,000 newtons). The fact that one method produced similar estimates in two different studies and that two different methods used in other studies produced substantially different estimates supports the idea that dinosaur bite force estimates are significantly influenced by the methodology used to produce them.

Choice A is incorrect because it inaccurately describes data from the table. The table does show that the studies by Meers and by Cost et al. used different estimation methods and produced very different ranges of estimated dinosaur bite force, which would support Barrett and Rayfield’s suggestion that different methodologies may produce significantly different estimates. However, the table doesn’t show that the study by Meers produced the lowest estimated maximum bite force while the study by Cost et al. produced the highest. In fact, the study by Meers estimated a maximum bite force of approximately 235,000 newtons, which is the highest of all the estimated maximums. Choice B is incorrect. Although the data from Gignac and Ericson’s study are accurately described, a single set of findings from one study using only one methodology can’t show that different methodologies may produce significantly different dinosaur bite force estimates, as Barrett and Rayfield suggest. Choice D is incorrect. Although the table shows that the maximum bite force estimated by Cost et al. was higher than that estimated by Bates and Falkingham, the difference is relatively small; in fact, both teams estimated a minimum bite force of approximately 35,000 newtons and a maximum bite force close to approximately 60,000 newtons. Because these findings demonstrate that a single methodology (muscular and skeletal modeling) produced similar overall results in two studies, the findings don’t support Barrett and Rayfield’s suggestion that different methodologies may produce significantly different dinosaur bite force estimates.

Question Difficulty: Hard

Question ID 1b9b29f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 1b9b29f1

A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that _____

Which choice most logically completes the text?

- A. temperature-sensitive accelerated flowering is unique to *A. thaliana*.
- B. *A. thaliana* increases ELF3 production as temperatures rise.
- C. ELF3 enables *A. thaliana* to respond to increased temperatures.
- D. temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

ID: 1b9b29f1 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of accelerated flowering in *A. thaliana* plants. The text indicates that *A. thaliana* plants show accelerated flowering at high temperatures. To investigate the mechanism for this accelerated flowering, biologists replaced the ELF3 protein in one group of *A. thaliana* plants with a similar protein found in another plant species that doesn’t show accelerated flowering. The team then compared these modified plants to *A. thaliana* plants that retained their original ELF3 protein. The text states that the two samples of plants showed no difference in flowering at 22° Celsius, but at 27° Celsius the unaltered plants with ELF3 showed accelerated flowering while the plants without ELF3 didn’t. If accelerated flowering at the higher temperature occurred in the *A. thaliana* plants with ELF3 but not in the plants without the protein, then ELF3 likely enables *A. thaliana* to respond to increased temperatures.

Choice A is incorrect because the text doesn’t mention whether any plants other than *A. thaliana* and stiff brome show temperature-sensitive flowering, so there is no support for the idea that this type of flowering is unique to *A. thaliana*. Choice B is incorrect because the text discusses the effects of ELF3 and not the production of it. There’s nothing in the text to suggest that the amount of ELF3 in *A. thaliana* varies with temperature. Choice D is incorrect. While the text states that there was no difference in the flowering of modified and unmodified *A. thaliana* plants at 22° Celsius, there’s no suggestion that *A. thaliana* only begins to flower at 22° Celsius; the text doesn’t mention a specific temperature threshold required for *A. thaliana* flowering.

Question Difficulty: Hard

Question ID 61228830

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 61228830

A heliograph is a semaphore device used for sending optical communications—usually in the form of Morse code—by reflecting flashes of sunlight off a mirror. Heliographs were used for rapid communication across expansive distances for military, surveying, and forestry purposes during the late nineteenth and early twentieth centuries, but they were largely effective only during the daytime, and the range of the device depended on factors such as the opacity of the air and line of sight. Therefore, heliographs were eventually replaced by technology that _____

Which choice most logically completes the text?

- A. worked on similar principles but was easier to produce and maintain.
- B. was not so constrained by environmental circumstances.
- C. could be used for more than military, surveying, or forestry purposes.
- D. enabled communication that didn’t require knowledge of Morse code.

ID: 61228830 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The passage explains that heliographs “were largely effective only during the daytime” and that “the range of the device depended on factors such as the opacity of the air and line of sight.” These constraints would have greatly limited the use of the heliograph, so we can infer that this would have been a reason to replace it with new technology.

Choice A is incorrect. The passage doesn’t discuss the production and maintenance of heliographs, so there’s no basis for this inference. Choice C is incorrect. The passage doesn’t mention any “other purposes” that a heliograph wouldn’t work for, so there’s no basis for this inference. Choice D is incorrect. The passage doesn’t mention knowledge of Morse code as a particular problem with the use of heliographs, so there’s no basis for this inference.

Question Difficulty: Hard

Question ID 2df56712

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 2df56712

Electronic music pioneer Wendy Carlos is credited with the music for three feature films: *A Clockwork Orange* (1971), *The Shining* (1980), and *Tron* (1982). However, her musical score for *A Clockwork Orange* is mostly made up of her arrangements of Ludwig van Beethoven’s work. Also, almost all the music that she and Rachel Elkind composed for *The Shining* was unused by director Stanley Kubrick. It did not appear in the film. Of the three films, *Tron* is the one in which audiences can hear the most of Carlos’s original compositions.

Which choice best states the main idea of the text?

- A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.
- B. The director of *The Shining* used most of the music that Carlos composed for it.
- C. Beethoven is widely considered to be a more important composer than Carlos.
- D. Carlos is a notable innovator among film composers in the 1970s and 1980s.

ID: 2df56712 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it best states the main idea of the text. The text indicates that Wendy Carlos scored three feature films: *A Clockwork Orange*, *The Shining*, and *Tron*. It also indicates that Carlos’s work on *A Clockwork Orange* consisted primarily of electronic arrangements of Beethoven compositions and that very little of what she and Rachel Elkind composed for *The Shining* was used in the film. But the soundtrack for *Tron* consists largely of music composed by Carlos, so it contains more of her original music than do the soundtracks for the other two films. Thus, the main idea of the text is that of the three films for which Carlos is credited, *Tron* features the most original music from her.

Choice B is incorrect because it directly contradicts the text’s claim that "very little of what she and Rachel Elkind composed for *The Shining* was used in the film." Choice C is incorrect because the text doesn’t address the perceived quality of different composers, instead, it discusses how much of Carlos’s original work was used in each of the three films she scored. Choice D is incorrect because it doesn’t relate to any of the major themes of the text: the work Wendy Carlos did for three feature films, how much of that work was composed by Carlos, and how much of her original work was ultimately used in the film.

Question Difficulty: Hard

Question ID af9e3240

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: af9e3240

Electra is a circa 420–410 BCE play by Sophocles, translated in 1870 by R.C. Jebb. Electra, who is in mourning for her dead father and her long-absent brother, is aware of the intensity of her grief but believes it to be justified: _____

Which quotation from *Electra* most effectively illustrates the claim?

- “O thou pure sunlight, and thou air, earth’s canopy, how often have ye heard the strains of my lament, the wild blows dealt
A. against this bleeding breast, when dark night fails!”
- “Send to me my brother; for I have no more the strength to bear up alone against the load of grief that weighs me down.”
B.
- “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied complaints,
C. while life is in me.”
- “But never will I cease from dirge and sore lament, while I look on the trembling rays of the bright stars, or on this light of
D. day.”

ID: af9e3240 Answer

Correct Answer: C

Rationale

Choice C is the best answer. Electra states that she “knows her own passion,” which shows that she’s aware of the intensity of her grief. But she also claims that the “causes are so dire”—meaning the reasons for her grief are so awful—that she can’t let it go, which shows that she believes her grief is justified.

Choice A is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice B is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice D is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of the intensity of her grief, but it doesn’t suggest that she has a legitimate reason for feeling that way.

Question Difficulty: Hard

Question ID 6536183b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 6536183b

In the mountains of Brazil, *Barbacenia tomentosa* and *Barbacenia macrantha*—two plants in the Velloziaceae family—establish themselves on soilless, nutrient-poor patches of quartzite rock. Plant ecologists Anna Abrahão and Patricia de Britto Costa used microscopic analysis to determine that the roots of *B. tomentosa* and *B. macrantha*, which grow directly into the quartzite, have clusters of fine hairs near the root tip; further analysis indicated that these hairs secrete both malic and citric acids. The researchers hypothesize that the plants depend on dissolving underlying rock with these acids, as the process not only creates channels for continued growth but also releases phosphates that provide the vital nutrient phosphorus.

Which finding, if true, would most directly support the researchers’ hypothesis?

Other species in the Velloziaceae family are found in terrains with more soil but have root structures similar to those of *B. tomentosa* and *B. macrantha*.

Though *B. tomentosa* and *B. macrantha* both secrete citric and malic acids, each species produces the acids in different proportions.

The roots of *B. tomentosa* and *B. macrantha* carve new entry points into rocks even when cracks in the surface are readily available.

B. tomentosa and *B. macrantha* thrive even when transferred to the surfaces of rocks that do not contain phosphates.

ID: 6536183b Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would support the researchers' hypothesis about the plants' dependence on dissolving rock. The text indicates that the roots of the two plant species grow directly into quartzite rock, where hairs on the roots secrete acids that dissolve the rock. The researchers hypothesize that the plants depend on this process because dissolving rock opens spaces for the roots to grow and releases phosphates that provide the plants with phosphorous, a vital nutrient. If the plants carry out this process of dissolving rock even when the rock already has spaces into which the roots could grow, that would support the researchers' hypothesis because it suggests that the plants are getting some advantage—such as access to phosphorous—from the action of dissolving rock. If the plants don't benefit from dissolving rock, they would be expected to grow in the cracks that already exist, as doing so would mean that the plants don't have to spend energy creating and secreting acids; if, however, the plants create new entry points by dissolving rock even when cracks already exist, that would support the hypothesis that they depend on dissolving rock for some benefit.

Choice A is incorrect because the existence of soil-inhabiting members of the Velloziaceae family with similar root structures to those of the two species discussed in the text wouldn't support the researchers' hypothesis that the species discussed in the text depend on dissolving rock. If other such members exist, that might suggest that the root structures can serve more functions than secreting acids to dissolve rock (since dissolving rock may not be necessary for plants living in soil), but that wouldn't suggest anything about whether the species discussed in the text benefit from dissolving rock. Choice B is incorrect because differences in the proportions of citric and malic acid secreted by the two species would be irrelevant to the hypothesis that the plants depend on dissolving rock. There's no information in the text to suggest that the proportion of each acid has any bearing on the process of dissolving rock or on any benefits the plants might receive from that process. Choice D is incorrect because if the two species thrive on rocks without phosphates, that would weaken the researchers' hypothesis that the plants depend on dissolving rock partly because dissolving rock gives them access to phosphates. If the plants can survive on rocks without getting a vital nutrient by dissolving those rocks, then either the nutrient isn't actually vital for those plants or they can get the nutrient in some way other than by dissolving rocks.

Question Difficulty: Hard

Question ID 96d1c1fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 96d1c1fe

Tatiana R. Feuerborn and colleagues analyzed the genomes of more than a hundred domesticated dogs from sites in Siberia dating from 11,000 years ago to the present. They found that the dogs constituted a genetically isolated population of Arctic breeds until approximately 2,000 years ago, at which point there was substantial interbreeding with Near Eastern dog breeds. Furthermore, beginning around 2,000 years ago, some sites contain artifacts consistent with a Near East origin, like glass beads, but the people show no evidence of having traveled extensively outside Siberia. From this, Feuerborn and colleagues concluded that around 2,000 years ago _____

Which choice most logically completes the text?

- A. dogs and artifacts like glass beads began to be transported from the Near East to Siberia.
- B. people from Siberia began to reach the Near East, where they acquired dogs and artifacts such as glass beads.
- C. glass beads and other artifacts from the Near East began to be exchanged for dogs from Siberia.
- D. dogs from the Near East began to be exchanged for glass beads and other artifacts from Siberia.

ID: 96d1c1fe Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically follows from the text's discussion of Feuerborn and colleagues' findings about sites in Siberia. The text states that genomic analysis of many domesticated dogs from Siberian sites dating from the last 11,000 years revealed that the dogs represented a genetically isolated population of breeds (that is, they didn't reproduce with dogs of any outside breeds) until about 2,000 years ago, when there is evidence that the dogs began interbreeding with breeds from the Near East. The text also indicates that around the same time, the sites began to contain glass beads and other artifacts seeming to be from the Near East, even though there is no evidence that the people of those sites traveled widely outside Siberia. Because there is evidence of Near East dog breeds and Near East artifacts at Siberian sites beginning about 2,000 years ago but no evidence of the people's travel to the Near East around that time, it's reasonable to infer that Feuerborn and colleagues concluded that dogs and artifacts like glass beads were transported to Siberia from the Near East, starting around 2,000 years ago.

Choice B is incorrect because the text explicitly states that there is no evidence that the people of the Siberian sites 2,000 years ago traveled extensively beyond Siberia, which they would've had to do to reach the Near East; therefore, it wouldn't make sense to conclude that the presence of Near East dog breeds and artifacts like glass beads suggests that people from Siberia began arriving in the Near East around 2,000 years ago. Choice C is incorrect because the text addresses the presence of Near East dog breeds (and evidence of interbreeding with Siberian dogs) and artifacts like glass beads at Siberian sites starting about 2,000 years ago, suggesting only that the dogs and artifacts started to arrive in Siberia around that time. The text doesn't suggest the purpose of the arrival of the Near East dogs and artifacts or give any indication of Siberian dog breeds being transported elsewhere through trade. Choice D is incorrect because the text addresses the presence of Near East dog breeds and artifacts like glass beads at Siberian sites starting about 2,000 years ago, suggesting only that the dogs and artifacts started to arrive in Siberia around that time; it doesn't suggest the purpose of this arrival and makes no mention of Siberian glass beads or other artifacts of Siberian origin.

Question Difficulty: Hard

Question ID 2644644a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 2644644a

In countries with right-hand traffic, drivers who want to make a left turn at a traffic intersection with stoplights have to wait for either a gap in oncoming traffic or a designated left-turn signal to turn green. At busy intersections, this often causes a backup of vehicles waiting to turn left or being prevented from proceeding by left-turning vehicles in front of them. Transportation researcher Vikash V. Gayah claims that in urban areas eliminating the option to turn left at busy intersections—both with and without dedicated left-turn signals—would improve traffic flow and, as a result, reduce overall travel times even if such a restriction would require drivers to sometimes travel a slightly longer distance.

Which finding, if true, would most directly support the researcher’s claim?

In a town that installed left-turn signals at all busy intersections, seven out of ten survey respondents agreed with the
A. statement “the streets in my community are easier to navigate by motor vehicle than before.”

A traffic study of intersections in a large city shows that on average drivers wait longer to make a left turn at intersections
B. without left-turn signals than at intersections with such signals.

After a city eliminated left turns at busy intersections, a package-delivery company reports that its drivers have been able
C. to reach more addresses in the city daily, on average, and therefore deliver more packages there annually.

Statistics reveal that school buses in a city that eliminated left turns at most intersections took on average two minutes
D. longer to complete their routes after the restriction took effect than they did before.

ID: 2644644a Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most directly support Vikash V. Gayah's claim that eliminating the option to turn left at busy intersections would improve traffic flow and reduce overall travel times. The text begins by describing a problem encountered by drivers in countries with right-hand traffic—namely that drivers wanting to make a left turn must wait for either gaps in oncoming traffic or for designated left-turn signals before proceeding. The resulting backup of vehicles causes increased traffic congestion at busy intersections that slows overall travel times. According to Gayah, eliminating left turns from busy intersections in urban areas would ease the congestion caused by vehicles waiting to turn left. If vehicles spend less time waiting at intersections for left turns, faster overall travel times would result even if some drivers would have to drive slightly longer distances to make the desired left turn. Drivers for package-delivery companies, who presumably spend most of the day driving to destinations across the city in which they are based, would likely provide a good indication of overall traffic patterns across the city. A finding that after a city had eliminated left turns at busy intersections, package-delivery companies were able to complete more daily deliveries on average—which implies faster travel times between package destinations—would therefore support the claim that overall travel times would decrease if left turns were eliminated at busy intersections.

Choice A is incorrect because a finding that a majority of survey respondents agreed with the statement that implementing left-turn signals at all busy intersections made navigating streets in their communities easier wouldn't support the claim that eliminating the option to turn left altogether at some busy intersections—both with and without dedicated left-turn signals—would improve the flow of traffic and overall traffic times. In fact, the text emphasizes that designated left-turn signals contribute to increased congestion because they require drivers to wait to turn left, which results in a backup of vehicles. Installing such signals at all busy intersections would thus potentially compound the problem of congestion, not improve it. Moreover, although making streets easier for drivers to navigate might indicate that left-turn signals are beneficial to drivers, it doesn't indicate that overall travel times would necessarily be reduced. Choice B is incorrect because a study concluding that drivers wait longer to make left turns at intersections without dedicated left-turn signals than at intersections with such signals wouldn't support the claim that eliminating the option to turn left altogether would reduce overall travel times. In fact, the finding would merely support the idea that installing left-turn signals would likely reduce the time drivers spend waiting at busy intersections that didn't previously have left-turn signals, not that overall travel times would be reduced. Choice D is incorrect because the finding that after a restriction eliminating left turns at most intersections took effect, school buses took longer to complete their routes than they did before the restriction was implemented would contradict rather than support the researcher's claim that eliminating left turns would reduce overall travel times.

Question Difficulty: Hard

Question ID b5e9f3c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: b5e9f3c2

Ancestral Puebloans, the civilization from which present-day Pueblo tribes descended, emerged as early as 1500 B.C.E. in an area of what is now the southwestern United States and dispersed suddenly in the late 1200s C.E., abandoning established villages with systems for farming crops and turkeys. Recent analysis comparing turkey remains at Mesa Verde, one such village in southern Colorado, to samples from modern turkey populations in the Rio Grande Valley of north central New Mexico determined that the latter birds descended in part from turkeys cultivated at Mesa Verde, with shared genetic markers appearing only after 1280. Thus, researchers concluded that _____

Which choice most logically completes the text?

- A. conditions of the terrains in the Rio Grande Valley and Mesa Verde had greater similarities in the past than they do today.
- B. some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.
- C. Indigenous peoples living in the Rio Grande Valley primarily planted crops and did not cultivate turkeys before 1280.
- D. the Ancestral Puebloans of Mesa Verde likely adopted the farming practices of Indigenous peoples living in other regions.

ID: b5e9f3c2 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text’s discussion of Ancestral Puebloans’ migration to the Rio Grande Valley. The text states that in the late 1200s C.E., the Ancestral Puebloan civilization abandoned villages in its original homeland, which included the Mesa Verde site. The text goes on to say that recent genetic analysis has demonstrated that the modern turkey population in the Rio Grande Valley descends partly from the ancient turkeys raised at Mesa Verde, and that the genetic markers shared by the two turkey populations first appeared at Mesa Verde only after 1280 C.E. Therefore, it can reasonably be concluded that some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried their agricultural practices—including the farming of turkeys—to their new home.

Choice A is incorrect because the text never compares the condition of the Rio Grande Valley’s terrain to that of Mesa Verde’s terrain, either in the present or in the past. Choice C is incorrect. Although genetic analysis has demonstrated that the modern turkey population in the Rio Grande valley descended in part from the turkey population raised by the Ancestral Puebloans of Mesa Verde before their migration to the valley in 1280, this finding doesn’t eliminate the possibility that Indigenous peoples living in the valley before 1280 might also have farmed turkeys. Choice D is incorrect. The text doesn’t consider the possibility that before their migration to the Rio Grande Valley after 1280, the Ancestral Puebloans of Mesa Verde might have adopted turkey farming from an outside Indigenous civilization in another region; instead, the text provides evidence suggesting that the Ancestral Puebloans brought turkey farming to another region—the Rio Grande Valley—after 1280.

Question Difficulty: Hard

Question ID 5eda42a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 5eda42a3

The following text is from Maggie Pogue Johnson’s 1910 poem “Poet of Our Race.” In this poem, the speaker is addressing Paul Laurence Dunbar, a Black author.

Thou, with stroke of mighty pen,

Hast told of joy and mirth,

And read the hearts and souls of men

As cradled from their birth.

The language of the flowers,

Thou hast read them all,

And e’en the little brook

Responded to thy call.

Which choice best states the main purpose of the text?

- A. To praise a certain writer for being especially perceptive regarding people and nature
- B. To establish that a certain writer has read extensively about a variety of topics
- C. To call attention to a certain writer’s careful and elaborately detailed writing process
- D. To recount fond memories of an afternoon spent in nature with a certain writer

ID: 5eda42a3 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main purpose of the text. In the first part of the text, the speaker addresses Paul Laurence Dunbar’s ability to understand people (he has “read the hearts and souls of men” and written of their “joy and mirth”). In the second part of the text, the speaker describes Dunbar’s thorough understanding of the natural world (he has read “the language of the flowers” and engaged with “the little brook”). Thus, the text mainly praises Dunbar for being especially perceptive about people and nature.

Choice B is incorrect because the speaker describes Dunbar as having read the “hearts and souls of men” and the “language of flowers” to convey Dunbar’s ability to comprehend people and nature, not to suggest that Dunbar has literally read any of these things or has read a great deal about them. Choice C is incorrect because the text notes how well Dunbar has made sense of the topics he’s written about but doesn’t address any specific parts of Dunbar’s writing process beyond the suggestion that he used a pen. Choice D is incorrect because the text focuses on Dunbar’s understanding of people and nature as expressed in his writing. Nothing in the text suggests that the speaker is recalling a particular afternoon actually spent in nature with Dunbar; even if there had been a shared experience, the text isn’t focused on reminiscing.

Question Difficulty: Hard

Question ID 010fa3f0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 010fa3f0

Simulated Change in Annual Aquifer Input and Irrigation Output if Precipitation Concentration Increases as Climate Models Predict

Baseline concentration of annual precipitation	% change in water entering aquifers	% change in surface water used for irrigation	% change in groundwater used for irrigation
Precipitation is currently somewhat concentrated	4.9	0.4	0.9
Precipitation is currently evenly distributed	11.0	9.0	7.9

Some climate models for the western United States predict that while total annual precipitation may remain unchanged from the present level, precipitation will become concentrated into fewer but more intense rain and snow events. University of Texas climate scientist Geeta Persad and her colleagues simulated how the amount of water entering aquifers and the amount being used for irrigation purposes would change if this were to occur. Persad and her colleagues concluded that concentration of precipitation into fewer events would result in a higher number of dry days, triggering more irrigation, but that this change in irrigation output is highly sensitive to the baseline concentration of precipitation that currently exists in an area.

Which choice best describes data from the table that support Persad and her colleagues’ conclusion?

- If baseline precipitation is somewhat concentrated, the amount of water being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater, whereas the amount of water entering aquifers will increase 11.0% if baseline

A. precipitation is evenly distributed.
- If baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas it will

B. increase 9.0% for surface water and 7.9% for groundwater if baseline precipitation is evenly distributed.
- If baseline precipitation is somewhat concentrated, the amount of water entering aquifers will increase 4.9%, while the

C. amount being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater.
- If baseline precipitation is somewhat concentrated, water use for irrigation will decline by a small amount, whereas it will

D. increase 11.0% for surface water and 9.0% for groundwater if baseline precipitation is evenly distributed.

ID: 010fa3f0 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it describes data from the table that support Persad and her colleagues' conclusion. The text explains that, according to some climate models, precipitation in the western United States will become concentrated into fewer, more intense rain and snow events. According to the text, Persad and her colleagues concluded that more irrigation will consequently be needed but that the change in irrigation output will be highly sensitive to, or greatly affected by, the baseline concentration of precipitation in an area. This conclusion is supported by data from the researchers' simulations of changes in annual irrigation output in two different scenarios—one in which an area's annual precipitation is already somewhat concentrated and one in which its annual precipitation is evenly distributed. The table shows that if baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas if baseline precipitation is evenly distributed, water use for irrigation will increase much more—9.0% for surface water and 7.9% for groundwater. This difference illustrates the researchers' conclusion that the amount of additional water needed for irrigation will vary greatly depending on how concentrated or spread out the annual precipitation in an area already is.

Choice A is incorrect because it compares changes in the amount of water being used for irrigation to changes in the amount of water entering aquifers. Persad and her colleagues' conclusion doesn't focus on changes to the amount of water entering aquifers; rather, the researchers' conclusion focuses on changes to irrigation output relative to how concentrated or spread out the annual precipitation in an area is. Choice C is incorrect because it supports only part of Persad and her colleagues' conclusion. According to the text, the researchers concluded that the concentration of precipitation into fewer events will trigger more irrigation but that this change in irrigation output will be highly sensitive to an area's baseline concentration of annual precipitation. The data in this choice support the idea that more irrigation will be needed, but to support the rest of the researchers' conclusion, additional data from the table are required to show that the increases in water use for irrigation will vary depending on how concentrated or spread out the annual precipitation in an area already is. Choice D is incorrect because data in the table indicate no declines in water use for irrigation, showing only increases in the form of positive values.

Question Difficulty: Hard

Question ID 4f9f8ea6

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 4f9f8ea6

Birds of many species ingest foods containing carotenoids, pigmented molecules that are converted into feather coloration. Coloration tends to be especially saturated in male birds’ feathers, and because carotenoids also confer health benefits, the deeply saturated colors generally serve to communicate what is known as an honest signal of a bird’s overall fitness to potential mates. However, ornithologist Allison J. Shultz and others have found that males in several species of the tanager genus *Ramphocelus* use microstructures in their feathers to manipulate light, creating the appearance of deeper saturation without the birds necessarily having to maintain a carotenoid-rich diet. These findings suggest that _____

Which choice most logically completes the text?

- A. individual male tanagers can engage in honest signaling without relying on carotenoid consumption.
- B. feather microstructures may be less effective than deeply saturated feathers for signaling overall fitness.
- C. scientists have yet to determine why tanagers have a preference for mates with colorful appearances.
- D. a male tanager’s appearance may function as a dishonest signal of the individual’s overall fitness.

ID: 4f9f8ea6 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of Shultz’s finding about male tanagers. The text explains that because carotenoids both contribute to deeply saturated feathers and offer health benefits, having deeply saturated feathers is usually “an honest signal” (a true indication) that a bird is generally fit. However, Shultz and others have found that certain male tanagers can appear to have deeply saturated feathers even if they haven’t consumed a diet rich in carotenoids, thanks to microstructures in their feathers that manipulate light. If those birds aren’t necessarily eating carotenoid-rich diets, they may actually be less fit than other birds that appear to have similarly saturated feathers; this suggests that a male tanager’s appearance may function as a dishonest signal, or a false indication, of the bird’s overall fitness.

Choice A is incorrect because Shultz’s finding suggests that some tanagers can signal fitness without consuming the carotenoids that contribute to fitness, thereby making those signals dishonest, not that tanagers can give honest signals of their fitness without consuming carotenoids. Choice B is incorrect because Shultz’s finding suggests that the microstructures in certain tanagers’ feathers can give a dishonest signal of fitness, not that the microstructures are less effective than actual pigmentation for signaling fitness. Whether the signal of fitness is honest or dishonest has no bearing on how effective the signal is: a signal is effective if potential mates behave as though it’s true, regardless of whether it’s actually true. Since there’s no information in the text about how potential mates respond to the dishonest signals of some tanagers, there’s no support for the idea that the dishonest signals are less effective than the honest signals. Choice C is incorrect because Shultz’s finding suggests that certain male tanagers may appear to be fitter than they actually are, not that scientists haven’t determined why tanagers prefer mates with colorful appearances.

Question Difficulty: Hard

Question ID 2c06139b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 2c06139b

Tadpole Body Mass and Toxin Production after Three Weeks in Ponds

Population density	Average tadpole body mass (milligrams)	Average number of distinct bufadienolide toxins per tadpole	Average amount of bufadienolide per tadpole (nanograms)	Average bufadienolide concentration (nanograms per milligram of tadpole body mass)
High	193.87	22.69	5,815.51	374.22
Medium	254.56	21.65	5,525.72	230.10
Low	258.97	22.08	4,664.99	171.43

Ecologist Veronika Bókony and colleagues investigated within-species competition among common toads (*Bufo bufo*), a species that secretes various unpleasant-tasting toxins called bufadienolides in response to threats. The researchers tested *B. bufo* tadpoles’ responses to different levels of competition by creating ponds with different tadpole population densities but a fixed amount of food. Based on analysis of the tadpoles after three weeks, the researchers concluded that increased competition drove bufadienolide production at the expense of growth.

Which choice uses data from the table to most effectively support the researchers’ conclusion?

The difference in average tadpole body mass was small between the low and medium population density conditions and

A. substantially larger between the low and high population density conditions.

Tadpoles in the low and medium population density conditions had substantially lower average bufadienolide

B. concentrations but had greater average body masses than those in the high population density condition.

Tadpoles in the high population density condition displayed a relatively modest increase in the average amount of bufadienolide but roughly double the average bufadienolide concentration compared to those in the low population

C. density condition.

Tadpoles produced approximately the same number of different bufadienolide toxins per individual across the population

D. density conditions, but average tadpole body mass decreased as population density increased.

ID: 2c06139b Answer

Correct Answer: B

Rationale

Choice B is the best answer. This data shows that the tadpoles in the high-density pond (meaning those with the most competition) didn't grow as big as the other two groups but produced more bufadienolide.

Choice A is incorrect. This doesn't fully support the conclusion. It doesn't include any data about bufadienolide production.

Choice C is incorrect. This doesn't fully support the conclusion. It doesn't include any data about growth. Choice D is

incorrect. This doesn't fully support the conclusion. It doesn't demonstrate that the tadpoles in the high-density pond produced more bufadienolide overall. The fact that they didn't produce more kinds of bufadienolide isn't relevant to the conclusion.

Question Difficulty: Hard

Question ID 01c1d9ee

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 01c1d9ee

Swahili Speakers in Three African Countries

Country	Approximate number of speakers (in millions)	Estimated % of population
Democratic Republic of the Congo	22	25
Kenya	55	100
Tanzania	61	100

Swahili is estimated to be the first language of up to 15 million people worldwide. It’s also an officially recognized language in Tanzania, Kenya, and the Democratic Republic of the Congo, which means these countries use Swahili in government documents and proceedings. But even in countries where almost everyone speaks Swahili, for many it isn’t their first language but is instead their second, third, or even fourth language.

Which choice most effectively uses data from the table to support the underlined claim?

Tanzania has approximately 61 million Swahili speakers, which is much more than the estimated total number of people
A. worldwide for whom Swahili is their first language.

Tanzania is estimated to have at most 15 million Swahili speakers, while the country’s total population is approximately
B. 61 million people.

Approximately 100 percent of the people who speak Swahili as their first language live in Kenya, which has a total
C. population of approximately 55 million people.

Approximately 100 percent of Kenya’s population speaks Swahili, while only about 25 percent of the Democratic Republic
D. of the Congo’s population speaks Swahili.

ID: 01c1d9ee Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to support the underlined claim. The text indicates that Swahili is the first language of up to 15 million people worldwide. The text goes on to claim, in the underlined portion, that even in countries where nearly everyone speaks Swahili, many of the language’s speakers don’t have Swahili as their first language. The table indicates that 61 million people in Tanzania, which amounts to 100 percent of the population, speak Swahili. If 61 million people in Tanzania speak Swahili, but only 15 million people worldwide have Swahili as their first language, that means there are many people in Tanzania who speak Swahili as a language other than their first language. This information about Swahili speakers in Tanzania therefore supports the claim that many Swahili speakers in countries where nearly everyone speaks Swahili speak it as a language other than their first language (such as their second, third, or fourth language).

Choice B is incorrect because it doesn’t accurately describe information in the table. According to the table, Tanzania has 61 million Swahili speakers, not at most 15 million Swahili speakers. Additionally, the table indicates that 100 percent of Tanzania’s population speak Swahili, which means that the number of Swahili speakers in the country and the country’s total population should be the same, not that they should differ by such a large amount. Choice C is incorrect because there’s no information in the table or the text that indicates where people who speak Swahili as their first language live. Although Kenya’s total population can be inferred from the table—if Kenya has 55 million Swahili speakers and 100% of Kenya’s population speak Swahili, then Kenya must have a population of 55 million people—nothing suggests that all the people who speak Swahili as their first language live in a single country, let alone that they all live in Kenya. Choice D is incorrect. Although the table does indicate that 100 percent of Kenya’s population and 25 percent of the Democratic Republic of the Congo’s population speak Swahili, this comparison is irrelevant to the claim that Swahili isn’t the first language of many of its speakers even in countries where almost everyone speaks Swahili. On its own, a difference in the proportions of the population who speak Swahili cannot reveal whether those Swahili speakers have Swahili as their first language or a subsequent language.

Question Difficulty: Hard

Question ID f2250478

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: f2250478

Among social animals that care for their young, such as chickens, macaque monkeys, and humans, newborns appear to show an innate attraction to faces and face-like stimuli. Elisabetta Versace and her colleagues used an image of three black dots arranged in the shape of eyes and a nose or mouth to test whether this trait also occurs in *Testudo* tortoises, which live alone and do not engage in parental care. They found that tortoise hatchlings showed a significant preference for the image, suggesting that _____

Which choice most logically completes the text?

- face-like stimuli are likely perceived as harmless by newborns of social species that practice parental care but as threatening by newborns of solitary species without parental care.
- researchers should not assume that an innate attraction to face-like stimuli is necessarily an adaptation related to social interaction or parental care.
- researchers can assume that the attraction to face-like stimuli that is seen in social species that practice parental care is learned rather than innate.
- newly hatched *Testudo* tortoises show a stronger preference for face-like stimuli than adult *Testudo* tortoises do.

ID: f2250478 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of the study by Versace and colleagues. The text indicates that newborn animals of some species are attracted to faces and to stimuli that resemble faces. These species, the text says, share two characteristics: they're social and they practice parental care, meaning that parents care for their young. The text goes on to describe Versace and colleagues' experiment, which showed that *Testudo* tortoises, which aren't social and don't practice parental care, were attracted to a stimulus that resembles a face. Since Versace and colleagues have shown that a species that isn't social and doesn't practice parental care nevertheless has the innate characteristic of being attracted to face-like stimuli, it follows that this characteristic shouldn't be assumed to be an adaptation related to social interaction or parental care.

Choice A is incorrect because the text indicates that the tortoise hatchlings, which are solitary and don't practice parental care, were attracted to the face-like stimuli, not that they perceived the stimuli as threatening. Choice C is incorrect because the phenomenon discussed in the text is an attraction to faces and face-like stimuli on the part of newborn animals, which can't show any learned characteristics since they were just born. Additionally, the text tells us that the tortoises Versace and colleagues studied aren't social and don't practice parental care, so any findings about those tortoises wouldn't be relevant to the question of whether an attraction to faces in social species that practice parental care is innate or learned. Choice D is incorrect because the text gives no indication that adult tortoises were tested on face-like stimuli and, if adults were in fact tested, no information about how they responded is provided. Since no information about adult tortoises' responses is provided, no conclusion comparing those responses to the responses of newly hatched tortoises can be supported.

Question Difficulty: Hard

Question ID 6e0e0de1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 6e0e0de1

Aerogels are highly porous foams consisting mainly of tiny air pockets within a solidified gel. These lightweight materials are often applied to spacecraft and other equipment required to withstand extreme conditions, as they provide excellent insulation despite typically being brittle and eventually fracturing due to degradation from repeated exposure to high heat. Now, Xiangfeng Duan of the University of California, Los Angeles, and colleagues have developed an aerogel with uniquely flexible properties. Unlike earlier aerogels, Duan’s team’s material contracts rather than expands when heated and fully recovers after compressing to just 5% of its original volume, suggesting that _____

Which choice most logically completes the text?

- the aerogel’s remarkable flexibility results from its higher proportion of air pockets to solidified gel as compared to other
- A. aerogels.
- the aerogel’s overall strength is greater than that of other insulators but its ability to withstand exposure to intense heat is
- B. lower.
- the aerogel will be more effective as an insulator for uses that involve gradual temperature shifts than for those that
- C. involve rapid heat increases.
- the aerogel will be less prone to the structural weakness that ultimately causes most other aerogels to break down with
- D. use.

ID: 6e0e0de1 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of aerogels. The text states that aerogels—highly porous foams—offer "excellent insulation" but typically break down after prolonged exposure to high heat. However, according to the text, Duan and colleagues developed an aerogel that "contracts rather than expands when heated" and recovers its original volume after this contraction. Thus, it is logical to conclude that Duan’s team’s aerogel material will be less prone to the structural weakness that caused earlier aerogels to break down.

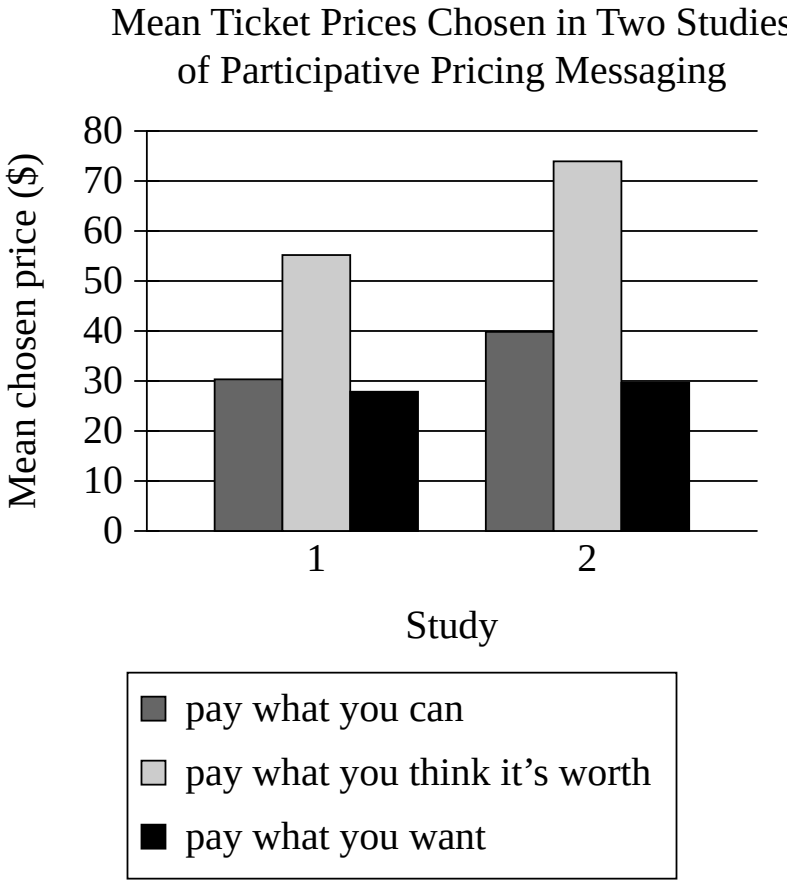
Choice A is incorrect. Although the text indicates that aerogels consist "mainly of tiny air pockets within a solidified gel," it doesn’t mention the number or proportion of air pockets to solidified gel in typical aerogels or in the aerogel developed by Duan’s team. Choice B is incorrect because the text suggests that the aerogel developed by Duan’s team has a higher, not a lower, ability to withstand exposure to intense heat due to its contraction and subsequent recovery. Choice C is incorrect. Although the text discusses temperature tolerances of aerogels and says that they offer "excellent insulation despite typically being brittle and eventually fracturing," it doesn’t discuss how different rates of temperature change can affect aerogels.

Question Difficulty: Hard

Question ID 3430be35

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 3430be35



Participative pricing, in which purchasers choose the prices they pay for products, can enable sellers to capitalize on the heterogeneous values consumers assign to the same goods and services, but doing so requires careful messaging. Annie Peng Cui and Jennifer Wiggins recruited 171 participants (ages 18–60) online for an initial study and 83 students (ages 18–31) at a state university for a second study to test the effect of three different messages—“pay what you can,” “pay what you think it’s worth,” and “pay what you want”—on how much participants would pay for concert tickets. Their results illustrate both the heterogeneity of consumer valuations and how sellers can benefit by prompting consumers to consider their own valuations: _____

Which choice most effectively uses data from the graph to complete the text?

- the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but
- A. when considering what they could afford to pay, the students tended to choose a lower price than did the other group.
- in all three messaging conditions, the group of participants recruited online tended to choose lower prices than did the students, but both groups tended to choose prices closest to the actual cost of the tickets when prompted to consider
- B. the tickets’ value.

the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but both groups tended to choose a higher price when considering the value of the tickets than when considering what they C. could afford or wanted to pay.

within each group of participants, there was wide variation in the value that individuals assigned to the concert tickets, D. but the students tended to assign a higher value to the tickets than did the more age-diverse group recruited online.

ID: 3430be35 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively uses data from the graph to complete the text about the effect of messaging on participative pricing. The graph shows mean ticket prices chosen by participants in response to three messages across two studies: Study 1, which the text indicates was conducted with an age-diverse group recruited online, and Study 2, which was conducted with student participants. The graph indicates that in the "pay what you think it's worth" condition, the mean price of the concert tickets in Study 2 was about \$74, which is greater than the mean price of about \$55 in Study 1. In other words, when participants were asked to consider their valuation of the tickets, the response was heterogeneous, or mixed. Moreover, according to the graph, both Study 1 and Study 2 show higher prices for the tickets under the "pay what you think it's worth" condition than they do under both the "pay what you can" and the "pay what you want" conditions. That is, the data suggest that both groups of participants named higher prices when considering the value of the tickets than when considering either what they could afford or wanted to pay, a finding that supports the idea that sellers can benefit when prompting consumers to consider their own valuations when they choose prices.

Choice A is incorrect because it contradicts information in the graph. Although the graph shows that students in Study 2 assigned a higher value to the tickets than did the age-diverse group in Study 1, which would support the idea that consumer valuations were heterogeneous, the graph shows that in the "pay what you can" (i.e., what you can afford) condition, the students in Study 2 assigned a higher price (about \$40), not a lower price, than the age-diverse group in Study 1 did (about \$30). Moreover, even if it were true that the students had assigned a lower price in this condition, it wouldn't support the result described in the text, only that the participants across the two studies had different ideas of what they can afford to pay. Choice B is incorrect. Although a finding that participants tended to choose prices that were closest to the actual ticket costs in the "pay what you think it's worth" condition would support the idea that sellers benefit by prompting consumers to think about their own valuations (since it's implied that sellers would lose money in the other conditions, where chosen prices were lower than the participants' valuations), neither the text nor the graph addresses how any of the prices chosen by the study participants relate to the tickets' actual market price. Choice D is incorrect. Although the wide variation in participant valuations would support the idea that consumer valuations tend to be heterogeneous, neither the text nor the graph provides any information from which to discern the relative levels of variance among the responses from participants in either study.

Question Difficulty: Hard

Question ID 7f293254

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 7f293254

Art collectives, like the United States- and Vietnam-based collective The Propeller Group or Cuba’s Los Carpinteros, are groups of artists who agree to work together: perhaps for stylistic reasons, or to advance certain shared political ideals, or to help mitigate the costs of supplies and studio space. Regardless of the reasons, art collectives usually involve some collaboration among the artists. Based on a recent series of interviews with various art collectives, an arts journalist claims that this can be difficult for artists who are often used to having sole control over their work.

Which quotation from the interviews best illustrates the journalist’s claim?

- “The first collective I joined included many amazingly talented artists, and we enjoyed each other’s company, but because
- A. we had a hard time sharing credit and responsibility for our work, the collective didn’t last.”
- “We work together, but that doesn’t mean that individual projects are equally the work of all of us. Many of our projects
- B. are primarily the responsibility of whoever originally proposed the work to the group.”
- “Having worked as a member of a collective for several years, it’s sometimes hard to recall what it was like to work alone
- C. without the collective’s support. But that support encourages my individual expression rather than limits it.”
- “Sometimes an artist from outside the collective will choose to collaborate with us on a project, but all of those projects
- D. fit within the larger themes of the work the collective does on its own.”

ID: 7f293254 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents the quotation that best illustrates the journalist’s claim. By indicating that a collective didn’t continue because it was hard to share credit and responsibilities within the group even though the company was enjoyable, the quotation shows that working collaboratively can be difficult for artists who are used to having complete control over their work.

Choice B is incorrect because the quotation indicates that members of a collective are able to collaborate together and have agreed on a fair way to manage their responsibilities; this doesn’t demonstrate the challenge of sharing control among members of a collective. Choice C is incorrect because the quotation highlights the support and encouragement of individual expression an artist experiences due to working in a collective; these positive aspects don’t demonstrate the challenge of sharing control among members of a collective. Choice D is incorrect because the quotation doesn’t address any challenges of sharing control among members of a collective; it simply indicates that artists sometimes choose to work with collectives without having to be a member. Therefore, the quotation doesn’t illustrate the journalist’s claim.

Question Difficulty: Hard

Question ID 4d38c2ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 4d38c2ea

Gorgets, or necklaces with large pendants, have been part of the ceremonial attire of tribes from the US Southeast for centuries. One of the oldest examples, the Fairfield Gorget, was found in Fairfield, Missouri, in 1958. Its overall design resembles that of other art from the region during the Mississippian period (900–1600 CE). Yet the image on the gorget is of a jaguar—a species whose range doesn’t extend to Missouri. Jaguar images are common in ancient Mexican art, and Mexico lies squarely in the species’ range. Therefore, some scholars argue that long-distance trade in Mexican art objects brought the imagery to Missouri, where a local artist could have adopted it and incorporated it into the Fairfield Gorget.

Which finding, if true, would most strongly support the underlined explanation?

- The Fairfield Gorget is dated not to the Mississippian period but instead to the earlier Woodland period, which ended
- A. around 900 CE.
- The range of the jaguar is shown to have expanded dramatically after the Mississippian period came to a close around
- B. 1600 CE.
- An ancient Mexican art object is found at a site that dates to the Mississippian period and is close to where the Fairfield
- C. Gorget was found.
- Certain works by present-day artists from Southeastern tribes reflect outside cultural influences, including contemporary
- D. Mexican art.

ID: 4d38c2ea Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most strongly support the underlined explanation that the jaguar imagery on the Fairfield Gorget could have been inspired by art objects brought to Missouri from Mexico. The text describes the Fairfield Gorget as one of the oldest examples of these necklaces, noting that it's similar to other art from the Mississippian period except for its image of a jaguar. The text goes on to explain that this is unusual because the jaguar's range didn't include Missouri. The Fairfield Gorget's artist likely wouldn't have encountered a live jaguar, but since such jaguar images were common in ancient Mexican art (and the jaguar's range did include Mexico), perhaps some of that art inspired the gorget's artist. Therefore, evidence that an ancient Mexican art object had been found at a site that was close to the Fairfield Gorget in both location and time period would strongly support the explanation that long-distance trade from Mexico took place during the Mississippian period and that jaguar imagery could have reached Missouri by those means.

Choice A is incorrect. The underlined explanation indicates that Mexican artistic influence preceded the creation of the Fairfield Gorget. An artifact from Mexico that appeared in Missouri only after the gorget was created cannot support such a claim. Choice B is incorrect. The underlined explanation pertains to whether the jaguar image on the Fairfield Gorget was suggested to the Mississippian artist who made it by encounters with depictions of jaguars in Mexican art. The appearance of jaguars in Missouri after the Mississippian period cannot provide evidence for such a claim. Choice D is incorrect. The underlined explanation pertains to influence that Mexican art might have had on Mississippian art, restricted to the period from 900 CE to 1600 CE. The actions of contemporary artists cannot provide evidence for centuries-old networks of artistic influence and exchange.

Question Difficulty: Hard

Question ID 5cf5c0d3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 5cf5c0d3

Credited Film Output of James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr

Individual	Years active	Number of films known and commonly credited
James Young Deer	1909–1924	33 (actor), 35 (director), 10 (writer)
Dark Cloud	1910–1920	35 (actor), 1 (writer)
Edwin Carewe	1912–1934	47 (actor), 58 (director), 20 (producer), 4 (writer)
Lillian St. Cyr (Red Wing)	1908–1921	66 (actor)

Some researchers studying Indigenous actors and filmmakers in the United States have turned their attention to the early days of cinema, particularly the 1910s and 1920s, when people like James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr (known professionally as Red Wing) were involved in one way or another with numerous films. In fact, so many films and associated records for this era have been lost that counts of those four figures’ output should be taken as bare minimums rather than totals; it’s entirely possible, for example, that _____

Which choice most effectively uses data from the table to complete the example?

- A. Dark Cloud acted in significantly fewer films than did Lillian St. Cyr, who is credited with 66 performances.
- B. Edwin Carewe’s 47 credited acting roles includes only films made after 1934.
- C. Lillian St. Cyr acted in far more than 66 films and Edwin Carewe directed more than 58.
- D. James Young Deer actually directed 33 films and acted in only 10.

ID: 5cf5c0d3 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it uses data from the table to effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums—that is, that we should assume that the individuals actually had higher outputs than those recorded. The table presents the years during which the individuals were active and the number of known films the individuals are credited in. The table indicates that Lillian St. Cyr has 66 film credits as an actor and that Edwin Carewe has 58 film credits as a director; it follows that if some films and records for the era were lost, it’s possible that Lillian St. Cyr acted in far more than 66 films and that Edwin Carewe directed more than 58 films.

Choice A is incorrect because it doesn’t effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums. Rather than addressing the idea that the individuals likely had higher outputs than those presented in the table, this choice simply compares data from the table to make the point that Dark Cloud has fewer credited acting roles than Lillian St. Cyr (35 and 66, respectively). Choice B is incorrect because it misrepresents data from the table, even though it may exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums by implying that Edwin Carewe acted in more than 47 films. The table indicates that Edwin Carewe was active from 1912 to 1934, meaning that his 47 credited acting roles were in films made before or during 1934, not after that time. Choice D is incorrect because it doesn’t effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums. Instead of addressing the idea that the individuals likely had higher outputs than those recorded, this choice suggests that James Young Deer actually acted in and directed fewer films than presented in the table (only 33 known films as a director instead of 35, and only 10 known films as an actor instead of 33).

Question Difficulty: Hard

Question ID 5cd55c77

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: 5cd55c77

During their larval phase, numerous species of coral reef fish are drawn toward areas where light is present. To better understand how artificial light at night (ALAN) might affect some coral reef fish, researchers explored the effect of exposure to low levels of ALAN on the reproductive success of the common clownfish (*Amphiprion ocellaris*). While exposure to low levels of ALAN had no significant effect on spawning frequency and egg fertilization in *A. ocellaris*, incubation in the presence of ALAN completely inhibited hatching. These findings suggest that _____

Which choice most logically completes the text?

- A. *ocellaris* that settle in areas with low levels of ALAN have significantly higher rates of successful egg fertilization than A. *ocellaris* that settle in areas without ALAN do.

the reproductive success of *A. ocellaris* would be at risk if they were to selectively settle in regions that are regularly B. exposed to low levels of ALAN.

the reproductive success of *A. ocellaris* is more greatly affected by the presence of low levels of ALAN during incubation C. than the reproductive success of other species of coral reef fish is.

the spawning frequency of *A. ocellaris* was more strongly affected by the presence of low levels of ALAN than egg D. fertilization was, though both were less affected than incubation.

ID: 5cd55c77 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of artificial light at night (ALAN) and the common clownfish (*A. ocellaris*). The text indicates that researchers found that when *A. ocellaris* is exposed to low levels of ALAN, spawning frequency and egg fertilization aren’t significantly affected but hatching is completely inhibited—that is, the eggs never hatch. Therefore, if *A. ocellaris* were to settle only in regions where the fish would be regularly exposed to low levels of ALAN, their eggs would stop hatching and their reproductive success would be at risk.

Choice A is incorrect because the text indicates that exposure to low levels of ALAN had no significant effect on egg fertilization for *A. ocellaris*, so there’s no reason to expect there would be any significant difference in rates of successful egg fertilization between areas with low levels of ALAN and areas without ALAN. Choice C is incorrect because the text doesn’t discuss the particular effects of low levels of ALAN on any species of coral reef fish besides *A. ocellaris*. For this reason, there’s no support in the text for the idea that the reproductive success of *A. ocellaris* is more greatly affected by the presence of low levels of ALAN than the reproductive success of other species of coral reef fish is. Choice D is incorrect. The text does indicate that *A. ocellaris* incubation was most strongly affected by low levels of ALAN, but it doesn’t indicate that there was a greater effect on spawning frequency than on egg fertilization; in fact, the text states that there was no significant effect on either.

Question Difficulty: Hard

Question ID b2e54b50

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: b2e54b50

Correlations Between Congestion Ratings and Features of the Crowd in Raters’ Immediate Vicinity

Crowd feature	Before obstacle	After obstacle	Overall
Density	0.8592	0.7308	0.7447
Velocity	−0.9357	−0.9518	−0.8587

Researcher Xiaolu Jia and colleagues monitored individuals’ velocity and the surrounding crowd density as a group of study participants walked through a space and navigated around an obstacle. Participants rated how congested it seemed before the obstacle, after the obstacle, and overall, and the researchers correlated those ratings with velocity and density. (Correlations range from −1 to 1, with greater distance from 0 indicating greater strength). The researchers concluded that the correlations with velocity are stronger than those with density.

Which choice best describes data from the table that support the researchers’ conclusion?

The correlation between congestion ratings before the obstacle and density is further from 0 than the correlation
A. between overall congestion rating and velocity is.

The correlation between congestion ratings before the obstacle and velocity is further from 0 than the correlation
B. between congestion overall and velocity is.

C. For each of the three ratings, the correlation with velocity is negative while the correlation with density is positive.

For each of the three ratings, correlations with velocity are further from 0 than the corresponding correlations with
D. density are.

ID: b2e54b50 Answer

Correct Answer: D

Rationale

Choice D is the best answer. The text tells us that the farther the correlation is from 0, the “stronger” it is (doesn’t matter if it’s negative or positive). The table shows that the correlations with velocity are farther from zero than the correlations with density, which supports the conclusion that the correlations with velocity are stronger.

Choice A is incorrect. This choice doesn’t support the conclusion. It makes an “apples to oranges” comparison by comparing density and velocity correlations across features instead of comparing them for each feature. Choice B is incorrect. This choice doesn’t support the conclusion. It doesn’t include the density correlations for comparison. Choice C is incorrect. This choice doesn’t support the conclusion. The text tells us that the farther the correlation is from 0, the “stronger” it is: it doesn’t matter for “strength” whether it’s negative or positive.

Question Difficulty: Hard

Question ID f495b554

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: f495b554

Silicon-based photovoltaic cells account for 95% of the cells used in solar panels worldwide despite converting an average of only 18–22% of the sunlight that reaches them. In a study addressing this relative inefficiency, a team led by Laura Miranda-Pérez demonstrated that the addition of a thin layer of the mineral perovskite—which captures the blue range of light in the solar spectrum, whereas silicon captures the red range—allows the cells to convert 29.5% or more of the Sun’s energy into usable electricity. Cells made with only perovskite, however, are no more efficient than silicon-based ones. It’s reasonable to conclude, then, that _____

Which choice most logically completes the text?

- photovoltaic cells with both silicon and perovskite are more efficient because they make use of more of the solar
- A. spectrum.
- B. photovoltaic cells with only perovskite and no silicon would likely convert more than 29.5% of the Sun’s energy.
- C. solar power will remain elusive until photovoltaic cells are replaced with a more practical technology.
- D. researchers need to evaluate whether other minerals like perovskite are as effective as perovskite seems to be.

ID: f495b554 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically completes the text’s discussion of the efficiency of various photovoltaic cells. The text indicates that Miranda-Pérez and her team found that adding a layer of perovskite to silicon-based cells increased the percentage of sunlight converted to usable electricity relative to the percentage converted to electricity by silicon-based cells. Using cells with only perovskite and no silicon, however, resulted in no improvement in efficiency over silicon-based cells. The text also states that perovskite captures light in the blue range of the solar spectrum and silicon captures light in the red range of the spectrum. If perovskite and silicon capture different parts of the solar spectrum and the combination of perovskite and silicon results in greater efficiency than the use of either perovskite or silicon alone does, then it is reasonable to infer that the improved efficiency of perovskite-silicon cells could be attributable to their making use of more of the solar spectrum than perovskite-based or silicon-based cells do.

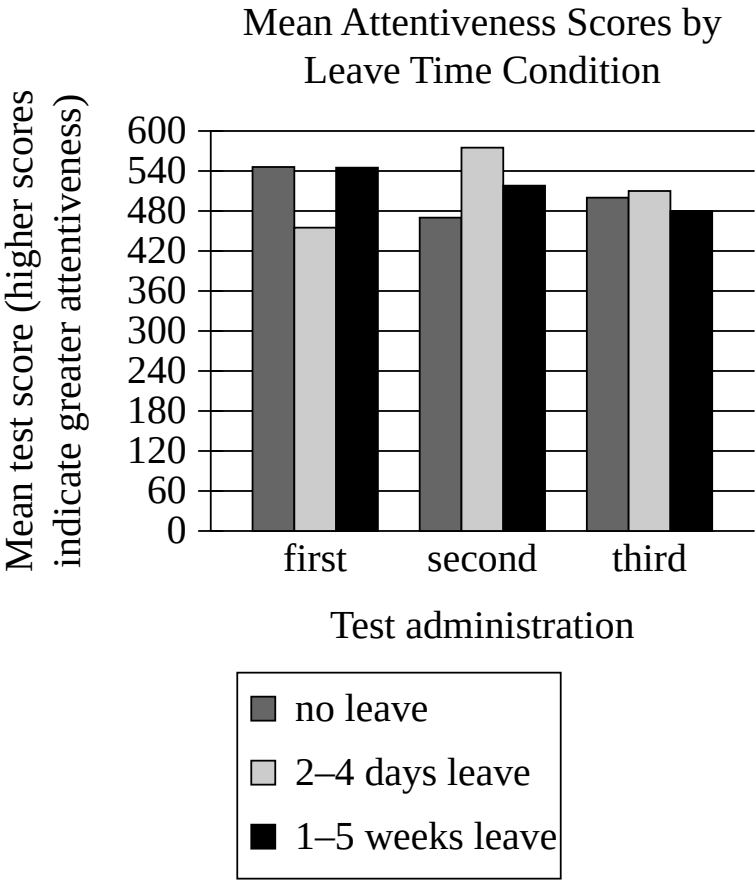
Choice B is incorrect because it is directly contradicted by the text, which says that silicon-based cells convert 18%–22% of the sunlight that reaches them to electricity and that cells with only perovskite are no more efficient than silicon-based cells are. Perovskite-only cells thus must convert 22% or less—not more than 29.5%—of the sunlight that reaches them to electricity. Choice C is incorrect because the text focuses on a possible improvement to photovoltaic cells and makes no mention of any alternatives to those cells, so there is no reason to conclude that photovoltaic cells must be replaced with some other technology. Additionally, the text presents solar power as something that currently exists, so it wouldn’t make sense to conclude that solar power will remain elusive. Choice D is incorrect because nothing in the text suggests that there is a need to evaluate the effectiveness of other minerals than perovskite. In fact, the text gives no indication that it is even possible to use other minerals than silicon and perovskite in photovoltaic cells. Instead, the text is focused on the effect of combining perovskite and silicon and the different parts of the solar spectrum that perovskite and silicon capture.

Question Difficulty: Hard

Question ID 7dab4d5d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 7dab4d5d



To investigate potential cognitive benefits of taking leave from work, psychologist Jan Packer and colleagues conducted a six-month study of Australian university staff members who took no leave from work during the study, took 2–4 days of leave, or took 1–5 weeks of leave. Tests of attentiveness were administered to participants three times during the study: at random for the no-leave staff, and for the rest, one week before their leave, one week following their return to work, and one week after the second test administration. After analyzing the results, the researchers concluded that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Which choice best describes data from the graph that support the researchers’ conclusion?

- In the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, but in the third test administration, those who took no leave had higher average scores than
- A. those who took 1–5 weeks of leave.
- In the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did
- B. those who took 1–5 weeks of leave and those who took no leave.
- In both the second and third test administrations, participants who took 2–4 days of leave had higher average
- C. attentiveness scores than did participants who took 1–5 weeks of leave.

In the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave.

ID: 7dab4d5d Answer

Correct Answer: C

Rationale

Choice C is the best answer because it describes data from the graph that support Jan Packer and colleagues' conclusion about the effect of leave time on the attentiveness of university employees. According to the text, the researchers' study design included a group of employees who took no leave, a group who took 2–4 days of leave, and a group who took 1–5 weeks of leave. The participants who took leave were tested for attentiveness one week before their leave (the first test administration), one week after their return to work (the second test administration), and two weeks after their return (the third test administration). The participants who took no leave were tested three times at random. The graph shows that at one week after their return to work, participants who took only 2–4 days of leave had an average attentiveness score of between 540 and 600, while participants who took 1–5 weeks of leave had an average score of between 480 and 540. At two weeks after their return to work, those who took only 2–4 days of leave had an average score of between 480 and 540, while those who took 1–5 weeks of leave had an average score of approximately 480. In other words, the graph shows that on both post-leave testing dates, participants with longer leave times had lower average attentiveness scores than those with shorter leave times. Since attentiveness is an indicator of cognitive functioning, these data confirm Packer and colleagues' conclusion that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Choice A is incorrect. The graph does show that in the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave and also shows that in the third test administration, those who took no leave had higher average scores than those who took 1–5 weeks of leave. But neither of these findings has a direct bearing on the researchers' conclusion, which concerns a comparison of participants who took 2–4 days of leave with those who took 1–5 weeks, rather than a comparison of either group with participants who took no leave. Choice B is incorrect. Although the graph does show that in the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did those who took 1–5 weeks of leave and those who took no leave, this test administration occurred before any participants went on leave; therefore, these results have no bearing on the researchers' conclusion about how the amount of leave taken by participants affected their cognitive functioning. Choice D is incorrect. Although the graph does show that in the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, the researchers' conclusion is about the effects of short leave compared with the effects of long leave, not the effects of short leave compared with the effects of no leave. These results are therefore irrelevant to the conclusion.

Question Difficulty: Hard

Question ID 014b3394

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 014b3394

Average Number and Duration of Torpor Bouts and Arousal Episodes for
Alaska Marmots and Arctic Ground Squirrels, 2008–2011

Feature	Alaska marmots	Arctic ground squirrels
torpor bouts	12	10.5
duration per bout	13.81 days	16.77 days
arousal episodes	11	9.5
duration per episode	21.2 hours	14.2 hours

When hibernating, Alaska marmots and Arctic ground squirrels enter a state called torpor, which minimizes the energy their bodies need to function. Often a hibernating animal will temporarily come out of torpor (called an arousal episode) and its metabolic rate will rise, burning more of the precious energy the animal needs to survive the winter. Alaska marmots hibernate in groups and therefore burn less energy keeping warm during these episodes than they would if they were alone. A researcher hypothesized that because Arctic ground squirrels hibernate alone, they would likely exhibit longer bouts of torpor and shorter arousal episodes than Alaska marmots.

Which choice best describes data from the table that support the researcher’s hypothesis?

- The Alaska marmots’ arousal episodes lasted for days, while the Arctic ground squirrels’ arousal episodes lasted less

A. than a day.
- The Alaska marmots and the Arctic ground squirrels both maintained torpor for several consecutive days per bout, on

B. average.
- C. The Alaska marmots had shorter torpor bouts and longer arousal episodes than the Arctic ground squirrels did.
- The Alaska marmots had more torpor bouts than arousal episodes, but their arousal episodes were much shorter than

D. their torpor bouts.

ID: 014b3394 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it describes data from the table that support the researcher’s hypothesis. According to the text, the researcher hypothesized that Arctic ground squirrels would exhibit longer torpor bouts and shorter arousal episodes than Alaska marmots do—or, put the other way, that the marmots would show shorter torpor bouts and longer arousal episodes than the ground squirrels do. The table shows data about torpor bouts and arousal episodes for the two species from 2008 to 2011. According to the table, the average duration of torpor bouts was 13.81 days for Alaska marmots, shorter than the average of 16.77 days for Arctic ground squirrels, and the average duration of arousal episodes was 21.2 hours for Alaska marmots, longer than the average of 14.2 hours for Arctic ground squirrels. Thus, the table supports the researcher’s hypothesis by showing that Alaska marmots had shorter bouts of torpor and longer arousal episodes than Arctic ground squirrels did.

Choice A is incorrect because it inaccurately describes data from the table and doesn’t support the researcher’s hypothesis. The table shows that the average duration of arousal episodes was less than a day for both Alaska marmots (21.2 hours) and Arctic ground squirrels (14.2 hours). Additionally, information about arousal episodes for Alaska marmots and Arctic ground squirrels isn’t sufficient to support a hypothesis involving comparisons of both arousal episodes and torpor bouts for those animals. Choice B is incorrect because it doesn’t support the researcher’s hypothesis, which involves comparisons of arousal episodes as well as torpor bouts for Alaska marmots and Arctic ground squirrels. Noting that both animals had torpor bouts lasting several days, on average, doesn’t address arousal episodes at all, nor does it reveal how the animals’ torpor bouts compared. Choice D is incorrect because it doesn’t support the researcher’s hypothesis. Although the table does show that Alaska marmots had more torpor bouts (12) than arousal episodes (11) and that their arousal episodes were much shorter than their torpor bouts (21.2 hours and 13.81 days, respectively), comparing data across only Alaska marmot behaviors isn’t sufficient to support a hypothesis about torpor and arousal behaviors of both Alaska marmots and Arctic ground squirrels.

Question Difficulty: Hard

Question ID 66fd248a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 66fd248a

“Poetry” is a 1919 poem by Marianne Moore. The poem highlights an ambivalence toward poetry as the speaker acknowledges its merits while also expressing a sense of displeasure, writing _____

Which quotation from “Poetry” most effectively illustrates the claim?

- A. “nor is it valid / to discriminate against ‘business documents and / school-books’; all these phenomena are important.”
- B. “One must make a distinction / however: when dragged into prominence by half poets, the result is not / poetry”
“when [poems] become so derivative as to become unintelligible, the / same thing may be said for all of us—that we / do
- C. not admire what / we cannot understand.”
- “Reading [poetry], however, with a perfect contempt for it, one discovers that there is in / it after all, a place for the
- D. genuine.”

ID: 66fd248a Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses a quotation from "Poetry" to illustrate the claim that the poem highlights an ambivalence, or a conflicted attitude, toward poetry. In the quotation, the speaker suggests that one might read poetry with "contempt," or disdain, for it, but even with this negative attitude one will find "a place for the genuine." Because the quotation expresses conflicting attitudes toward poetry, it effectively illustrates the speaker’s ambivalence in discussing the merits and displeasure of reading poetry.

Choice A is incorrect because it doesn’t mention poetry or show ambivalence. Choice B is incorrect. Although the idea of "half poets" may seem to relate to ambivalence, the speaker mentions only negative attitudes toward certain works and the quotation therefore lacks a contrasting positive or neutral attitude that would be needed to indicate ambivalence. Choice C is incorrect because the speaker mentions only negative attitudes toward certain works and the quotation therefore lacks a contrasting positive or neutral attitude that would be needed to indicate ambivalence.

Question Difficulty: Hard

Question ID 9c407117

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 9c407117

A student performs an experiment testing her hypothesis that a slightly acidic soil environment is more beneficial for the growth of the plant *Brassica rapa parachinensis* (a vegetable commonly known as choy sum) than a neutral soil environment. She plants sixteen seeds of choy sum in a mixture of equal amounts of coffee grounds (which are highly acidic) and potting soil and another sixteen seeds in potting soil without coffee grounds as the control for the experiment. The two groups of seeds were exposed to the same growing conditions and monitored for three weeks.

Which finding, if true, would most directly weaken the student’s hypothesis?

- The choy sum planted in the soil without coffee grounds were significantly taller at the end of the experiment than the
- A. choy sum planted in the mixture of soil and coffee grounds.
- The choy sum grown in the soil without coffee grounds weighed significantly less at the end of the experiment than the
- B. choy sum grown in the mixture of soil and coffee grounds.
- The choy sum seeds planted in the soil without coffee grounds sprouted significantly later in the experiment than did the
- C. seeds planted in the mixture of soil and coffee grounds.
- Significantly fewer of the choy sum seeds planted in the soil without coffee grounds sprouted plants than did the seeds
- D. planted in the mixture of soil and coffee grounds.

ID: 9c407117 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it describes an experimental outcome that would most directly weaken the student’s hypothesis. According to the text, the student hypothesizes that *Brassica rapa parachinensis* (choy sum) will benefit more from acidic soil than it will from neutral soil. The text then explains that the student planted 16 choy sum seeds in potting soil with coffee grounds added to increase acidity and another 16 seeds in soil without coffee grounds as a control (a group identical to the experimental group except for the experimental modification being tested). If the hypothesis were correct, the plants in the more acidic soil-and-coffee-grounds mixture would grow faster than those in the control group. However, choice A proposes a scenario in which the plants in soil without coffee grounds were “significantly taller” than those in the more acidic mixture—an outcome that weakens the hypothesis that higher acidity is beneficial to the plants’ growth.

Choice B is incorrect. If the choy sum planted in the neutral soil produced less plant matter and therefore weighed less than the choy sum planted in the acidic soil-and-coffee-grounds mixture, this finding would strengthen the student’s hypothesis, not weaken it. Choice C is incorrect. If seeds planted in neutral soil (without coffee grounds) sprouted significantly later than seeds planted in the acidic soil-and-coffee-grounds mixture, this finding would strengthen, not weaken, the student’s hypothesis that acidic soil benefits choy sum. Choice D is incorrect. If seeds planted in the neutral soil (without coffee grounds) sprouted significantly fewer plants than seeds planted in the acidic soil-and-coffee-grounds mixture did, this finding would strengthen, not weaken, the student’s hypothesis that choy sum benefits from acidic soil.

Question Difficulty: Hard

Question ID 6d69ab93

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 6d69ab93

Initially observed in 2017, the interstellar object ‘Oumuamua is the first object of its kind to be seen in our solar system. Researchers have been puzzled because its acceleration cannot be entirely explained by the gravitational pull of nearby bodies: there must be a nongravitational influence on its velocity and trajectory. Some previously suggested explanations for this nongravitational acceleration involve mechanisms that are unlikely or unrealistic, such as geometric effects from ‘Oumuamua being potentially composed of several spatially separated bodies. Now, Jennifer Bergner and colleagues propose that the nongravitational acceleration is due to the gaseous expulsion of entrapped hydrogen from ‘Oumuamua’s water-rich icy body.

Which statement, if true, would most strongly support the claim made by Bergner and colleagues about the cause of ‘Oumuamua’s acceleration?

- Existing proposed models of outgassing from ‘Oumuamua include the direct conversion of nitrogen or carbon monoxide from a solid to a gaseous state without becoming liquid, but these models have theoretical or observational
- A. inconsistencies.
- ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration that would be caused by the release of hydrogen gas resulting from the processing of water ice (H₂O), but the interstellar object’s observable properties can be
- B. explained if it has a significant component of molecular hydrogen ice (H₂).
- Since nongravitational accelerations of interstellar objects are several orders of magnitude weaker than gravitational accelerations, deviation from behavior that could be fully attributed to gravitational pull has been detected on a limited
- C. number of objects similar to ‘Oumuamua.
- Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, ‘Oumuamua experienced warming sufficient to alter its icy structure
- D. and allow for outgassing.

ID: 6d69ab93 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents findings that, if true, would support the claim made by Bergner and colleagues that the nongravitational acceleration of 'Oumuamua is due to the expulsion of entrapped hydrogen. The text first introduces the observation of a unique interstellar object named 'Oumuamua and goes on to explain that the object exhibited nongravitational acceleration that could not be fully attributed to the expected cause: gravitational pull of nearby celestial bodies. The text concludes by stating that Bergner and colleagues claim that the nongravitational acceleration is caused by expulsion of hydrogen gas from 'Oumuamua's water-rich icy body. To support this claim requires evidence that hydrogen gas could be present within 'Oumuamua at all, which this answer choice presents: cosmic radiation can result in embedded pockets of hydrogen gas in water ice. Additionally, evidence that this gas can be released from such a body is required to fully support the claim, which this answer choice goes on to provide: 'Oumuamua experienced sufficient warming as it traveled through the solar system to alter its icy structure and release the hydrogen gas. Thus, this answer choice provides the best evidence to support Bergner and colleagues' claim.

Choice A is incorrect because this answer choice concerns faults with previous models of outgassing from 'Oumuamua of carbon monoxide and nitrogen, which would not support a claim regarding hydrogen outgassing. Furthermore, inconsistencies in other models would not provide evidence in support of a different model or explanation. Choice B is incorrect. The evidence presented in this answer choice would weaken the claim proposed by Bergner and colleagues that the nongravitational acceleration of 'Oumuamua is caused by the expulsion of hydrogen gas because this answer choice suggests that there is evidence that refutes this claim: 'Oumuamua's trajectory is inconsistent with a nongravitational acceleration caused by the release of hydrogen gas. Furthermore, the remaining portion of this answer choice is unrelated to the claim. Choice C is incorrect because the claim being made by Bergner and colleagues concerns the expulsion of entrapped hydrogen gas, but this answer choice is concerned solely with the differences in magnitude of gravitational and nongravitational acceleration, which would not support Bergner's claim. Furthermore, this answer choice discusses interstellar objects similar to 'Oumuamua, but the text states that 'Oumuamua is the first observed object of its kind in our solar system, so evidence from other, similar bodies would not be available.

Question Difficulty: Hard

Question ID f27559d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: f27559d4

Volunteering, or giving time for a community service for free, is a valuable form of civic engagement because helping in a community is also good for society as a whole. In a survey of youths in the United States, most young people said that they believe volunteering is a way to help people on an individual level. Meanwhile, only 6% of the youths said that they think volunteering is a way to help fix problems in society overall. These replies suggest that _____

Which choice most logically completes the text?

- A. many young people think they can volunteer only within their own communities.
- B. volunteering may be even more helpful than many young people think it is.
- C. volunteering can help society overall more than it can help individual people.
- D. many young people may not know how to find ways to volunteer their time.

ID: f27559d4 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of volunteering. The text asserts that volunteering benefits both the community in which one volunteers and society as a whole. It then states that in a survey of young people, a majority of respondents said that volunteering helps individuals, but only 6% of respondents said that volunteering helps society as a whole. If volunteering does in fact help society as a whole, as the text says, but only 6% of young people believe it does, then it’s reasonable to conclude that volunteering is more helpful than many young people think it is.

Choice A is incorrect because the text discusses young people’s beliefs about the benefits of volunteering, not where young people believe they are able to volunteer. Nothing in the text suggests that many young people believe they are only able to volunteer in their own communities. Choice C is incorrect. Although the text indicates that volunteering is beneficial for society as a whole, nothing in the text suggests that volunteering can benefit society more than it can benefit individual people. The text doesn’t compare the benefits to society with the benefits to individuals. Choice D is incorrect because the text discusses young people’s beliefs about the benefits of volunteering, not how to find volunteering opportunities. There’s nothing in the text to suggest that many young people don’t know how to volunteer.

Question Difficulty: Hard

Question ID d1539546

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: d1539546

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh’s interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____

Which choice most logically completes the text?

- A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.
- B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs’ location.
- C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.
- D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

ID: d1539546 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The text says that crab burrowing in the pannes enables plants to grow there again. It also says that crab burrowing at the edges of the marsh speeds up marsh loss. This suggests that burrowing crabs can either help or hurt marshes, depending on where they’re located.

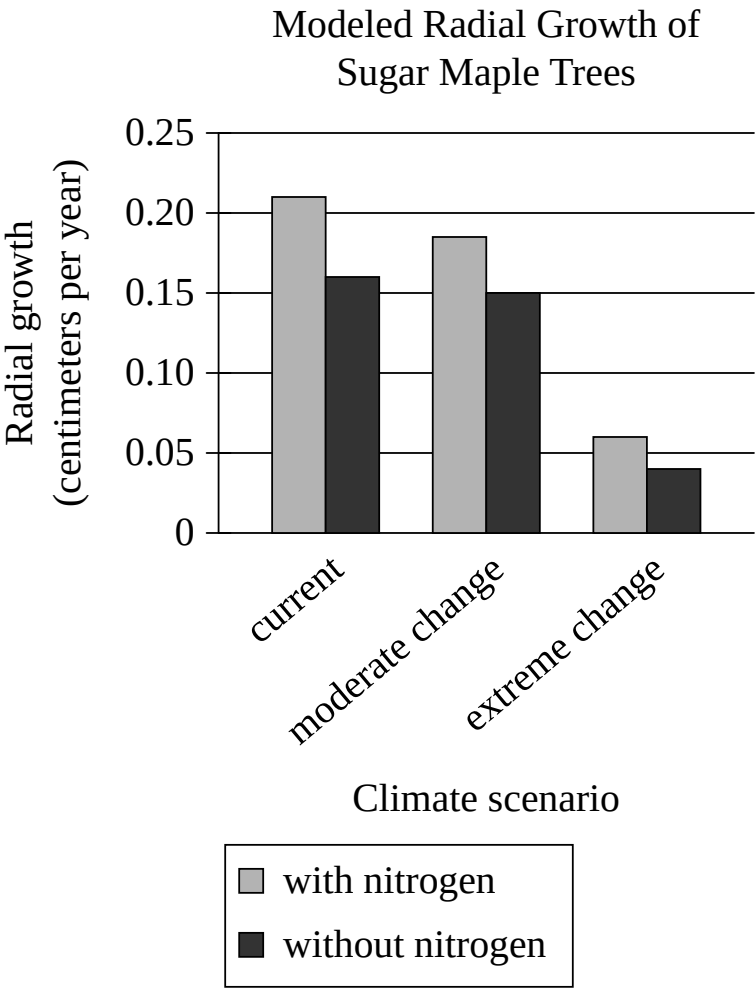
Choice A is incorrect. This inference isn’t supported. The text never discusses pannes of different sizes. Choice C is incorrect. This inference isn’t supported. The text never suggests that crabs do more burrowing in the pannes (the areas with standing water) than they do at the edges. Rather, the text says that the burrowing that happens in the pannes is beneficial, while the burrowing that happens at the edges is harmful. Choice D is incorrect. This inference isn’t supported. The text never discusses the intensity of crab burrowing, nor does it discuss the size of the panne relative to the size of the marsh.

Question Difficulty: Hard

Question ID 627d93e3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 627d93e3



Inés Ibáñez and colleagues studied a forest site in which some sugar maple trees receive periodic fertilization with nitrogen to mimic the broader trend of increasing anthropogenic nitrogen deposition in soil. Ibáñez and colleagues modeled the radial growth of the trees with and without nitrogen fertilization under three different climate scenarios (the current climate, moderate change, and extreme change). Although they found that climate change would negatively affect growth, they concluded that anthropogenic nitrogen deposition could more than offset that effect provided that change is moderate rather than extreme.

Which choice best describes data from the graph that support Ibáñez and colleagues’ conclusion?

Growth with nitrogen under the current climate exceeded growth with nitrogen under moderate change, but the latter
A. exceeded growth without nitrogen under extreme change.

Growth without nitrogen under the current climate exceeded growth without nitrogen under moderate change, but the
B. latter exceeded growth with nitrogen under extreme change.

Growth with nitrogen under moderate change exceeded growth without nitrogen under moderate change, but the latter
C. exceeded growth without nitrogen under extreme change.

Growth with nitrogen under moderate change exceeded growth without nitrogen under the current climate, but the latter
D. exceeded growth with nitrogen under extreme change.

ID: 627d93e3 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it describes data from the graph that support Ibáñez and colleagues' conclusion that increasing anthropogenic nitrogen deposition can compensate for the negative effect of climate change on tree growth if that change is moderate but not if it's extreme. The bar graph shows the growth of sugar maple trees with and without nitrogen fertilization under three different climate-change scenarios: current conditions, a moderate change, and an extreme change. According to the graph, radial growth without nitrogen fertilization is projected to be about 0.16 centimeters (cm) under current conditions, 0.15 cm under a moderate change, and 0.04 cm under an extreme change. The graph also shows that with nitrogen fertilization, growth is projected to be about 0.18 centimeters under a moderate change but only about 0.06 centimeters under an extreme change. Thus, the data in the graph support the researchers' conclusion by showing greater growth for a moderate change using nitrogen fertilization than they do either under current conditions without nitrogen fertilization or under an extreme change with nitrogen fertilization.

Choice A is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of anthropogenic nitrogen, but only if change to the climate is moderate and not if it's extreme. To support this would require comparing radial growth without nitrogen fertilization under current climate conditions to the growth with nitrogen fertilization under both moderate and extreme changes. This choice mentions only growth with nitrogen fertilization under current climate conditions and moderate change and growth without nitrogen fertilization under an extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change. Choice B is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of atmospheric nitrogen, but only if change to the climate is moderate and not if it's extreme. The support needed would compare radial growth under current climate conditions without nitrogen fertilization to the growth with nitrogen fertilization under moderate and extreme changes. This choice mentions only growth without nitrogen fertilization under current conditions and moderate change and growth with nitrogen fertilization under extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change. Choice C is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of atmospheric nitrogen, but only if change to the climate is moderate and not if it's extreme. The support needed would compare radial growth without adding nitrogen under current climate conditions to the growth with nitrogen fertilization under moderate and extreme changes. This choice mentions only the growth with and without nitrogen fertilization under moderate climate change and growth without nitrogen fertilization under extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change.

Question Difficulty: Hard

Question ID df9c5a1d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: df9c5a1d

Juvenile Plants Found Growing on Bare Ground and in Patches of Vegetation for Five Species

Species	Bare ground	Patches of vegetation	Total	Percent found in patches of vegetation
<i>T. moroderi</i>	9	13	22	59.1%
<i>T. libanitis</i>	83	120	203	59.1%
<i>H. syriacim</i>	95	106	201	52.7%
<i>H. squamatum</i>	218	321	539	59.6%
<i>H. stoechas</i>	11	12	23	52.2%

Alicia Montesinos-Navarro, Isabelle Storer, and Rocío Perez-Barrales recently examined several plots within a diverse plant community in southeast Spain. The researchers calculated that if individual plants were randomly distributed on this particular landscape, only about 15% would be with other plants in patches of vegetation. They counted the number of juvenile plants of five species growing in patches of vegetation and the number growing alone on bare ground and compared those numbers to what would be expected if the plants were randomly distributed. Based on these results, they claim that plants of these species that grow in close proximity to other plants gain an advantage at an early developmental stage.

Which choice best describes data from the table that support the researchers’ claim?

- A. For all five species, less than 75% of juvenile plants were growing in patches of vegetation.
- B. The species with the greatest number of juvenile plants growing in patches of vegetation was *H. stoechas*.

For *T. libanitis* and *T. moroderi*, the percentage of juvenile plants growing in patches of vegetation was less than what

C. would be expected if plants were randomly distributed.

For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what

D. would be expected if plants were randomly distributed.

ID: df9c5a1d Answer

Correct Answer: D

Rationale

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it’s advantageous for plants at an early stage of development to grow in patches of vegetation.

Choice A is incorrect because the statement that less than 75% of juvenile plants were found growing in patches of vegetation, while true, doesn’t clearly support the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. Saying that less than 75% of plants were found in patches doesn’t indicate how the percentage growing in patches compares with the percentage that would be expected to grow in patches on the basis of chance alone, which is the information necessary to evaluate whether the claim in the text has support in the table. Put another way, if the percentage of plants found growing in patches was 15% or less, it would be true that less than 75% were found in patches, but the data would in fact weaken the claim in the text, not strengthen it, since the data would show that growing in patches wasn’t advantageous. Choice B is incorrect because only 12 plants of this species were found growing in patches, which was the lowest number of any species, not the greatest number. Additionally, even if it were true that this species had the greatest number of plants growing in patches, the finding would be irrelevant to the claim that plants of all five species gained an advantage by growing in close proximity to other plants. Choice C is incorrect because 59.1% of the plants of these species were found growing in patches, which is a far greater percentage, not a lower percentage, than what would be expected if plants were randomly distributed (around 15%). Additionally, if it were true that the percentage of plants growing in patches was lower for these species than what would be expected from chance alone, that finding would weaken, not strengthen, the claim that growing in patches is advantageous.

Question Difficulty: Hard

Question ID d2e0cba5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: d2e0cba5

In a study of new technology adoption, Davit Marikyan et al. examined negative disconfirmation (which occurs when experiences fall short of one’s expectations) to determine whether it could lead to positive outcomes for users. The team focused on established users of “smart home” technology, which presents inherent utilization challenges but tends to attract users with high expectations, often leading to feelings of dissonance. The researchers found that many users employed cognitive mechanisms to mitigate those feelings, ultimately reversing their initial sense of disappointment.

Which choice best states the main idea of the text?

- Research suggests that most users of smart home technology will not achieve a feeling of satisfaction given the utilization challenges of such technology.
- Although most smart home technology is aimed at meeting or exceeding users’ high expectations, those expectations in general remain poorly understood.
- Research suggests that users with high expectations for a new technology can feel content with that technology even after experiencing negative disconfirmation.
- Although negative disconfirmation has often been studied, little is known about the cognitive mechanisms shaping users’ reactions to it in the context of new technology adoption.

ID: d2e0cba5 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The passage begins by describing the study, and concludes with its results: "many users employed cognitive mechanisms to mitigate those feelings [of negative disconfirmation]."

Choice A is incorrect. This is the opposite of what the text claims. Rather, the researchers found that "many users" reversed "their initial sense of disappointment." Choice B is incorrect. This choice doesn’t reflect the text. What the expectations of smart home tech users are is not discussed. Choice D is incorrect. This choice doesn’t reflect the text. How often these topics have been studied is not mentioned.

Question Difficulty: Hard

Question ID cf3acc50

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: cf3acc50

Compiled in the late 1500s largely through the efforts of Indigenous scribes, *Cantares Mexicanos* is the most important collection of poetry in Classical Nahuatl, the principal language of the Aztec Empire. The poems portray Aztec society before the occupation of the empire by the army of Spain, and marginal notes in *Cantares Mexicanos* indicate that much of the collection’s content predates the initial invasion. Nonetheless, some of the poems contain inarguable references to beliefs and customs common in Spain during this era. Thus, some scholars have concluded that _____

Which choice most logically completes the text?

- A. while its content largely predates the invasion, *Cantares Mexicanos* also contains additions made after the invasion.
although those who compiled *Cantares Mexicanos* were fluent in Nahuatl, they had limited knowledge of the Spanish language.
- B. language.
- C. before the invasion by Spain, the poets of the Aztec Empire borrowed from the literary traditions of other societies.
the references to beliefs and customs in Spain should be attributed to a coincidental resemblance between the societies
- D. of Spain and the Aztec Empire.

ID: cf3acc50 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text. The text explains that the *Cantares Mexicanos* contains poems about the Aztec Empire from before the Spanish invasion. Furthermore, it indicates that notes in the collection attest that some of these poems predate the Spanish invasion, while some customs depicted are likely Spanish in origin. The implication is that some poems were composed before the invasion but the references to Spanish customs could have come about only after the invasion, and thus that the collection includes content that predates the invasion and also content from after the invasion.

Choice B is incorrect because the text clearly indicates that the collection is in Nahuatl, not Spanish, so the compilers’ unfamiliarity with Spanish is irrelevant to whether the collection contains material composed after the Spanish invasion. Choice C is incorrect because the text mentions only the Aztec Empire and Spain: there is no information about the relationship of Aztec literature to any traditions other than its own or Spain’s. Choice D is incorrect because the text states that some of the poems make “inarguable references” to common Spanish customs, which conflicts with the idea that these references can reasonably be attributed to mere coincidence.

Question Difficulty: Hard

Question ID 9abc3ba5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 9abc3ba5

“Gestures” in painting are typically thought of as bold, expressive brushstrokes. In the 1970s, American painter Jack Whitten built a 12-foot (3.7-meter) tool he named the “developer” to apply paint to an entire canvas in one motion, resulting in his series of “slab” paintings from that decade. Whitten described this process as making an entire painting in “one gesture,” signaling a clear departure from the prevalence of gestures in his work from the 1960s. Some art historians claim this shift represents “removing gesture” from the process. Therefore, regardless of whether using the developer constitutes a gesture, both Whitten and these art historians likely agree that _____

Which choice most logically completes the text?

- A. any tool that a painter uses to create an artwork is capable of creating gestures.
- B. Whitten’s work from the 1960s exhibits many more gestures than his work from the 1970s does.
- C. Whitten became less interested in exploring the role of gesture in his work as his career progressed.
- D. Whitten’s work from the 1960s is much more realistic than his work from the 1970s is.

ID: 9abc3ba5 Answer

Correct Answer: B

Rationale

Choice B is the best answer. Whitten thinks the tool made “one gesture” paintings, while historians think the tool “removed gesture” from the process completely. But putting that debate aside, both Whitten and the historians would agree that the paintings he made with the tool in the ’70s have way fewer gestures than his paintings from the ’60s, in which gestures are “prevalent,” meaning widely and extensively present.

Choice A is incorrect. This inference isn’t supported. The text only discusses the “developer”—it never mentions other tools. Choice C is incorrect. This inference isn’t supported. If anything, the text suggests the opposite: that Whitten became more interested in exploring the role of gesture in his work as his career progressed, as his earlier paintings had many gestures, and his ’70s paintings only had “one gesture.” Choice D is incorrect. This inference isn’t supported. The text never discusses the “realism” of Whitten’s art.

Question Difficulty: Hard

Question ID 7c9a65bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 7c9a65bb

Optical tweezers are specialized scientific tools—particularly useful in biology and medicine—that use high-powered beams of light to trap and manipulate minuscule particles for study. Use of the tool has led to several scientific and medical breakthroughs over the last few decades, but the particles are often under prolonged exposure to the intense heat of the light beams. To overcome the risk of overheating, and thereby damage, researchers sometimes attach nano-sized glass beads to particles, allowing the light to focus on the beads instead of the particles.

Based on the text, what is one advantage of attaching glass beads to particles when using optical tweezers?

- A. It decreases the time it takes for the optical tweezers to locate and capture the particles.
- B. It facilitates the maneuvering of particles without directly heating the particles themselves.
- C. It allows researchers to use weaker light beams to manipulate particles.
- D. It adds a material to which particles can transfer any heat absorbed from the optical tweezers’ light beam.

ID: 7c9a65bb Answer

Correct Answer: B

Rationale

Choice B is the best answer. The text says that the glass beads get the "focus" of the light beams so that the particles don’t overheat. From this, we can infer that the beads allow the particles to be manipulated without being directly heated by the light beams.

Choice A is incorrect. The text never says that attaching the glass beads saves time in any way. Choice C is incorrect. The text never says that attaching the glass beads allows researchers to use weaker light beams. Choice D is incorrect. The text doesn’t say that the particles can transfer heat to the glass beads—rather, it says the heat from the light focuses on the glass beads instead of the particles.

Question Difficulty: Hard

Question ID 378c66d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 378c66d5

A member of the Otomi, an Indigenous people in Central Mexico, Octavio Medellín immigrated to the United States as a child, and his sculpture bears the impress of traditions on both sides of the border: US-based modernist sculpture, Mexican modernist painting, Otomi art, and the ancient sculpture of other Mexican Indigenous peoples, including the Maya. In his 1950 masterpiece *History of Mexico*, Medellín fuses these influences into a style so idiosyncratic that it resists efforts to view his work through the lens of nationality or cultural identity. Artists, he insisted, should strive for individual expression, even as they draw inspiration from their heritage and the communities where they live and work.

Which quotation from an art critic most directly challenges the underlined claim in the text?

- “Although a number of ancient Indigenous artistic traditions pictured human forms in profile, the forms populating the surface of *A History of Mexico* suggest a specifically Maya influence.”

“In *A History of Mexico*, the synthesis of ancient and modernist traditions functions as a stylistic parallel to the work’s subject matter: a survey of centuries of Mexican history.”

“Many critics focus on Indigenous influences in *A History of Mexico* and other key works by Medellín to the exclusion of influences from non-Indigenous art.”

“While *A History of Mexico* features modernist motifs, it relies primarily on angular human forms in profile—a staple of Maya sculpture—and thus invites classification as Indigenous art.”

ID: 378c66d5 Answer

Correct Answer: D

Rationale

Choice D is the best answer. This critic challenges the claim by arguing that *A History of Mexico* is not so idiosyncratic (unique) as to resist classification because its use of Maya-style human profiles actually “invites classification as Indigenous art.” Therefore, according to this critic, the work can be viewed, at least partially, through a lens of national or cultural identity.

Choice A is incorrect. While it describes the Maya influence on a particular aspect of *A History of Mexico*, this quotation doesn’t push back on Medellín’s “idiosyncratic” (unique) style, nor does it try to categorize the entire work into a single cultural tradition. Choice B is incorrect. This choice doesn’t directly challenge the underlined claim, but rather supports it. It suggests that the work effectively blends a variety of artistic products to create a single work that can’t be defined by any one tradition. Instead, the work recalls many centuries of history and culture all at once. Choice C is incorrect. This choice supports rather than challenges the underlined claim. The quotation argues that other critics focus too much on Indigenous influences on the artwork instead of viewing all of the influences equally—that they should instead be viewing the work as an idiosyncratic whole instead of through one or two narrow cultural lenses.

Question Difficulty: Hard

Question ID 6762772f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 6762772f

Artificial leaves are a developing renewable energy technology that mimics the process of photosynthesis in plants. These devices are silicon-based solar cells coated in chemical catalysts that activate reactions that split water molecules into hydrogen and oxygen gas. The technology, while generating lots of interest, is not yet commercially viable as a large-scale energy source. To meet this challenge, scientists from many fields are researching ways to store, transport, and distribute the energy the devices produce while other scientists are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

- Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an
- A. energy source.
- The recent increase in the commercial use of artificial leaves as an energy source has encouraged many scientists to
- B. research ways to improve the technology.
- Artificial leaves split water molecules into oxygen and hydrogen gas using catalysts more efficiently than plants do using
- C. the process of photosynthesis.
- Artificial leaves were developed to mimic the natural process of photosynthesis in plants in order to store energy for long-
- D. term commercial use.

ID: 6762772f Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. According to the text, artificial leaves are a renewable energy technology that’s in development and generating interest. The text goes on to state that artificial-leaf technology "is not yet commercially viable as a large-scale energy source" and that scientists from several fields are doing additional research to refine the technology. Thus, the main idea is that continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.

Choice B is incorrect. While the text does state that scientists from many fields are researching artificial leaves, they’re doing so to address the fact that the leaves are not yet commercially viable, rather than in response to an increase in commercial use. Choice C is incorrect. Although the text does explain that artificial leaves help split water molecules into oxygen and hydrogen gas using catalysts, it doesn’t argue that this process is more efficient than photosynthesis. This is merely a detail of the text and not the main idea, which is about the additional research needed to improve the cost and efficiency of these artificial leaves. Choice D is incorrect. Although the text indicates that artificial leaves mimic photosynthesis in plants, this is an incidental detail in support of the main idea that additional research is needed to make artificial leaves commercially viable.

Question Difficulty: Hard

Question ID 22b3da87

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 22b3da87

During the Bourbon Restoration in France (1814–1830), the right to vote required in part that a person paid at least 300 francs in direct taxes to the government. The four most common taxes (the *quatre vieilles*) were levied on real estate (both land and buildings); the doors and windows in taxpayer homes; the rental values of homes; and the businesses of artisans and merchants. (Foreign investments were either exempt from taxation or taxed lightly.) Although relatively few people paid the tax on real estate, it was the main means of voter qualification and accounted for over two-thirds of government receipts during this period, suggesting that during the Bourbon Restoration _____

Which choice most logically completes the text?

- A. those people who had the right to vote most likely had substantial holdings of French real estate.
- B. the voting habits of French artisans and merchants were effective in reducing tax burdens on businesses.
- C. the number of doors and windows in French residences was kept to a minimum but increased after 1830.
- D. French people with significant foreign investments were unlikely to have the right to vote.

ID: 22b3da87 Answer

Correct Answer: A

Rationale

Choice A is the best answer. We’re told that people needed to pay “at least 300 francs in direct taxes” to be able to vote. We’re also told that, while “relatively few people paid the tax on real estate,” real estate taxes were both the main way people qualified to vote and the main source of revenue for the government. Based on this, we can infer that those who did qualify to vote likely had significant French real estate holdings.

Choice B is incorrect. The passage doesn’t mention the voting habits of artisans and merchants nor any reduction in tax burdens on businesses, so there’s no basis for this inference. Choice C is incorrect. Although we know that doors and windows were taxed during the Bourbon Restoration, we don’t have enough information to infer if doors and windows increased after this time. Choice D is incorrect. Although we know that foreign investments were only minimally taxed, we don’t have enough information to determine if those with significant foreign investments were unlikely to have voting rights. For example, it’s possible that those with significant foreign investments were likely to also be people with significant domestic investments which they did pay taxes on, so we don’t have the information necessary to make this inference.

Question Difficulty: Hard

Question ID 303537cf

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: 303537cf

The following text is adapted from Lewis Carroll’s 1889 satirical novel *Sylvie and Bruno*. A crowd has gathered outside a room belonging to the Warden, an official who reports to the Lord Chancellor.

One man, who was more excited than the rest, flung his hat high into the air, and shouted (as well as I could make out) “Who roar for the Sub-Warden?” Everybody roared, but whether it was for the Sub-Warden, or not, did not clearly appear: some were shouting “Bread!” and some “Taxes!”, but no one seemed to know what it was they really wanted.

All this I saw from the open window of the Warden’s breakfast-saloon, looking across the shoulder of the Lord Chancellor. “What can it all mean?” he kept repeating to himself. “I never heard such shouting before—and at this time of the morning, too! And with such unanimity!”

Based on the text, how does the Lord Chancellor respond to the crowd?

- A. He asks about the meaning of the crowd’s shouting, even though he claims to know what the crowd wants.
- B. He indicates a desire to speak to the crowd, even though the crowd has asked to speak to the Sub-Warden.
- C. He expresses sympathy for the crowd’s demands, even though the crowd’s shouting annoys him.
- D. He describes the crowd as being united, even though the crowd clearly appears otherwise.

ID: 303537cf Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a statement about how the Lord Chancellor responds to the crowd that is supported by the text. The text indicates that the people in the crowd are roaring and shouting “Bread!” or “Taxes!” and presents them as not knowing what they really want. The Lord Chancellor’s response is to ask what their shouting means but also to observe that they’re shouting with “unanimity,” or total agreement. Clearly, this isn’t the case, which supports the statement that the Lord Chancellor describes the crowd as being united even though it’s not.

Choice A is incorrect because it isn’t supported by the text. Although the text indicates that the Lord Chancellor asks about the meaning of the crowd’s shouting, it doesn’t suggest that he knows what the crowd really wants. Choice B is incorrect because the text doesn’t suggest that the Lord Chancellor wants to speak to the crowd. Furthermore, the text doesn’t indicate that the crowd wants to hear from the Sub-Warden. Although the crowd roars when asked “Who roar for the Sub-Warden?” it’s unclear what the roaring means. Choice C is incorrect because the text doesn’t suggest that the Lord Chancellor knows of or sympathizes with the crowd’s demands. In addition, the text doesn’t indicate that the crowd’s shouting annoys the Lord Chancellor, just that it causes him to keep repeating “What can it all mean?”

Question Difficulty: Hard

Question ID 35ec767c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 35ec767c

Corn-Related Vocabulary in Various Southeastern Languages

Language family	Word (language)	English translation	Proposed origin in vocabulary of the Totozoquean language family
Muskogean	tanchi' (Chickasaw); tanchi (Choctaw); vce (Muscogee, pronounced "uh-chi")	corn	no
Iroquoian	se-lu (Cherokee)	corn	no
Caddoan	-k'as- (Caddo)	dried corn	yes
Chitimacha	k'asma (Chitimacha)	corn	yes

In Caddo, a language from what is now the US Southeast, vocabulary pertaining to corn cultivation resembles equivalent vocabulary in the Totozoquean language family in Mexico. This resemblance is perhaps attributable to cultural contact: such words could have entered Caddo through the intermediary of the neighboring but unrelated Chitimacha language, concurrent with the dissemination of corn itself from Mexico into the Southeast after 700 CE. That the vocabulary pertaining to domestic crops accompanies them as they diffuse into new regions is an established phenomenon globally. Crops may also be decoupled from vocabulary altogether: corn cultivation became ubiquitous among the Southeastern tribes, yet _____

Which choice most effectively uses data from the table to complete the statement?

- the origins of vocabulary pertaining to the crop vary across languages in the region, with the words for corn in Cherokee
- A. and the Muskogean languages showing no demonstrable relationship to Totozoquean vocabulary.
- the region is linguistically diverse, being home not only to Chitimacha and Caddo, but also to the Muskogean language
- B. family (including Chickasaw, Choctaw, and Muscogee) and to one Iroquoian language (Cherokee).
- corn-related vocabulary underwent changes when entering other, unrelated languages, as can be seen by the divergence
- C. of the Caddo word from the Chitimacha word it originated in.
- words for corn in the languages of the Muskogean family evolved from a common root, with the Muscogee word having
- D. lost certain consonant sounds still present in the Chickasaw and Choctaw words.

ID: 35ec767c Answer

Correct Answer: A

Rationale

Choice A is the best answer. The table provides evidence that the words for corn in the Cherokee and Muskogean languages appear unrelated to those in Caddo language, which is described as closely related to the Totozoquean language family. This supports the claim that corn cultivation spread across the Southeast without necessarily spreading Totozoquean vocabulary along with it.

Choice B is incorrect. This choice emphasizes the diversity of Southeastern languages in general, but we’re specifically looking for information about words associated with corn. Choice C is incorrect. This choice shows how words for corn can change and evolve, but we’re looking for an example that shows how some words for corn can be completely unrelated. Choice D is incorrect. This choice shows words for corn that share a common root, but we’re looking for an example that shows how some words for corn can be unrelated.

Question Difficulty: Hard

Question ID 24c1b7e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 24c1b7e4

Percentage Point Changes in US Federal Outlays Relative to GDP by Congressional Status

Period	Congressional status	Change in total outlays	Change in nondefense outlays	Change in defense outlays
1981–1988	divided	–0.4	–1.3	0.9
1975–1976	divided	2.7	3.0	–0.3
1977–1980	undivided	0.3	0.6	–0.3
1964–1968	undivided	1.9	1.4	0.5
1969–1974	divided	–1.8	2.1	–3.9

Economist Steve H. Hanke has shown that divided US Congresses—which occur when one party holds the majority in the House of Representatives and another holds the majority in the Senate—tend to accompany reductions in total federal outlays (spending) relative to gross domestic product (GDP), which Hanke interprets to reflect decreases in government size. Hanke calculated the percentage point change in total outlays (encompassing nondefense and defense outlays) for consecutive US Congresses. Hanke has pointed to his calculations as evidence that a divided Congress may be a “necessary but not sufficient condition” for a decrease in government size to occur.

Which choice best describes data from the table that support the underlined claim?

- The periods of undivided Congresses were associated with increases in nondefense outlays, whereas all the periods of

A. divided Congresses except one were associated with reductions in defense outlays.
- All the periods of divided Congresses were associated with reductions in total outlays, although two periods were also

B. associated with increases in nondefense outlays.
- The periods of undivided Congresses were associated with increases in total outlays, whereas all the periods of divided

C. Congresses were associated with reductions in either nondefense outlays or defense outlays.
- All the periods of divided Congresses except one were associated with reductions in total outlays, whereas the periods of

D. undivided Congresses were associated with increases in total outlays.

Correct Answer: D

Rationale

Choice D is the best answer. The claim is that divided Congresses are necessary but insufficient—that is, we need divide Congresses, but they are not enough—to decrease government size, as measured by total federal outlays. This choice accurately expresses the supporting data from the “change in total outlays” part of the graph. Within the data set, divided Congresses sometimes decreased total outlays, but undivided ones never did.

Choice A is incorrect. The claim is only about government size, as measured by total federal outlays—defense and nondefense outlays aren’t relevant. Choice B is incorrect. The claim is only about government size as measured by total federal outlays—nondefense outlays aren’t relevant. Choice C is incorrect. The claim is only about government size as measured by total federal outlays—specific information about defense or nondefense outlays isn’t relevant.

Question Difficulty: Hard

Question ID 56f477fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 56f477fb

Distribution of Ecosystem Services Affected by Invasive Species by Service Type

Region (Overall)	Provisioning (75%)	Regulating (21%)	Cultural (4%)
West	73%	27%	0%
North	88%	12%	0%
South	79%	14%	7%
East	83%	6%	11%
Central	33%	67%	0%

To assess the impact of invasive species on ecosystems in Africa, Benis N. Egoh and colleagues reviewed government reports from those nations about how invasive species are undermining ecosystem services (aspects of the ecosystem on which residents depend). The services were sorted into three categories: provisioning (material resources from the ecosystem), regulating (natural processes such as cleaning the air or water), and cultural (nonmaterial benefits of ecosystems). Egoh and her team assert that countries in each region reported effects on provisioning services and that provisioning services represent the majority of the reported services.

Which choice best describes data from the table that support Egoh and colleagues’ assertion?

- Provisioning services represent 73% of the services reported for the West region and 33% of those for the Central region,
A. but they represent 75% of the services reported overall.
- None of the percentages shown for provisioning services are lower than 33%, and the overall percentage shown for
B. provisioning services is 75%.
- C. Provisioning services are shown for each region, while no cultural services are shown for some regions.
- The greatest percentage shown for provisioning services is 88% for the North region, and the least shown for
D. provisioning services is 33% for the Central region.

ID: 56f477fb Answer

Correct Answer: B

Rationale

Choice B is the best answer. The finding that all the regions reported at least some effects on provisioning services supports the first part of the assertion. And the fact that provisioning services comprise 75% of the reported services overall supports the second part of the assertion.

Choice A is incorrect. This choice doesn't fully support the assertion. It doesn't include the finding that all the regions (not just Central and West) reported at least some effects on provisioning services. Choice C is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall. Choice D is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall.

Question Difficulty: Hard

Question ID 783d1388

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 783d1388

The Souls of Black Folk is a 1903 book by W.E.B. Du Bois. In the book, Du Bois suggests that upon hearing Black folk songs, he felt an intuitive and sometimes unexpected sense of cultural recognition: _____

Which quotation from *The Souls of Black Folk* most effectively illustrates the claim?

A. “[Black folk music] still remains as the singular spiritual heritage of the nation and the greatest gift of the Negro people.”

“Ever since I was a child these songs have stirred me strangely. They came out of the South unknown to me, one by one, and yet at once I knew them as of me and of mine.”

“Caricature has sought again to spoil the quaint beauty of the music, and has filled the air with many debased melodies which vulgar ears scarce know from the real. But the true Negro folk-song still lives in the hearts of those who have heard them truly sung and in the hearts of the Negro people.”

“The songs are indeed the siftings of centuries; the music is far more ancient than the words, and in it we can trace here and there signs of development.”

ID: 783d1388 Answer

Correct Answer: B

Rationale

Choice B is the best answer because the quotation from *The Souls of Black Folk* illustrates the claim that Du Bois felt a sense of cultural recognition when he heard Black folk songs. In the quotation, Du Bois explains that for his entire life, Black folk songs “stirred [him] strangely.” Even though they originated in the South, a region he wasn’t familiar with, he knew the songs “as of me and of mine.” That is, he identified strongly with them and associated them with his community. Therefore, Du Bois felt an intuitive sense of cultural recognition when he heard Black folk songs.

Choice A is incorrect. Although the quotation considers the cultural and spiritual value of Black folk music, it doesn’t establish that this music inspired in Du Bois a sense of cultural recognition. Choice C is incorrect because this quotation addresses the cultural survival of Black folk songs despite attempts to caricature, or parody, them, not Du Bois’s sense of cultural connection to them. Choice D is incorrect because the quotation indicates that the Black folk songs and music are old, “the siftings of centuries,” instead of addressing how Du Bois felt when he heard the songs.

Question Difficulty: Hard

Question ID 0045c234

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 0045c234

Given that stars and planets initially form from the same gas and dust in space, some astronomers have posited that host stars (such as the Sun) and their planets (such as those in our solar system) are composed of the same materials, with the planets containing equal or smaller quantities of the materials that make up the host star. This idea is also supported by evidence that rocky planets in our solar system are composed of some of the same materials as the Sun.

Which finding, if true, would most directly weaken the astronomers’ claim?

- Most stars are made of hydrogen and helium, but when cooled they are revealed to contain small amounts of iron and
- A. silicate.
- B. A nearby host star is observed to contain the same proportion of hydrogen and helium as that of the Sun.
- Evidence emerges that the amount of iron in some rocky planets is considerably higher than the amount in their host
- C. star.
- The method for determining the composition of rocky planets is discovered to be less effective when used to analyze
- D. other kinds of planets.

ID: 0045c234 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would weaken the astronomers’ claim about the makeup of host stars and their planets. The text explains that because stars and planets begin forming from the same gas and dust, astronomers believe planets should be composed of the same materials as their host stars, but in equal or smaller quantities. The finding that the amount of iron in some rocky planets is much higher than the amount in their host star would weaken the astronomers’ claim because it would show that some planets contain the same material as their host star, but in higher quantities.

Choice A is incorrect because a finding only about the makeup of stars, whether they’ve cooled or not, would provide no information about the makeup of planets. Thus, it wouldn’t have any bearing on the claim that planets and their host stars are composed of the same materials in differing quantities. Choice B is incorrect because a finding about two host stars having similar proportions of certain materials wouldn’t provide any information about the makeup of planets. Thus, it wouldn’t be relevant to the claim that planets and their host stars are composed of the same materials in differing quantities. Choice D is incorrect because the text indicates that the astronomers’ claim is based on a fact—that stars and planets begin forming from the same gas and dust in space—which would remain true regardless of the effectiveness of a method for analysis of compositions. The text does cite analysis of rocky planets in our solar system and the Sun, but only as a single piece of evidence that is consistent with the claim and not as the source of the claim; the finding that the method used for that analysis is less effective in other scenarios wouldn’t weaken a claim that’s based on knowledge of how stars and planets initially form.

Question Difficulty: Hard

Question ID a3fb5e77

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: a3fb5e77

Some animal-behavior studies involve observing wild animals in their natural habitat, and some involve capturing wild animals and observing them in a laboratory. Each approach has advantages over the other. In wild studies, researchers can more easily presume that the animals are behaving normally, and in lab studies, researchers can more easily control factors that might affect the results. But if, for example, the results from a wild study and a lab study of Western scrub-jays (*Aphelocoma californica*) contradict each other, one or both of the studies must have failed to account for some factor that was relevant to the birds’ behavior.

Which choice best states the main idea of the text?

- When the results of a natural-habitat study and those from a lab study of a wild animal such as the Western scrub-jay
- A. conflict, the study in the natural habitat is more likely than the lab study to have accurate results.
- Studying wild animals such as the Western scrub-jay in both their natural habitat and lab settings is likely to yield
- B. conflicting results that researchers cannot fully resolve.
- Wild animals such as the Western scrub-jay can be effectively studied in their natural habitat and in the lab, but each
- C. approach has drawbacks that could affect the accuracy of the findings.
- Differing results between natural-habitat and lab studies of wild animals such as the Western scrub-jay are a strong
- D. indication that both of the studies had design flaws that affected the accuracy of their results.

ID: a3fb5e77 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most accurately states the main idea of the text. The text begins by explaining that wild animals can be studied in their natural habitat or in a laboratory setting, with each setting offering unique advantages to researchers. The text then highlights an instance in which Western scrub-jays were studied in both settings but with conflicting results, indicating that one or both studies may have failed to account for the disadvantages of its research setting. Thus, the main idea of the text is that while wild animals can be effectively studied in natural or lab settings, there are drawbacks to each that need to be considered to ensure accurate results.

Choice A is incorrect because the text does not position one study setting (natural or lab) as superior to the other; rather, the text states that each one has advantages over the other. Choice B is incorrect. The text provides a hypothetical example of two studies in different environments with conflicting results, but this single example cannot be extrapolated to a general claim about the likelihood that results of studies in different environments will conflict. Additionally, the text does not assert anything about how researchers can or cannot resolve conflicting study results. Choice D is incorrect because the text does not state that discrepancies between natural-habitat and lab-based animal behavior studies are due to both of the designs being flawed. Rather, the text states that the conflict in results can be the consequence of one or both of the studies having failed to account for some factor.

Question Difficulty: Hard

Question ID 4fc9a13a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 4fc9a13a

The novelist Toni Morrison was the first Black woman to work as an editor at the publishing company Random House, from 1967 to 1983. A scholar asserts that one of Morrison’s likely aims during her time as an editor was to strengthen the presence of Black writers on the list of Random House’s published authors.

Which finding, if true, would most strongly support the scholar’s claim?

- The percentage of authors published by Random House who were Black rose in the early 1970s and stabilized

A. throughout the decade.
- Black authors who were interviewed in the 1980s and 1990s were highly likely to cite Toni Morrison’s novels as a

B. principal influence on their work.
- The novels written by Toni Morrison that were published after 1983 sold significantly more copies and received wider

C. critical acclaim than the novels she wrote that were published before 1983.
- Works that were edited by Toni Morrison during her time at Random House displayed stylistic characteristics that

D. distinguished them from works that were not edited by Morrison.

ID: 4fc9a13a Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents a finding that, if true, would support the scholar’s claim about Toni Morrison’s likely goal of strengthening the presence of Black writers on Random House’s list of published authors. The text explains that Morrison was the first Black woman to be an editor for Random House and that she was an editor there from 1967 to 1983. If it were true that Random House published a higher percentage of works by Black authors throughout the 1970s—during most of Morrison’s time working there—than it had previously published, that would suggest that Morrison may have made a deliberate effort to strengthen the presence of Black authors on the list of Random House’s published authors, thus supporting the scholar’s claim.

Choice B is incorrect because the scholar’s claim is about Morrison’s work as an editor at a publishing company and her likely effort to strengthen the presence of Black writers on that company’s list of published authors. It might be true that Black authors interviewed in the 1980s and 1990s often cited Morrison’s novels as an influence on their work, but that finding would simply suggest something about how those authors approached their work; it wouldn’t show that Morrison intended to increase the number of Black writers among the published authors specifically at Random House. Choice C is incorrect because the scholar’s claim is about Morrison’s work as an editor at a publishing company, not about her work as a novelist. Therefore, a finding that Morrison’s novels published after 1983 sold more copies and were more widely acclaimed than her earlier novels would have no bearing on the claim that as an editor Morrison made an effort to ensure that more Black writers were present on Random House’s list of published authors. Choice D is incorrect. Although the text discusses Morrison’s work as an editor at Random House, the scholar’s claim focuses on Morrison’s likely effort in that role to increase the number of Black writers present on Random House’s list of published authors, not on the influence she may have had on the content of the works she edited. Without knowing whether Morrison’s stylistic influence led to more publications or if Morrison applied her influence specifically to works by Black writers, the finding that works edited by Morrison could be identified by stylistic characteristics would have no bearing on the claim that Morrison intended to strengthen the presence of Black writers among the published authors at Random House.

Question Difficulty: Hard

Question ID 5d20f560

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 5d20f560

Conventional theories of rhetoric hold that presenting information as coming from credentialed experts increases that information’s credibility. When communications researcher Sungkyoung Lee and her colleagues tested messages seeking volunteers for clinical trials, however, they found that participants in their study judged recruitment messages from former trial volunteers as significantly more credible than messages from doctors (i.e., credentialed experts). One reason for this may be that the doctors’ status as credentialed experts wasn’t ignored but rather was outweighed by participants’ views of the experiential relevance of the two types of messengers; that is, participants may have reacted the way they did because _____

Which choice most logically completes the text?

- messages from former trial volunteers depicted clinical trials as being more positive experiences than did messages from doctors.
- participants did not have enough experience to evaluate the credibility of the doctors’ messages but did have enough experience to evaluate the credibility of former trial volunteers’ messages.
- the fact that former trial volunteers went through the same experience that participants were contemplating while doctors did not was more important to participants than the doctors’ status as credentialed experts was.
- participants regarded the experiences of both the doctors and former trial volunteers as relevant to the subject of clinical trials but were skeptical of the doctors’ status as credentialed experts.

ID: 5d20f560 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents the conclusion that most logically follows from the text's discussion of how participants considered messages from former trial volunteers and doctors. The text first establishes that information coming from credentialed experts typically increases the credibility of that information. However, the text goes on to describe a situation that goes against this expectation: the messages from previous trial volunteers were judged as more credible than the messages from licensed doctors when recruiting clinic trial participants. The text then goes on to speculate as to why this may have been the case, stating that participants likely considered the messages from the former trial volunteers to be more convincing than the messages from the doctors because the former trial volunteers were perceived as having undergone the same experience that the participants were considering. It is reasonable to infer, then, that participants regarded the former trial volunteers' direct experience as more important to their decision-making than the doctors' status as credentialed experts.

Choice A is incorrect because the text does not provide any indication as to the content or the tone of the messages provided by either the doctors or the former trial volunteers regarding the clinical trials. Choice B is incorrect. Since the study participants had not themselves participated in the clinical trial, they could not rely on their experience to evaluate the credibility of either the former trial volunteers or the doctors. Choice D is incorrect. While the text does establish that the participants likely considered the messages from both the doctors and the former trial volunteers, it does not suggest that the participants doubted the doctors' credentials. Rather, the text speculates that participants may have given the experiences of the former trial volunteers more importance than the doctors' credentials when considering the recruitment messages.

Question Difficulty: Hard

Question ID cae97f58

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: cae97f58

Mosses can struggle in harsh desert conditions because these plants require enough sunlight for photosynthesis but not so much that they risk drying out. Researchers Jenna Ekwealor and Kirsten M. Fisher found several species of *Syntrichia caninervis*, a type of desert moss, growing under quartz crystals in California’s Mojave Desert. To evaluate whether these semitransparent rocks benefited the moss, the researchers compared the shoot tissue, a measure of plant growth, of *S. caninervis* when growing on the soil surface versus when the moss was growing under the quartz rocks. They found that the shoot tissue was 62% longer for moss growing under the quartz as compared to moss on the soil surface, suggesting that

Which choice most logically completes the text?

- A. *S. caninervis* is one of the few types of moss that can survive under semitransparent rocks.
- B. quartz crystals do not transmit the necessary sunlight for photosynthesis in *S. caninervis*.
- C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.
- D. quartz crystals are capable of supporting *S. caninervis* growth if the crystals are not too thin.

ID: cae97f58 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text. The text explains that while desert moss species need sufficient sunlight for photosynthesis, exposure to sunlight can also affect the plants negatively by drying them out. Ekwealor and Fisher’s team found specimens of *S. caninervis* moss growing under quartz crystals that were semitransparent, allowing some but not all sunlight to pass through them, and the shoot tissue of these specimens was longer than that of *S. caninervis* specimens found growing on the surface that were unprotected by such crystals. Since, as the text explains, the length of shoot tissue is an indicator of plant growth, and since greater exposure to sunlight results in greater loss of moisture, it can be inferred that *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.

Choice A is incorrect because the text doesn’t mention another moss species besides *S. caninervis* or discuss whether other such species are able to grow under transparent crystals, as *S. caninervis* is. Choice B is incorrect because, as the text explains, specimens of *S. caninervis* were found growing under quartz crystals and exhibited more plant growth than specimens growing on the soil surface. This wouldn’t have been the case if the crystals transmitted insufficient light for the moss’s photosynthesis. Choice D is incorrect. The text contrasts the growth of *S. caninervis* specimens found beneath semitransparent quartz crystals with that of specimens found on the soil surface absent such crystals, but it doesn’t make a comparison among specimens growing beneath crystals of different degrees of thickness or consider how the relative thickness of the crystals affects the growth of the moss.

Question Difficulty: Hard

Question ID 8391a002

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 8391a002

Black beans (*Phaseolus vulgaris*) are a nutritionally dense food, but they are difficult to digest in part because of their high levels of soluble fiber and compounds like raffinose. They also contain antinutrients like tannins and trypsin inhibitors, which interfere with the body’s ability to extract nutrients from foods. In a research article, Marisela Granito and Glenda Álvarez from Simón Bolívar University in Venezuela claim that inducing fermentation of black beans using lactic acid bacteria improves the digestibility of the beans and makes them more nutritious.

Which finding from Granito and Álvarez’s research, if true, would most directly support their claim?

- When cooked, fermented beans contained significantly more trypsin inhibitors and tannins but significantly less soluble
- A. fiber and raffinose than nonfermented beans.
- Fermented beans contained significantly less soluble fiber and raffinose than nonfermented beans, and when cooked, the
- B. fermented beans also displayed a significant reduction in trypsin inhibitors and tannins.
- When the fermented beans were analyzed, they were found to contain two microorganisms, *Lactobacillus casei* and *Lactobacillus plantarum*, that are theorized to increase the amount of nitrogen absorbed by the gut after eating
- C. beans.
- Both fermented and nonfermented black beans contained significantly fewer trypsin inhibitors and tannins after being
- D. cooked at high pressure.

ID: 8391a002 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that would best support Granito and Álvarez’s claim that fermenting black beans makes them easier to digest and more nutritious. The text indicates that high levels of soluble fiber and raffinose in black beans make the beans hard to digest and that tannins and trypsin inhibitors make it harder for the body to extract nutrients from the beans. If it were found that fermenting the beans significantly reduces their levels of soluble fiber, raffinose, trypsin inhibitors, and tannins when cooked, this would directly support the claim that fermentation improves the digestibility of the beans and makes them more nutritious.

Choice A is incorrect because the text indicates that trypsin inhibitors and tannins interfere with the body’s ability to extract nutrients from black beans; if fermentation and cooking were found to increase these antinutrients, fermented beans would likely be less nutritious than unfermented ones, not more nutritious (as Granito and Álvarez claim). Choice C is incorrect because the text doesn’t address the idea that greater nitrogen absorption in the gut has an effect on a food’s digestibility or level of nutrition, so the discovery of the presence of microorganisms that may increase nitrogen absorption wouldn’t provide relevant support for the claim that fermentation makes black beans easier to digest and more nutritious. Choice D is incorrect because Granito and Álvarez’s claim focuses on the effect of fermenting black beans, but the finding that nonfermented black beans also have fewer trypsin inhibitors and tannins when cooked at high pressure would suggest that the role of the cooking method could be significant when it comes to nutrition; further, the finding wouldn’t address the beans’ digestibility.

Question Difficulty: Hard

Question ID 124fdcd7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 124fdcd7

Many archaeologists will tell you that categorizing excavated fragments of pottery by style, period, and what objects they belong to relies not only on standard criteria, but also on instinct developed over years of practice. In a recent study, however, researchers trained a deep-learning computer model on thousands of images of pottery fragments and found that it could categorize them as accurately as a team of expert archaeologists. Some archaeologists have expressed concern that they might be replaced by such computer models, but the researchers claim that outcome is highly unlikely.

Which finding, if true, would most directly support the researchers’ claim?

- In the researchers’ study, the model was able to categorize the pottery fragments much more quickly than the archaeologists could.
- In the researchers’ study, neither the model nor the archeologists were able to accurately categorize all the pottery fragments that were presented.
- A survey of archaeologists showed that categorizing pottery fragments limits the amount of time they can dedicate to other important tasks that only human experts can do.
- A survey of archaeologists showed that few of them received dedicated training in how to properly categorize pottery fragments.

ID: 124fdcd7 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text explains that although archaeologists hold that categorizing pottery fragments relies on both objective criteria and instinct developed through direct experience, researchers have found that a computer model can categorize the fragments with the same degree of accuracy as the humans can—a finding that has caused some archaeologists to worry that their own work won't be needed any longer. If survey results indicate that categorizing pottery fragments limits the amount of time archaeologists can dedicate to other important tasks that only human experts can do, that would mean that computer models aren't able to do all of the important things archaeologists do, thus supporting the researchers' claim that computer models are unlikely to replace human archaeologists.

Choice A is incorrect because if it were true that the computer model could categorize the pottery fragments much more quickly than the archaeologists could, that would weaken the researchers' claim that archaeologists are unlikely to be replaced by certain computer models, since it would demonstrate that the models could conduct the archaeologists' work not only with equal accuracy but also at a faster pace. Choice B is incorrect because the inability of both the computer model and the archaeologists to accurately categorize all of the pottery fragments presented wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text indicates that some archaeologists are worried because the computer model's accuracy is equal to their own, and that could be the case whether both were perfectly accurate or were unable to achieve complete accuracy. Choice D is incorrect because survey results showing that few archaeologists received special training in properly categorizing pottery fragments wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The amount of special training in categorizing pottery fragments that archaeologists have received has no direct bearing on whether computer models' success at categorizing fragments will lead to the models replacing the archaeologists.

Question Difficulty: Hard

Question ID 03701ef3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 03701ef3

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps’ burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow’s measurements both at a moment in time and throughout the entire burrow-building process, something that’s impossible with resin casting because _____

Which choice most logically completes the text?

- A. it can only be used on burrows below a certain size.
- B. it does not allow for multiple castings of the same burrow over time.
- C. the casting process takes more time than *A. bellulus* takes to construct a burrow.
- D. the process of recovering the model distorts the resin’s shape.

ID: 03701ef3 Answer

Correct Answer: B

Rationale

Choice B is the best answer. Since resin casting “inevitably requires destroying the burrow,” it would be impossible to make multiple castings of the same burrow over time.

Choice A is incorrect. The passage doesn’t discuss size requirements for completing resin casting, so there’s no basis for this inference. Choice C is incorrect. The passage never mentions how long the casting process takes nor how long *A. bellulus* takes to construct a borrow, so there’s no basis for this inference. Choice D is incorrect. The passage never states that recovering the model distorts the resin’s shape, only that it destroys the burrow. Therefore, there’s no basis for this inference.

Question Difficulty: Hard